Rational Dynamics Notes

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Prerequisites

From Complex Analysis

Definition (d-fold map)**.** We say a map $f : V \to W$ is a d-*fold map* if, for every w in W, the equation $f(z) = w$ has *exactly* d solutions in V (counting multiple solutions by their multiplicity).

Theorem (Inverse Function Theorem in $\mathbb{C}$)**.** *Let $f : U \to \mathbb{C}$ be a holomorphic function on an open set $U \subseteq \mathbb{C}$, and let $z_0 \in U$. If $f'(z_0) \neq 0$, then there exist open neighborhoods $V \subseteq U$ of z_0 and W of $f(z_0)$ such that the restriction $f : V \to W$ is a bijection.*
Furthermore, the local inverse function $f^{-1} : W \to V$ is holomorphic, and its derivative satisfies:

$$(f^{-1})'(f(z_0)) = \frac{1}{f'(z_0)}.$$

Theorem (Maximum Modulus Theorem)**.** *Let $D \subseteq \mathbb{C}$ be a domain (an open, connected set) and let $f : D \to \mathbb{C}$ be a holomorphic function. If there exists a point $z_0 \in D$ such that $|f(z_0)| \geqslant |f(z)|$ for all $z \in D$, then f must be a constant function on D.*

Lemma (Schwarz's Lemma)**.** *Let $C = \{z \in \mathbb{C} \mid |z| < 1\}$ be the open unit disk, and let $f : C \to C$ be a holomorphic function such that $f(0) = 0$. Then, for all $z \in C$, we have $|f(z)| \leqslant |z|$ and $|f'(0)| \leqslant 1$.*
Furthermore, if $|f(z)| = |z|$ for some non-zero $z \in C$, or if $|f'(0)| = 1$, then f is a rotation, meaning $f(z) = \lambda z$ for some constant $\lambda \in \mathbb{C}$ with $|\lambda| = 1$.

Theorem (Rouché's Theorem)**.** *Let* $D \subseteq \mathbb{C}$ *be a domain, and let* γ *be a simple closed contour in* D *whose interior is also entirely contained in* D*. If* f *and* g *are holomorphic functions on* D *satisfying*

$$|f(z) - g(z)| < |f(z)|$$

for all points z *on* γ*, then* f *and* g *have the exact same number of zeros strictly inside* γ*, counting multiplicities.*

Theorem (Hurwitz's Theorem)**.** *If a sequence of univalent analytic maps* f_n *on a domain* D *converges uniformly to* f*, then* f *is either constant or univalent in* D*.*

From Topology

To support subsequent global dynamic formulations, we briefly review standard topological notions adapted to our setting.

Basic concepts

Definition (Topology)**.** Let X be a set. A *topology* on X is a collection τ of subsets of X that satisfies the following three properties:

- The empty set $\emptyset$ and the set X itself belong to τ.
- The union of any arbitrary collection of sets in τ also belongs to τ.
- The intersection of any finite collection of sets in τ also belongs to τ.

Definition (Topological space)**.** A *topological space* is a pair (X, τ) consisting of a set X and a topology τ on X. When the topology τ is clear from the context, we often simply refer to the space X.

Definition (Open set)**.** Let (X, τ) be a topological space. A subset $U \subseteq X$ is called an *open set* if it belongs to the topology τ, that is, if $U \in \tau$.

Definition (Closed set)**.** A subset $K \subseteq X$ of a topological space is called a *closed set* if its complement, $X - K$, is an open set.

Definition (Closure)**.** Let A be a subset of a topological space X. The *closure* of A, denoted by $\overline{A}$, is the intersection of all closed sets containing A. Equivalently, it is the smallest closed set containing A.

Definition (Dense set)**.** A subset A of a topological space X is called *dense* in X if its closure is the entire space, that is, if $\overline{A} = X$.

Definition (Interior)**.** Let A be a subset of a topological space X. The *interior* of A, denoted by A^o, is the union of all open sets contained within A. Equivalently, it is the largest open set contained in A.

Definition (Boundary)**.** Let A be a subset of a topological space X. The *boundary* (or frontier) of A, denoted by ∂A, is the set of points that are in the closure of A but not in the interior of A. Thus, $\partial A = \overline{A} - A^o$.

Definition (Accumulation points and derived set)**.** Let $A \subseteq X$. A point $x \in X$ is an *accumulation point* (or limit point) of A if every neighborhood of x contains at least one point of A distinct from x. The set of all accumulation points of A is called the *derived set* of A and is denoted by A'.

Definition (Isolated points)**.** Let $A \subseteq X$. A point $x \in A$ is an *isolated point* of A if it is not an accumulation point of A; equivalently, if there exists a neighborhood of x that contains no other points of A besides x.

Definition (Neighborhood)**.** A subset N of a topological space X is a *neighborhood* of a point $x \in X$ if there exists an open set U such that $x \in U \subseteq N$. If N itself is an open set, it is frequently referred to as an *open neighborhood*.

Metric spaces

Definition (Metric or distance)**.** Let X be a non-empty set. A *metric* (or *distance*) (on X) is a mapping

$$\begin{aligned} \sigma : X \times X &\to [0, \infty) \\ (x, y) &\mapsto \sigma(x, y) \end{aligned}$$

verifying the following for every $x, y, z \in X$:

- $\sigma(x, y) = 0 \iff x = y$;
- $\sigma(x, y) = \sigma(y, x)$; and
- $\sigma(x, z) \leqslant \sigma(x, y) + \sigma(y, z)$.

Definition (Metric space)**.** A metric space is a tuple (X, σ) where X is a non-empty set and σ is a metric on X.

Definition (Distance from a point to a set)**.** Let (X, σ) be a metric space, $A \subseteq X$ a non-empty set, and $x \in X$. The *distance* between the point x and the set A is defined as

$$\sigma(x, A) = \inf_{y \in A} \sigma(x, y).$$

Definition (Distance between two sets)**.** Let A and B be non-empty subsets of a metric space (X, σ). The *distance* between A and B is defined as

$$\sigma(A, B) = \inf_{\substack{x \in A \\ y \in B}} \sigma(x, y).$$

Definition (Diameter of a set)**.** Let A be a non-empty subset of a metric space (X, σ). The *diameter* of A is defined as

$$\operatorname{diam}(A) = \sup_{x,y \in A} \sigma(x, y).$$

Definition (Ball)**.** Let (X, σ) be a metric space, $x \in X$, and $r > 0$. The *open ball* (or simply the *ball*) of radius r centered at x is the set

$$B(x, r) = \{y \in X \mid \sigma(x, y) < r\}.$$

Similarly, the *closed ball* of radius r centered at x is the set

$$\overline{B}(x, r) = \{y \in X \mid \sigma(x, y) \leqslant r\}.$$

Definition (Bounded set)**.** A subset A of a metric space (X, σ) is *bounded* if its diameter is finite, that is, if $\operatorname{diam}(A) < \infty$. Equivalently, A is bounded if it is contained in a ball (open or closed).

Definition (Open set)**.** A subset U of a metric space (X, σ) is *open* if, for every point $x \in U$, there exists a radius $r > 0$ such that the ball $B(x, r)$ is entirely contained in U.

Compactness

Definition (Compact Set)**.** A set K is *compact* if every open cover of K contains a finite subcover. In a metric space, this is equivalent to *sequential compactness*, where every sequence in K has a convergent subsequence whose limit is in K.

By the Heine-Borel theorem, compactness for a subset $K \subseteq \mathbb{C}$ is equivalent to K being closed and bounded. However, because the extended plane $\mathbb{C}_\infty$ is itself already compact (being homeomorphic to the sphere S^2), bounds are redundant. The transformation from $\mathbb{C}$ to $\mathbb{C}_\infty$ represents a standard one-point or Alexandroff compactification.

Definition (Compact set in $\mathbb{C}_\infty$)**.** A subset $K \subseteq \mathbb{C}_\infty$ is compact if and only if it is closed in the chordal topology.

Connectedness

Definition (Connected Set)**.** A set A is *connected* if it cannot be partitioned by two disjoint, non-empty open sets V and W such that $A \subseteq V \cup W$.

Definition (Component of a set)**.** Given a set B, we say a subset A of B is a component of B if it is a maximal connected subset.

Definition (Path-Connected Set)**.** A set $A \subseteq \mathbb{C}_\infty$ is *path-connected* if for every pair of points $z_0, z_1 \in A$, there exists a continuous map (a path) $\gamma : [0,1] \to A$ such that $\gamma(0) = z_0$ and $\gamma(1) = z_1$.

In general topology, path-connectedness is a strictly stronger condition than connectedness; every path-connected set is connected, but the converse is false (as exemplified by the Topologist's Sine Curve).

Definition (Domain)**.** An open, connected subset $D \subseteq \mathbb{C}_\infty$ is called a *domain*.

The regular geometry of open sets yields an equivalence vital to complex dynamics.

Theorem (An open set is connected $\iff$ is path-connected)**.** *Let* $U \subseteq \mathbb{C}_\infty$ *be an open set. Then* U *is connected if and only if it is path-connected.*

Consequently, any domain D is automatically path-connected. Thus, domains could also be defined as open, path-connected subsets of $\mathbb{C}_\infty$.

Jordan Curves

To properly interpret the connectivity of domains and properties of the Fatou and Julia sets, we recall the basic topology of curves on the Riemann sphere

$\mathbb{C}_\infty$.

Definition (Jordan Arc and Jordan Curve)**.** A *Jordan arc* is the image of a continuous, injective map $\gamma : [0,1] \to \mathbb{C}_\infty$. A *Jordan curve* (or a simple closed curve) is the image of a continuous map $\gamma : [0,1] \to \mathbb{C}_\infty$ that is injective on $[0,1)$ and satisfies $\gamma(0) = \gamma(1)$.

The fundamental global property of Jordan curves is their ability to divide the sphere into distinct regions, a result which is heavily utilized in complex dynamics when defining domain connectivity.

Theorem (Jordan Curve Theorem)**.** *Let γ be a Jordan curve in $\mathbb{C}_\infty$. Then its complement $\mathbb{C}_\infty - \gamma$ consists of exactly two disjoint connected components, each of which is an open domain having γ as its common boundary.*

While the Jordan Curve Theorem guarantees exactly two components, complex dynamics frequently requires these components to be topologically well-behaved. This regular geometry is guaranteed by a stronger two-dimensional extension.

Theorem (Schoenflies Theorem for the Sphere)**.** *Let γ be a Jordan curve in $\mathbb{C}_\infty$. Then there exists a homeomorphism $\Phi : \mathbb{C}_\infty \to \mathbb{C}_\infty$ that maps γ directly onto the standard equatorial circle. Consequently, the two components of $\mathbb{C}_\infty - \gamma$ are homeomorphic to open discs, and their closures are homeomorphic to closed discs (topological discs).*

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Chapter 1

Examples

In this chapter we introduce some of the main ideas in iteration theory by discussing a variety of simple examples. The discussions involve only elementary mathematics, and our sole objective is to illustrate and stress those features that will be met in a general context later.

1.1 Introduction

Proposition 1.1 ($z_n \xrightarrow[n\to\infty]{} w \implies w$ is a fixed point)**.** *Let $z_0 \in \mathbb{C}_\infty$, R be a rational function in $\mathbb{C}_\infty$, and $z_n = R^n(z_0)$. Then if z_n converges when n goes to infinity, $\lim_{n\to\infty} z_n = w$ is a fixed point of R.*

Proof. Since R is continuous (everywhere where it is defined, i.e., the denominator is not zero) because it is a rational function, the following is true:

$$w = \lim_{n\to\infty} z_n = \lim_{n\to\infty} R(z_{n-1}) = R\left(\lim_{n\to\infty} z_{n-1}\right) = R(w).$$

Then, $w = R(w)$, so it is a fixed point. □

Intuition: Let's see why the derivative can work for determining if a point is attracting or repelling. Since $|R(z) - \zeta| = |R(z) - R(\zeta)| \simeq |R'(\zeta)||z - \zeta|$ if z is close enough to ζ, we have that if $|R'(\zeta)| < 1$, $z_n \to \zeta$, and if $|R'(\zeta)| > 1$, z_n will be initially repelled from ζ. (A rigorous contradiction proof follows in the notes ensuring z_{n+1} cannot arbitrarily stay closer than z_n if repelled).

1.2 Iteration of Möbius Transformations

Definition 1.1 (Möbius Transformation)**.** A *Möbius transformation* is a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ of the form

$$R(z) = \frac{az + b}{cz + d},$$

where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$.

Theorem 1.2 (Classification of Möbius Transformations)**.** *Given* R *a Möbius transformation, then either it has:*

(i) One fixed point, ζ*, and for every* $z \in \mathbb{C}$*,* $R^n(z) \to \zeta$*.*

(ii) Two fixed points, ζ_1 *and* ζ_2*, and for every* $z \in \mathbb{C}_\infty$ *either:*

(a) $R^n(z) \to \zeta_1$

(b) $R^n(z) \to \zeta_2$

(c) $R^n(z)$ *moves cyclically over a finite set of points.*

(d) $\{R^n(z) : n \in \mathbb{N}_0\}$ *forms a dense subset of a circle.*

Proof. First, observe that R cannot have 0 fixed points. Solving $R(z) = z$:

$$\frac{az + b}{cz + d} = z \implies cz^2 + (d - a)z - b = 0$$

This is a quadratic equation, so it always has 1 or 2 solutions in $\mathbb{C}_\infty$ (if $c = 0$, ∞ is a fixed point).
Case 1: One fixed point. Suppose the fixed point is ∞. Then $R(z) = z + \beta$ for some $\beta \in \mathbb{C}$:

$$R(\infty) = \infty \implies c = 0 \implies R(z) = \frac{a}{d}z + \frac{b}{d} = \alpha z + \beta.$$

We also want R to have no other fixed point:

$$R(z) = z \iff \alpha z + \beta = z \iff (\alpha - 1)z + \beta = 0,$$

hence β has to be $\neq 0$ (otherwise $z = 0$ is a fixed point) and $\alpha = 1$ (otherwise $z = \beta/(\alpha - 1)$ is a fixed point).

If R has a general fixed point ζ, we can define the Möbius transformation S by using the conjugation $g(z) = \frac{1}{z-\zeta}$ and $S = g \circ R \circ g^{-1}$. Since conjugation preserves the number of fixed points, S has exactly one fixed point at ∞, meaning S is a translation $S(z) = z + \beta$. Thus, $S^n(z) \to \infty$, which implies $R^n(z) \to \zeta$.
Case 2: Two fixed points. Suppose the fixed points are 0 and ∞ (it can be shown similarly to the previous case). Then $R(z) = kz$ for some $k \in \mathbb{C} \setminus \{0, 1\}$. Analyzing $R^n(z) = k^n z$:

- If $|k| < 1$, then $R^n(z) \to 0$.
- If $|k| > 1$, then $R^n(z) \to \infty$.
- If $|k| = 1$ and k is a root of unity, it cycles over a finite set of points.
- If $|k| = 1$ and k is not a root of unity, the subset will be dense in a circle $|z| = |z_0|$.

For a general map with two fixed points ζ_1, ζ_2, we conjugate via $g(z) = \frac{z-\zeta_1}{z-\zeta_2}$ to place the fixed points at 0 and ∞, yielding the same behavioral classification. □

1.3 Iteration of $z \mapsto z^2$

Definition 1.2 (Forward and Backward Invariant)**.** Given a set $E \subseteq \mathbb{C}_\infty$ and a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$, we say E is:

(a) *Forward invariant* if $R(E) \subseteq E$.

(b) *Backward invariant* if $R^{-1}(E) \subseteq E$.

Definition 1.3 (Periodic Point)**.** Given a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ and $z \in \mathbb{C}_\infty$, we say z is *periodic with period* k if $R^k(z) = z$.

In the Julia set of $z \mapsto z^2$ (which is $J = \{z : |z| = 1\}$), two close points may behave radically differently. And it has this properties:

(i) Given I an arc in J, $\exists N \in \mathbb{N}_0$ such that $\forall n \geqslant N$, $R^n(I) = J$.

(ii) $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is dense in J.

(iii) $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is finite.

1.4 Tchebychev Polynomials

Theorem 1.3 (Characterization of Tchebychev polynomials)**.** *Suppose that* T *is a polynomial of degree* k*, where* $k \geqslant 2$*. Then, the interval* $[-1, 1]$ *is both forward and backward invariant under* T *if and only if* T *is* T_k *or* $-T_k$*.*

Proof. See [7, pp. 11–12] □

1.5 Iteration of $z \mapsto z^2 - 1$

For polynomials like $P(z) = z^2 - 1$, points outside a certain radius escape to infinity ($z \in F_\infty \implies P^n(z) \to \infty$).

$P(F_0) = F_{-1}$, $P(F_{-1}) = F_0$.

$P(\Omega) \to \ldots, F_0, F_{-1}, F_0, \ldots$, Ω a Fatou set $\neq F_\infty$.

1.6 Iteration of $z \mapsto z^2 + c$

Recall Vieta's formulas. Given a polynomial $P(x) = ax^2 + bx + c$ with roots r_1, r_2:

$$r_1 + r_2 = -\frac{b}{a}, \quad r_1 r_2 = \frac{c}{a}.$$

In general, given a polynomial of degree n, $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$

$$\sum_{i=1}^{n} r_i = -\frac{a_{n-1}}{a_n}, \quad \prod_{i=1}^{n} r_i = (-1)^n \frac{a_0}{a_n},$$

and

$$\sum_{1 \leqslant i < j \leqslant n}^{n} r_i r_j = \frac{a_{n-2}}{a_n}, \quad \sum_{1 \leqslant i < j < k \leqslant n}^{n} r_i r_j r_k = \frac{a_{n-3}}{a_n}, \quad \cdots$$

A second degree polynomial can be transformed into a composition of two affine transformations and z^2:

$$P(z) = z - z^2 = -(z^2 - z) = -\left(z^2 - z + \frac{1}{4} - \frac{1}{4}\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right).$$

Then defining

$$h(z) = z - \frac{1}{2} \quad , \quad g(z) = z^2 \quad , \quad f(z) = -\left(z - \frac{1}{4}\right) = \frac{1}{4} - z,$$

we have

$$P(z) = f \circ g \circ h(z) = f \circ g\left(z - \frac{1}{2}\right) = f\left(\left(z - \frac{1}{2}\right)^2\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right) = -(z^2 - z).$$

1.7 Iteration of $z \mapsto z + 1/z$

We consider the dynamics of the rational map defined by:

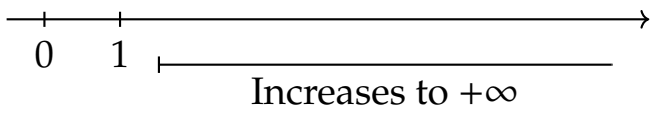
$$R(z) = z + \frac{1}{z}$$

Real Axis:

For real values $z = x \in \mathbb{R}$ with $x > 1$:

$$R(x) = x + \frac{1}{x} > x.$$

The sequence of iterates increases monotonically towards $+\infty$.

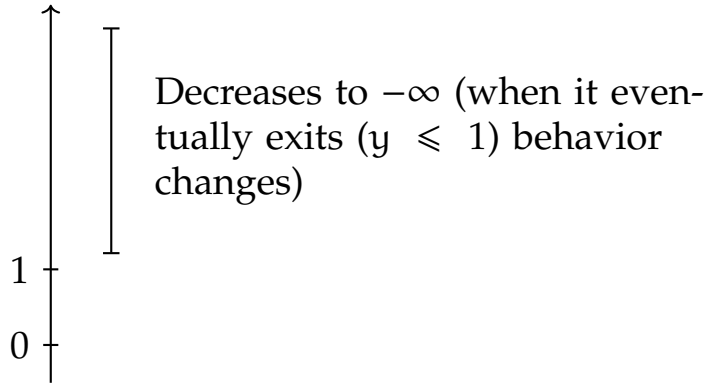


Imaginary Axis:

For purely imaginary values $z = iy \in i\mathbb{R}$ with $y > 1$:

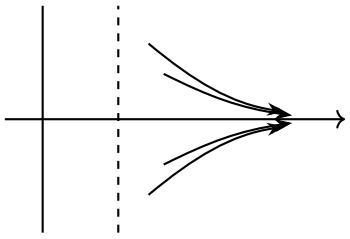
$$R(iy) = iy + \frac{1}{iy} = i\left(y - \frac{1}{y}\right).$$

Since $y > 1$, $y - \frac{1}{y} < y$. Thus, the imaginary part decreases towards 0 as long as $y > 1$. Once it eventually exits this regime ($y \leqslant 1$), the behavior changes completely.



Region $x > 1$:

In the half-plane where $\mathrm{Re}(z) > 1$, the trajectories flow outwards towards the right along curves that flatten asymptotically.



Analyzing the Conjugate Map:

We can understand the behavior at infinity by conjugating via $g(z) = \frac{1}{z}$, which maps the neighborhood of ∞ to a neighborhood of the origin. Let

$S = g \circ R \circ g^{-1}$. Since $g^{-1}(z) = \frac{1}{z}$, we have:

$$S(z) = g\left(R\left(\frac{1}{z}\right)\right) = \frac{1}{\frac{1}{z} + z} = \frac{z}{1 + z^2},$$

where $g = \frac{1}{z}$. See figure 1.1 for the action of R near ∞.

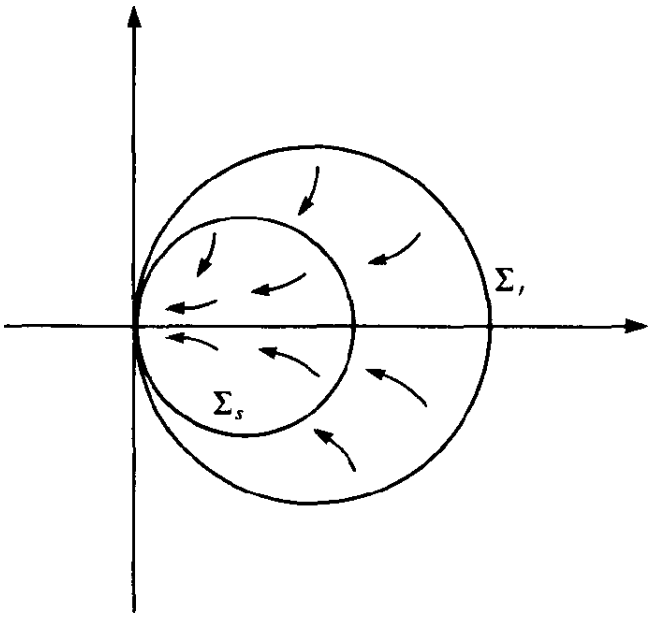


Figure 1.1: Action of $z \mapsto z + 1/z$ near ∞.

1.8 Iteration of $z \mapsto 2z - 1/z$

For $R(z) = 2z - 1/z$, we find the fixed points:

- $\zeta_1 = 1$ (repelling),
- $\zeta_2 = -1$ (repelling),
- $\zeta_3 = \infty$ (attracting).

1.9 Newton's Approximation

To find the roots of a polynomial function f in $\mathbb{R}$, the Newton-Raphson method uses the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Given f in $\mathbb{C}$, we can define the rational function:

$$R(z) = z - \frac{f(z)}{f'(z)}.$$

And the analysis of these kinds of functions (rational functions), falls within the scope of this text.

Note that, in $\mathbb{R}$ that kind of approximation works, but in $\mathbb{C}_\infty$ ... the conditions of z_0 to converge to a root are hard and have to do with the Julia set of R (which for polynomials f with degree 3 or more is usually a complex fractal).

Properties of the Newton method:

(i) The roots of f correspond with fixed points of R (proved in the book [7]).

(ii) Successive iterates of R have a tendency to converge to an attracting fixed point of R.

Thus, if ζ is a zero of f which gives rise to an attracting fixed point of F, then,

$$z_n = F^n(z_0) \xrightarrow[n\to\infty]{} \zeta$$

for at least initial guesses z_0 of ζ sufficiently accurate.

When z_0 is far from ζ the problem of deciding if $z_n \to \zeta$ is much harder.

In the book ([7]) there is analysis of polynomials of degree 2 with distinct roots.

A formulation of a theorem for polynomial with real distinct roots: z_n converges to a root of P for essentially every z_0 (except from a set with area 0)

1.10 General Remarks

For typical rational functions R, the extended complex plane $\mathbb{C}_\infty$ is divided into J (Julia set) and F (Fatou set).

- F preserves proximity.
- J is in every sense chaotic.
- J and F are both forward and backward invariant.

Chapter 2

Rational Maps

We present here some of the elementary properties of rational maps. A point ∞ is adjoined to the complex plane and a rational map of degree d is considered as a d-fold map of the extended complex plane onto itself. The spherical and chordal metrics are introduced, the important notions of conjugacy, valency, fixed points and critical points are discussed, and the Riemann-Hurwitz relation is proved.

2.1 The Extended Complex Plane

Definition 2.1 (Extended Complex Plane)**.** The *extended complex* plane is the union

$$\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\},$$

where ∞ is an "abstract" point outside $\mathbb{C}$.

Now let's define a metric in $\mathbb{C}_\infty$ that handles all points with equal ease and makes ∞ lose any special significance: *the chordal metric*.

Let's start by defining the *Riemann Sphere*. For that, we identify $\mathbb{C}$ with the horizontal plane

$$\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_3 = 0\}$$

in $\mathbb{R}^3$. Let S be the unit sphere in $\mathbb{R}^3$ centered at the origin, and $N = (0, 0, 1)$ be the north pole.

We now project each point in $\mathbb{C}$ to a point in S by taking the intersection of the unit sphere S with the line passing through N and the point $(x_1, x_2, 0)$. An illustration of this projection is given in Figure 2.1.

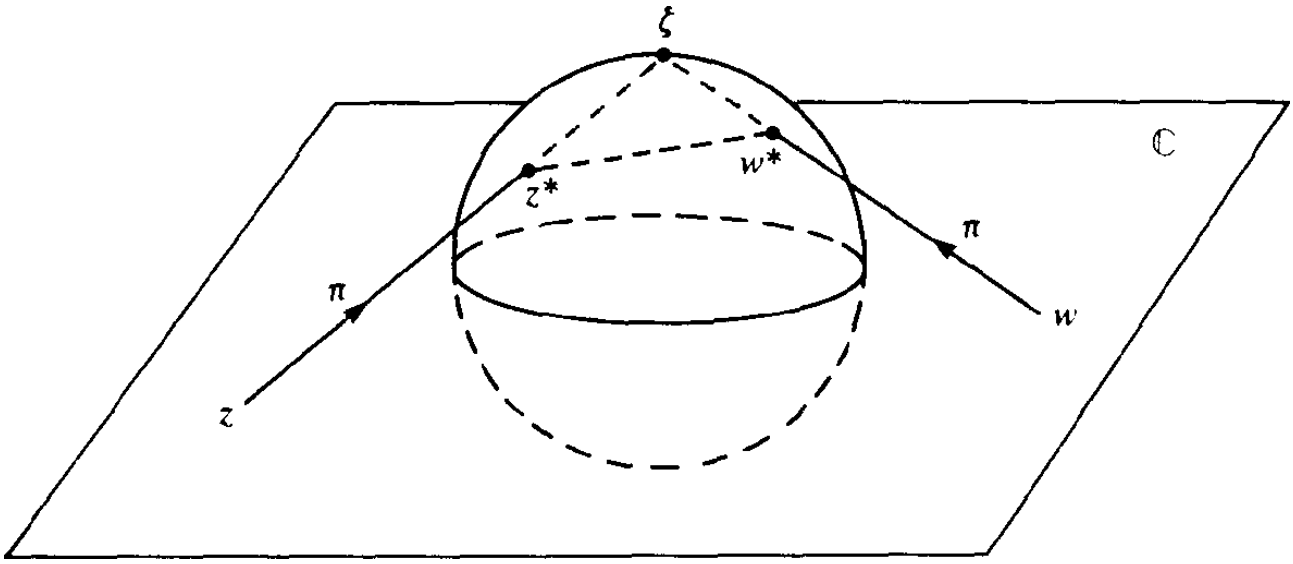


Figure 2.1: Stereographic projection.

We identify this projection as:

$$\pi : \mathbb{C}_\infty \longrightarrow S,$$

where for $z = x + iy \in \mathbb{C}$:

$$\pi(z) = \left(\frac{2x}{1+|z|^2}, \frac{2y}{1+|z|^2}, \frac{|z|^2-1}{1+|z|^2} \right),$$

and $\pi(\infty) = (0, 0, 1)$.

Definition 2.2 (Chordal Metric)**.** Given the projection π from $\mathbb{C}_\infty$ to S, we define the *chordal metric*, $\sigma : \mathbb{C}_\infty \times \mathbb{C}_\infty \implies \mathbb{R}^+ \cup \{0\}$, as the Euclidean distance from $\pi(z)$ to $\pi(w)$:

$$(z, w) \mapsto \sigma(z, w) = |\pi(z) - \pi(w)|.$$

Explicitly, when $z, w \in \mathbb{C}$:

$$\sigma(z, w) = \frac{2|z-w|}{(1+|z|^2)^{1/2}(1+|w|^2)^{1/2}}.$$

For the point at infinity, we have:

$$\sigma(z, \infty) = \lim_{w \to \infty} \sigma(z, w) = \frac{2}{(1+|z|^2)^{1/2}}.$$

Remark 2.1 ($z \mapsto 1/z$ is an isometry). $\sigma(z, w) = \sigma(z^{-1}, w^{-1})$, i.e., the map $z \mapsto \frac{1}{z}$ is an isometry.

Remark 2.2 (Rotation in $S \mapsto$ isometry in $\mathbb{C}_\infty$). Any rotation $\varphi : S \implies S$ of the sphere induces a map $\pi^{-1}\varphi\pi : \mathbb{C}_\infty \implies \mathbb{C}_\infty$ which is an isometry. These isometries are Möbius transformations of the form:

$$z \mapsto \frac{az - \bar{c}}{cz + \bar{a}},$$

where $|a|^2 + |c|^2 = 1$.

The chordal metric handles the point at infinity with ease, but it measures distance through the three-dimensional space containing the Riemann sphere. If we restrict our measurements strictly to the two-dimensional curved surface of the sphere, we obtain the spherical metric.

Definition 2.3 (Spherical Metric)**.** Given the stereographic projection $\pi : \mathbb{C}_\infty \to S$, the *spherical distance* $\sigma_0(z, w)$ between two points in $\mathbb{C}_\infty$ is defined as the Euclidean length of the shortest path (an arc of a great circle) connecting their projections $\pi(z)$ and $\pi(w)$ on the surface of S.

If the chord joining the projected points subtends an angle θ at the origin of the sphere, then by definition the spherical distance is exactly $\sigma_0(z, w) = \theta$. By using the trigonometric connection to the chordal metric, we can derive explicit formulas directly in terms of the complex coordinates for $z, w \in \mathbb{C}$:

$$\sigma(z, w) = 2 \sin\left(\frac{\theta}{2}\right) = 2 \sin\left(\frac{\sigma_0(z, w)}{2}\right).$$

Thus,

$$\begin{aligned}\sigma_0(z, w) &= 2 \arcsin\left(\frac{\sigma(z, w)}{2}\right) \\ &= 2 \arcsin\left(\frac{|z - w|}{(1 + |z|^2)^{1/2}(1 + |w|^2)^{1/2}}\right).\end{aligned}$$

For the point at infinity, taking the limit as $w \to \infty$ yields:

$$\sigma_0(z, \infty) = 2 \arcsin\left(\frac{1}{(1 + |z|^2)^{1/2}}\right).$$

Remark 2.3 ($\sigma \sim \sigma_0$). The chordal and spherical metrics are topologically equivalent ($\sigma(z,w) = 2\sin(\theta/2)$). Furthermore, they satisfy the following inequalities:

$$\frac{2}{\pi}\sigma_0(z,w) \leqslant \sigma(z,w) \leqslant \sigma_0(z,w)$$

Proof. The inequalities follow directly from applying the elementary real-variable inequality $\frac{2\theta}{\pi} \leqslant \sin\theta \leqslant \theta$ on the interval $0 \leqslant \theta \leqslant \frac{\pi}{2}$ to the angle $\theta = \frac{\sigma_0(z,w)}{2}$. □

Because these metrics differ by at most a bounded scale factor of $\frac{2}{\pi}$, we can use σ and σ_0 interchangeably throughout our study of rational dynamics whenever global constants are allowed to change.

Remark 2.4 (Spherical length). Given a continuously differentiable curve $\gamma : I \implies \mathbb{C}_\infty$, its *spherical length* is given by:

$$\int_\gamma \frac{2|dz|}{1+|z|^2}.$$

2.2 Rational Maps

Definition 2.4 (Rational Map)**.** A *rational map* is a function of the form

$$R(z) = \frac{P(z)}{Q(z)},$$

where $P(z)$ and $Q(z)$ are polynomials, not both being the zero polynomial.

- If $P = 0 \implies R = 0$.
- If $Q = 0 \implies R = \infty$.
- If $P(z_0) \neq 0$ and $Q(z_0) = 0$ for some $z_0 \in \mathbb{C}$, then $R(z_0) = \infty$.
- We define $R(\infty) = \lim_{z\to\infty} R(z)$.

- If $P(z_0) = 0$ and $Q(z_0) = 0$, strictly speaking $R(z_0)$ is not defined, but in this case we may cancel the common factors:

$$P(z) = (z - z_0)^m P_1(z)$$
$$Q(z) = (z - z_0)^n Q_1(z)$$

where $P_1(z_0) \neq 0$ and $Q_1(z_0) \neq 0$. Then:

$$R(z) = \frac{P_1(z)}{Q_1(z)}(z - z_0)^{m-n}.$$

From now on, we will consider that this has been done, i.e., P and Q are coprime. Thus, given R, the polynomials P and Q are uniquely determined up to scalar multiplication. This allows us to define the degree of a rational map.

Definition 2.5 (Degree of a Rational Map, $\deg(R)$)**.** Given a rational map $R = P/Q$ where P and Q are coprime polynomials, the *degree of* R is defined as:

$$\deg(R) = \max\{\deg(P), \deg(Q)\}$$

where $\deg(\cdot)$ is the usual degree of a polynomial. We will consider that if $R = \alpha \in \mathbb{C}_\infty$, then $\deg(R) = 0$.

Proposition 2.1 ($\deg(RS) = \deg(R)\deg(S)$)**.** *Given two rational function* R *and* S*, the degree of* RS *is* $\deg(R)$ *times* $\deg(S)$*. That is,*

$$\deg(RS) = \deg(R)\deg(S).$$

Remember that by writing RS *we mean composition (*$R \circ S$*) not multiplication. However,* $\deg(R)\deg(S)$ *is a multiplication.*

Proof. See [7, p. 32]. □

Corollary 2.1.1 ($\deg(R^n) = [\deg(R)]^n$)**.** *Given a rational function* R*, the degree of* R^n *is the* n*-th power of* $\deg(R)$*. That is,*

$$\deg(R^n) = [\deg(R)]^n.$$

Now, let's remember the definitions of *holomorphic*, *meromorphic* and *analytic* functions in $\mathbb{C}$ and see how they translate to $\mathbb{C}_\infty$ with the chordal metric.

Definition 2.6 (Holomorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \implies \mathbb{C}$ is *holomorphic* in D if the derivative of f exists at each point in D.

Definition 2.7 (Meromorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \implies \mathbb{C}_\infty$ is *meromorphic* in D if each point in D has a neighborhood on which either f or $1/f$ is holomorphic.

Definition 2.8 (Pole of f)**.** Given a meromorphic function f in $D \subseteq \mathbb{C}$, the *poles of* f are the points $w \in D$ where $f(w) = \infty$ ($\lim_{z \to w} f(z) = \infty$).

Remark 2.5 (f meromorphic $\implies$ $1/f$ holomorphic at f's poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then $1/f$ is holomorphic near w and $\frac{1}{f(w)} = 0$.

Definition 2.9 (f is defined near ∞)**.** A function f is said to be *defined near* ∞ (or *in a neighborhood of* ∞) if it is defined on some set $\{z \in \mathbb{C} \mid |z| > r\} \cup \{\infty\}$.

Definition 2.10 (Holomorphic or Meromorphic at ∞)**.** Given a function f defined near ∞, f is *holomorphic* (or *meromorphic*) *at* ∞ if $f(1/z)$ is holomorphic (or meromorphic) near 0.

Definition 2.11 (Analytic in $D \subseteq \mathbb{C}_\infty$)**.** Given a map $f : D_1 \implies D_2$ between any two subdomains of $\mathbb{C}_\infty$, we say f is *analytic* if it is holomorphic or meromorphic at each point in D_1.

Remark 2.6 (f meromorphic $\implies$ f continuous at its poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then (using the chordal metric and its topology), f is continuous at the pole. This is because

$z \mapsto 1/z$ is a σ-isometry:

$$\sigma(f(z), f(w)) = \sigma\left(\frac{1}{f(z)}, \frac{1}{f(w)}\right) = \sigma\left(\frac{1}{f(z)}, 0\right) = \frac{2|1/f(z)|}{(1+|1/f(z)|^2)^{1/2}} \xrightarrow{z \Longrightarrow w} 0$$

Summing up, we have that the concepts of *holomorphic*, *meromorphic* and *analytic* functions translate well from $\mathbb{C}$ with the Euclidean metric, to $\mathbb{C}_\infty$ with the chordal or spheric metric. Hence, making ∞ lose any special significance it may have had in $\mathbb{C}$.

Proposition 2.2 (Polynomials are analytic in $\mathbb{C}_\infty$)**.** *Given a polynomial* $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ *where* $n > 0$ *and* $a_n \neq 0$*. Then,* P *is analytic in* $\mathbb{C}_\infty$*.*

Proof. This proof serves to illustrate what holomorphic and meromorphic mean. We have to show that P is holomorphic or meromorphic at each point in $\mathbb{C}_\infty$. We have that P is analytic in $\mathbb{C}$ because it's a polynomial, thus, we only have to show that P is holomorphic or meromorphic at ∞. We'll see that it's meromorphic at ∞, i.e., $P(1/z)$ is meromorphic at 0, i.e., $1/P(1/z)$ is holomorphic at 0. And we can see that

$$\frac{1}{P(1/z)} = \frac{z^n}{a_0 z^n + \cdots + a_n}$$

is differentiable at 0, therefore, P is meromorphic at ∞. □

In this proof we have also seen that all polynomials have a pole in ∞, that is, $P(\infty) = \infty$. We will see that this means that ∞ is a fixed point for all polynomials.

Theorem 2.3 (Analytic in $\mathbb{C}_\infty \iff$ rational map)**.** *The rational maps are precisely the analytic maps of* $\mathbb{C}_\infty$ *into itself.*

Definition 2.12 (d-fold map)**.** We say a map $f : V \to W$ is a d*-fold map* if, for every w in W, the equation $f(z) = w$ has *exactly* d solutions in V (counting multiple solutions by their multiplicity).

Theorem 2.4 (Rational maps are d-fold maps in $\mathbb{C}_\infty$)**.** *Given* R *a rational function of positive degree* d*. Then* R *is a* d*-fold map of* $\mathbb{C}_\infty$ onto *itself.*

Proof. See [7, pp. 31–32]. □

Remark 2.7 (Zeros and poles of rational maps). Given $R = P/Q$ a rational function of positive degree d, where P and Q are coprime polynomials with $\deg(P) = n$ and $\deg(Q) = m$. Then, the zeros of R are the zeros of P, and the poles of R are the zeros of Q. Moreover, the number of zeros and poles of R at ∞ (counting multiplicities) are given by the following cases:

(i) if $n = m$, then all the zeros and poles of R are in $\mathbb{C}$;

(ii) if $n > m$, then the number of poles at ∞ is $n - m$;

(iii) and if $n < m$, then the number of zeros at ∞ is $m - n$.

2.3 The Lipschitz Condition

Theorem 2.5 (Rational maps satisfy a Lipschitz condition)**.** *A rational map* R *satisfies some* Lipschitz condition

$$\sigma_0(R(z), R(w)) \leqslant M\sigma_0(z, w)$$

on the complex sphere.

Proof. See [7, p. 33]. □

Theorem 2.6 (Möbius transformations explicit Lipschitz condition)**.** *The Möbius map*

$$g(z) = \frac{az + b}{cz + d}, \quad ad - bc = 1,$$

satisfies the Lipschitz condition

$$\sigma_0(g(z), g(w)) \leqslant \|g\|^2 \sigma_0(z, w).$$

Proof. See [7, pp. 33–34]. □

Theorem 2.7 (Möbius transformations family explicit uniform Lipschitz condition)**.** *Let* m *be any positive number. Then the Möbius transformations* g *which satisfy*

$$\begin{aligned} \sigma(g(0),\ g(1)) &\geqslant m, \\ \sigma(g(1),\ g(\infty)) &\geqslant m, \\ \sigma(g(\infty),g(0)) &\geqslant m \end{aligned}$$

also satisfy the uniform Lipschitz condition

$$\sigma(g(z), g(w)) \leqslant \frac{\pi}{m^3} \sigma(z, w).$$

Proof. See [7, pp. 34–35]. □

Definition 2.13 (Complementary components of a closed curve γ)**.** Let γ be a *closed curve* on the sphere. The complement of γ is a union of mutually disjoint domains, which we call the *complementary components* of γ.

Definition 2.14 (Exterior and interior of a closed curve γ contained in a open hemisphere)**.** Let γ be a *closed curve* that lies in some *open hemisphere*. Then exactly one of the complementary components contains a hemisphere; we denote this by $E(\gamma)$ and call it the *exterior* of γ. The other complementary components of γ are said to be the *interior components* of γ.

Theorem 2.8 (Interior components mapping theorem for rational maps)**.** *Let* R *be a rational function. Then there is a positive number* δ *such that if* γ *is any closed curve of* σ_0*-diameter less than* δ*, then the image* $R(\Omega)$ *of an interior component* Ω *of* γ *does not meet the exterior* $E(R(\gamma))$ *of* $R(\gamma)$*.*

Proof. See [7, pp. 35–36]. □

This result almost says that if the curve γ is sufficiently small, then the interior components of γ map to the interior components of the image curve $R(\gamma)$: however, the images of the interior components may also contain points of $R(\gamma)$, so it is necessary to express this a little more carefully in terms of the exterior of the image curve (Figure 2.2).

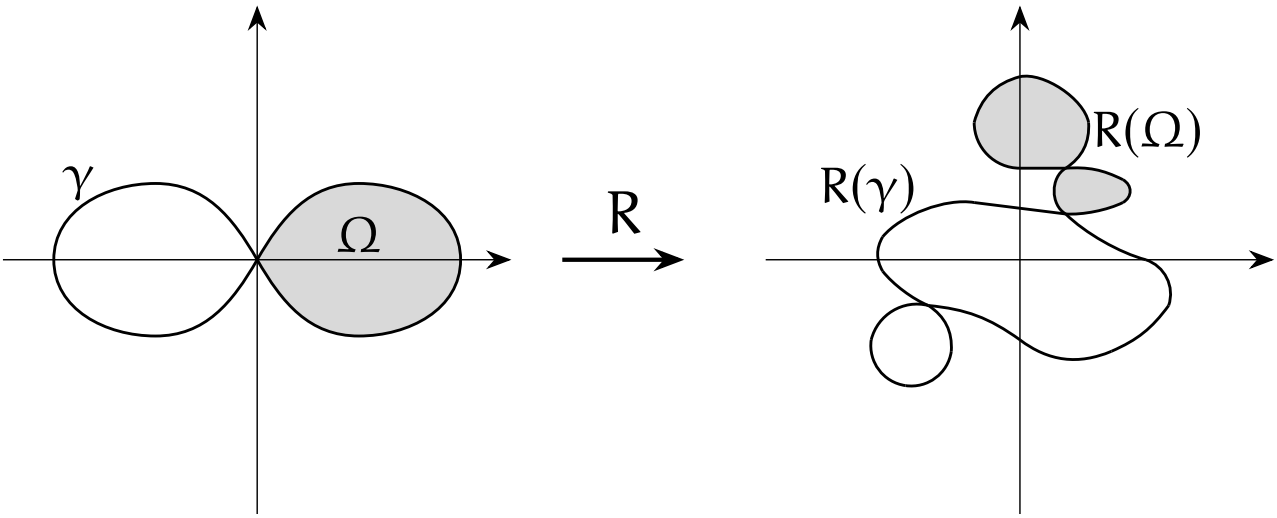


Figure 2.2: If the curve γ is sufficiently small ($\mathrm{diam}_{\sigma_0}(\gamma) < \delta$), the image of its interior component Ω under a rational function R is entirely contained within the bounded regions defined by the image curve $R(\gamma)$, never entering the exterior $E(R(\gamma))$.

2.4 Conjugacy

Definition 2.15 (Conjugate rational maps)**.** Given two rational maps R and S, we say R and S are *conjugate* if there is some Möbius map g with

$$S = gRg^{-1}.$$

Remark 2.8 (Conjugacy is an equivalence relation). Conjugacy is an equivalence relation, and the equivalence classes are *conjugacy classes* of rational maps.

Remark 2.9 (Invariant properties of conjugacy). Given two conjugate rational maps R and S, where $S = gRg^{-1}$. Then, the following are true:

(i) $\deg(R) = \deg(S)$;

(ii) $S^n = gR^n g^{-1}$ (this means that we can transfer a problem concerning R to a (possibly simpler) problem concerning a conjugate S);

(iii) and S fixes $g(z)$ if and only if R fixes z (conjugacy respects fixed points).

All of these observations are elementary; nevertheless, they are important because, in general, we shall not bother to distinguish between conjugate rational functions.

As a simple illustration of the use of conjugacy, let us characterize the polynomials within the class of rational maps.

Remark 2.10 (Characterization of polynomials). A rational map R is a polynomial if and only if R has a pole at ∞ and no poles in $\mathbb{C}$; more succinctly, if and only if $R^{-1}\{\infty\} = \{\infty\}$.

Theorem 2.9 (Characterization of the conjugacy class of the polynomials)**.** *A non-constant rational map R is conjugate to a polynomial if and only if there is some w in $\mathbb{C}_\infty$ with $R^{-1}\{w\} = \{w\}$.*

2.5 Valency

Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Then f has a Taylor expansion at z_0, say

$$f(z) = f(z_0) + a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \dots$$

where $a_k \neq 0$ and the positive integer k is uniquely determined by the condition that the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{(z - z_0)^k}$$

exists, is finite, and is non-zero.

Definition 2.16 (Valency of f at $z_0 \in \mathbb{C}$)**.** We denote this integer k by $\nu_f(z_0)$ and call it the *valency*, or *order*, of f at z_0.

Remark 2.11 (Valency is a local property). Note that when defining the valency, we've used the Taylor expansion at z_0. This expansion is defined *near the point* z_0, that is, valency is defined locally in a neighborhood of a point.

Remark 2.12 ($\nu_f(z_0)$ is the number of solutions of $f(z) = f(z_0)$). Of course, the valency of f is the number of solutions of $f(z) = f(z_0)$ at z_0 (restricting f to a neighborhood near z_0).

Proposition 2.10 (Chain Rule for Valency)**.** *The valency function ν satisfies the important* Chain Rule*:*

$$\nu_{fg}(z_0) = \nu_f(g(z_0))\nu_g(z_0),$$

where f and g are non-constant holomorphic functions near $g(z_0)$ and z_0 respectively; and z_0, $g(z_0)$, and $fg(z_0)$ are all in $\mathbb{C}$.

Proof. See [7, pp. 37–38]. □

Remark 2.13 (Local injectivity in terms of valency). Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Observe that f is injective in some neighborhood of z_0 if and only if $\nu_f(z_0) = 1$.

Remark 2.14 (Composing with injective functions preserves valency). Because of the previous remark (2.13), the Chain Rule shows that the va-

lency is preserved under a pre-application, and a post-application, of an injective function.
Explicitly, if fg is defined near z with g injective near z, and if hf is defined near z with h injective near $f(z)$, then:

$$\nu_{fg}(z) = \nu_f(g(z)),$$
$$\nu_{hf}(z) = \nu_f(z).$$

The properties of valency under injective functions enable us to extend our definition of $\nu_f(z_0)$ to the case where $z_0 = \infty$, or $f(z_0) = \infty$ (or both).

Definition 2.17 (Valency at ∞)**.** Given any function f that is non-constant and holomorphic near the point ∞ in $\mathbb{C}_\infty$ or z_0 in $\mathbb{C}_\infty$ such that $f(z_0) = \infty$ (or both). Then, we can select any Möbius maps g and h which map z_0 and $f(z_0)$ respectively to points in $\mathbb{C}$ and then define $\nu_f(z_0)$ by

$$\nu_f(z_0) = \nu_F(g(z_0)),$$

where $F = hfg^{-1}$.

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus $\nu_f(z_0)$ is defined independently of the choice of g and h.

Definition 2.18 (Univalent map)**.** An analytic map $f : D \to \mathbb{C}_\infty$ is said to be *univalent* in D if it is injective there.

Remark 2.15 (f univalent $\implies$ $\nu_f(z) = 1 \forall z \in D$). If f is univalent in D, then $\nu_f(z) = 1$ for every z in D.

The converse *need not hold*, being univalent is a global property (the whole domain of definition of the function), but valency is a local property (a neighborhood of z_0).

Example 2.1. The map $z \mapsto z^2$ of $\mathbb{C} - \{0\}$ onto itself has valency 1 everywhere, but it is not injective in $\mathbb{C} - \{0\}$.

Remark 2.16 (Rational maps are d-fold maps via valency). We've seen that rational maps are d-fold maps (Theorem 2.4), now we can write this in terms of valency. Given a non-constant rational map R of degree d, then for any z_0 in $R^{-1}\{w\}$, $\nu_R(z_0)$ is the number of solutions of $R(z) = w$ at z_0, and so for each w in the complex sphere,

$$\sum_{z \in R^{-1}\{w\}} \nu_R(z) = \deg(R).$$

2.6 The Inverse Function Theorem

Now that we have established the local behavior of analytic maps via valency, we can formalize when these maps can be locally inverted. In the classical complex plane, this is governed by the derivative.

Theorem 2.11 (Inverse Function Theorem in $\mathbb{C}$)**.** *Let $f : U \to \mathbb{C}$ be a holomorphic function on an open set $U \subseteq \mathbb{C}$, and let $z_0 \in U$. If $f'(z_0) \neq 0$, then there exist open neighborhoods $V \subseteq U$ of z_0 and W of $f(z_0)$ such that the restriction $f : V \to W$ is a bijection.*
Furthermore, the local inverse function $f^{-1} : W \to V$ is holomorphic, and its derivative satisfies:

$$(f^{-1})'(f(z_0)) = \frac{1}{f'(z_0)}.$$

Let us now bridge this classical theorem with our structural definition of valency at a point.

Remark 2.17 (Local injectivity in terms of derivative and valency in $\mathbb{C}$). By examining the Taylor expansion of a non-constant holomorphic function f near $z_0 \in \mathbb{C}$:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

we see that $f'(z_0) \neq 0$ if and only if the first non-zero coefficient appears at index $k = 1$. By Definition 2.16, this means $\nu_f(z_0) = 1$.
Thus, the Inverse Function Theorem in $\mathbb{C}$ (Theorem 2.11) states that f possesses a local holomorphic inverse near $z_0 \in \mathbb{C}$ if and only if $\nu_f(z_0) =$

1.

Just as we did with valency, we can exploit the fact that composing with injective functions preserves local bijectivity to extend this theorem to cases where $z_0 = \infty$, $f(z_0) = \infty$, or both.

Theorem 2.12 (Inverse Function Theorem in $\mathbb{C}_\infty$)**.** *Let $f : D_1 \to D_2$ be a non-constant analytic map between two subdomains of $\mathbb{C}_\infty$, and let $z_0 \in D_1$. Then, we can select any two Möbius transformations g and h which map z_0 and $f(z_0)$ respectively to points in $\mathbb{C}$. Let $F = hfg^{-1}$ be the localized map defined near $g(z_0) \in \mathbb{C}$. If $F'(g(z_0)) \neq 0$, then f is a local bijection from a σ-neighborhood of z_0 onto a σ-neighborhood of $f(z_0)$. Consequently, it possesses a local analytic inverse f^{-1} near $f(z_0)$.*

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus the theorem can be generalized to $\mathbb{C}_\infty$ from $\mathbb{C}$ independently of the choice of g and h.

Remark 2.18 (Local injectivity in terms of localized derivative and valency in $\mathbb{C}_\infty$). Let f, g, h, and F be defined as in Theorem 2.12. By expanding the non-constant holomorphic map F into a Taylor series near the point $w_0 = g(z_0) \in \mathbb{C}$:

$$F(w) = F(w_0) + F'(w_0)(w - w_0) + \frac{F''(w_0)}{2!}(w - w_0)^2 + \cdots$$

we observe that $F'(g(z_0)) \neq 0$ if and only if the first non-zero coefficient after the constant term appears at index $k = 1$.
By your definition of extended valency, this means $\nu_F(g(z_0)) = \nu_f(z_0) = 1$. Thus, the Inverse Function Theorem in $\mathbb{C}_\infty$ (Theorem 2.12) establishes a universal rule across the entire Riemann sphere: an analytic map f admits a local analytic inverse near a point $z_0 \in \mathbb{C}_\infty$ if and only if $\nu_f(z_0) = 1$.

2.7 Fixed Points

Definition 2.19 (Fixed point of f)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. A point ζ in D is a *fixed point* of f if $f(\zeta) = \zeta$.

The fixed points of a mapping play an important role in much of our later work, and they will be studied and classified in Chapter 6.

Proposition 2.13 (Fixed points of rational maps)**.** *Given by two coprime polynomials* P *and* Q *and the rational map* $R = P/Q$*. Then,*

(i) R *fixes* ∞ *if and only if* $\deg(P) > \deg(Q)$*;*

(ii) and R *fixes* $\zeta \in \mathbb{C}$ *if and only if* $P(\zeta) = \zeta Q(\zeta)$*; that is, the* fixed points of R in $\mathbb{C}$ *are the solutions of*

$$P(z) - zQ(z) = 0.$$

Proof. See [7, p. 39]. □

We count fixed points the same way we count zeros, that is, counting multiplicities.

Definition 2.20 (Number of fixed points at $\zeta \in \mathbb{C}$)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. The number of fixed points of f at a point ζ in $\mathbb{C}$ is given by the valency of the function $f(z) - z$ at ζ, i.e., $\nu_{f(z)-z}(\zeta)$.

Example 2.2. The map $f(z) = z + z^3$ has three fixed points at 0. This is because the fixed points of f are the zeros of $f(z) - z$, and

$$f(z) - z = z^3,$$

which has valency 3 at 0 ($z^3 = f(0) - 0 = 0$ has three solutions). This is not to be confused with the valency of f at 0, which is 1. In fact, $f(z) = f(0)$ has only one solution at 0 ($f(z) = z(1 + z^2)$).

To count the number of fixed points at ∞, we need

Lemma 2.14 ($\zeta \in \mathbb{C}$ fixed by $f \implies \zeta$ fixed by "conjugate" in $\mathbb{C}$)**.** *Let ζ in $\mathbb{C}$ be a fixed point of an analytic map f, and let φ be any map that is analytic, injective and finite in some neighborhood of ζ. Then, $\varphi f \varphi^{-1}$ has the same number of fixed points at $\varphi(\zeta)$ as f has at ζ.*

Proof. See [7, pp. 39–40]. □

Definition 2.21 (Number of fixed points at ∞)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$, $\infty \in D$. Then we can select an analytic map φ, that is injective and finite in some neighborhood of ∞, and then define the number of fixed points of f at ∞ by the number of fixed points of $\varphi f \varphi^{-1}$ at $\varphi(\infty)$.

The essential observation is that if we repeat this procedure using different maps φ, then the resulting answer will be the same; thus, the number of fixed points of f at ∞ is defined independently of the choice of φ.

With this definition, Lemma 2.14 implies

Theorem 2.15 (Number of fixed points is preserved under conjugacy)**.** *Given two conjugate rational maps R and S, where $S = gRg^{-1}$, g is a Möbius transformation. Then, the number of fixed points of R at ζ is the same as the number of fixed points of S at $g(\zeta)$.*

Theorem 2.16 (Number of fixed points of a rational map)**.** *Given a rational map R of degree $d \geqslant 1$. Then, the number of fixed points of R in $\mathbb{C}_\infty$ is exactly $d + 1$ (counting multiplicities).*

Proof. See [7, p. 40]. □

Definition 2.22 (Multiplier of a fixed point in $\mathbb{C}$)**.** Given a rational map R and a fixed point ζ in $\mathbb{C}$ of R. We define the *multiplier* of R at ζ, denoted

by $m(R, \zeta)$, as the complex number given by the derivative:

$$m(R, \zeta) = R'(\zeta).$$

Remark 2.19 (Multiplier in $\mathbb{C}$ is preserved under conjugation). Note that since the derivative is invariant under a conjugation, the multiplier is also invariant under a conjugation (provided the corresponding point is also in $\mathbb{C}_\infty$).

Definition 2.23 (Multiplier of the fixed point ∞)**.** Given a rational map R with a fixed point at ∞. Then, we can select a Möbius map g that maps ∞ to some point in $\mathbb{C}$, and then define the multiplier of R at ∞ by the multiplier of gRg^{-1} at $g(\infty)$:

$$m(R, \infty) = m(gRg^{-1}, g(\infty)).$$

Note that, by the preceding remarks, this definition is independent of the choice of g.

Remark 2.20 (Multiplier is preserved under conjugation). We now have defined a *conjugation invariant multiplier* $m(R, \zeta)$ of any rational map R at any of its fixed points.

Example 2.3 (Multiplier at ∞ for a rational map)**.** Let R be a rational map of the form

$$R(z) = \frac{a_0 + a_1 z + \cdots + a_n z^n}{b_0 + b_1 z + \cdots + b_m z^m},$$

where $a_n b_m \neq 0$ and $n > m$, so R fixes ∞. By definition, $m(R, \infty)$ is the derivative of the conjugate map $S(z) = 1/R(1/z)$ at the origin. A direct computation shows:

$$S'(0) = \begin{cases} b_m/a_n & \text{if } n = m+1, \\ 0 & \text{if } n > m+1. \end{cases}$$

On the other hand, computing the derivative $R'(z)$ and taking the limit

gives:

$$R'(\infty) = \lim_{z\to\infty} R'(z) = \begin{cases} a_n/b_m & \text{if } n = m+1, \\ \infty & \text{if } n > m+1. \end{cases}$$

Remark 2.21 (Formula for the multiplier of a rational map at a fixed point in $\mathbb{C}_\infty$). These results show that the multiplier $m(R,\zeta)$ of a rational map R at fixed point ζ is given by

$$m(R,\zeta) = \begin{cases} R'(\zeta) & \text{if } \zeta \neq \infty; \\ \frac{1}{R'(\infty)} & \text{if } \zeta = \infty. \end{cases}$$

Example 2.4. For example, if $R(z) = 3z$, then ∞ is an attracting fixed point of R (this is geometrically clear) and in this case, $m(R,\infty) = 1/3$.

We conclude this section by analyzing the behavior of the number of fixed points under iteration near a fixed point of f.

Theorem 2.17 (Iterates near the fixed point 0)**.** *Given an analytic function f that in a neighborhood of the origin has the expansion*

$$f(z) = az + b_1 z^{r+1} + \cdots,$$

where $a \neq 0$, $b_1 \neq 0$, and $r \geqslant 1$. Then its n-th iterate, in that neighborhood, has the form

$$f^n(z) = a^n z + b_n z^{r+1} + \cdots,$$

where

$$b_n = a^{n-1} b_1 \left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right).$$

Proof. See [7, p. 42]. □

From this theorem, we immediately obtain several important corollaries regarding the behavior of coefficients and the multiplicity of fixed points under iteration.

Corollary 2.17.1 ($a = 1 \implies b_n = nb_1$)**.** *If* $a = 1$*, then* $b_n = nb_1$.

Proof. This is a direct consequence of the formula for b_n:

$$b_n = a^{n-1}b_1\left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right) = b_1(1 + 1 + \cdots + 1) = nb_1.$$

□

Corollary 2.17.2 ($b_n = 0 \iff a^r \neq 1$ and $a^{nr} = 1$)**.** $b_n = 0$ *if and only if* $a^r \neq 1$ *and* $a^{nr} = 1$.

Proof. The terms inside the parentheses in the formula for b_n form a geometric series. Thus, we can write:

$$b_n = a^{n-1}b_1\frac{1 - a^{nr}}{1 - a^r}.$$

For b_n to be zero, the numerator of that fraction must be zero, but the denominator must not be zero (to avoid a $0/0$ situation, which we already showed in the previous corollary (Corollary 2.17.1) that it evaluates to n).

$$\begin{aligned} 1 - a^{nr} = 0 &\implies a^{nr} = 1, \\ 1 - a^r \neq 0 &\implies a^r \neq 1. \end{aligned}$$

Thus, $b_n = 0$ if and only if $a^r \neq 1$ and $a^{nr} = 1$. □

Remark 2.22 (f^n has a single fixed point at the origin $\implies$ f has a single fixed point at the origin). If $a^n \neq 1$, then $a \neq 1$. Thus, if f^n has a single fixed point at the origin, then so does f.

Proof. If $a = 1$, then $a^n = 1$ for all n. Thus, if $a^n \neq 1$, then $a \neq 1$.
If $a \neq 1$, the function $f(z) - z$ looks like $(a - 1)z + b_1z^{r+1} + \cdots = z((a - 1) + b_1z^r + \cdots)$, where $a - 1 \neq 0$. Thus, $f(z)$ has a single fixed point at the origin.

If f^n has a single fixed point at the origin, then we can only factor out one z from $f^n(z) - z$, giving us $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$, where $a^n - 1 \neq 0$.
Combining these three results, we have that if f^n has a single fixed point at the origin, then $a^n \neq 1$, which implies $a \neq 1$, thus f has a single fixed point at the origin. □

Corollary 2.17.3 (Number of fixed points at 0 of the iterates)**.** *The iterate f^n has at least as many fixed points at the origin as f has. Furthermore, if f^n has strictly more fixed points at the origin than f, then $a \neq 1$ but $a^n = 1$.*

Proof. Let's begin by showing that f^n has at least as many fixed points at the origin as f has. We have two cases:

- If $a \neq 1$: $f(z) - z$ has a single fixed point at the origin. Since $f^n(0) = 0$, 0 must be a fixed point of f^n of multiplicity at least 1.
- If $a = 1$: then $f(z) - z = b_1 z^{r+1} + \cdots$, so 0 is a fixed point of f of multiplicity $r + 1$ ($b_1 \neq 0$). On the other hand, by Corollary 2.17.1, $f^n(z) - z = n b_1 z^{r+1} + \cdots$, so 0 is a fixed point of f^n of multiplicity $r + 1$ (the same as for f). Thus, if $a = 1$, then f^n has as many fixed points at the origin as f has.

Now, let's show that if f^n has strictly more fixed points at the origin than f, then $a \neq 1$ but $a^n = 1$. From our analysis above, if $a = 1$, the multiplicities are exactly equal, so for f^n to have strictly more fixed points at the origin than f, we cannot have $a = 1$. Thus, $a \neq 1$.
If $a \neq 1$, then f has a single fixed point at the origin. For f^n to have strictly more fixed points at the origin than f, the term $(a^n - 1)$ from $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$ must vanish so that a higher power of z takes over. Thus, we must have $a^n = 1$. □

2.8 Critical Points

This section is devoted to the notion of critical points and critical values even though their full significance for iteration will not be apparent until Chapter 9 of [7].

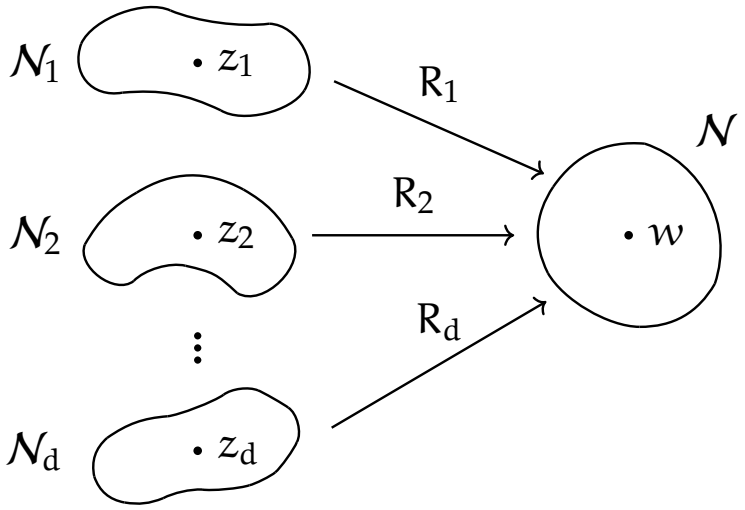


Figure 2.3: Bijective restrictions of R for a non-critical value w.

Definition 2.24 (Critical point of a rational map)**.** Given a rational map R. A point z in $\mathbb{C}_\infty$ is a *critical point* of R if R fails to be injective in any neighborhood of z.

Remark 2.23 (Critical points in terms of valency). Observe that if R is not constant, critical points are precisely the points at which $\nu_R(z) > 1$.

Definition 2.25 (Critical value of a rational map)**.** Given a rational map R. A point w in $\mathbb{C}_\infty$ is a *critical value* of R if it is the image of some critical point; that is, if $w = R(z)$ for some critical point z of R.

Remark 2.24 (Bijective restrictions of R^{-1} for non-critical values). Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, $R^{-1}\{w\}$ consists of precisely d distinct points, say $z_1, z_2, \ldots, z_d$. As none of the z_j are critical points, there are neighborhoods of N of w, and $N_1, \ldots, N_d$ of $z_1, \ldots, z_d$ respectively, with R acting as a bijection from each N_j onto N (Figure 2.3). It follows that for each j, the restriction R_j of R to N_j ($R : N_j \to N$) has an inverse

$$R_j^{-1} : N \to N_j.$$

Definition 2.26 (Branch of R^{-1} at a non-critical value w)**.** Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, we call each of the inverse maps $R_j^{-1} : N \to N_j$ a *branch* of R^{-1} at w.

By applying the Inverse Function Theorem (Theorem 2.11), we know that R is injective on some neighborhood of any point in $\mathbb{C}$ at which R' has neither a zero nor a pole: thus for all but a finite set of z, $\nu_R(z) = 1$ (recall that locally injective is the same as having valency 1 (Remark 2.13)) and, consequently,

$$\sum_{z \in \mathbb{C}_\infty} [\nu_R(z) - 1] < \infty.$$

This sum gives us a measure of the number of multiple roots of R (and the difficulties in defining R^{-1}), and its actual value is given by the *Riemann-Hurwitz relation*.

Theorem 2.18 (Riemann-Hurwitz relation)**.** *Given a non-constant rational map* R*. Then,*

$$\sum_{z \in \mathbb{C}_\infty} [\nu_R(z) - 1] = 2 \deg(R) - 2.$$

Proof. See [7, p. 44]. □

The general term in the sum is positive only when z is a critical point of R so, as the valency is an integer, this provides us with an estimate of the number of critical points of R. For a non-constant polynomial P, we have $\nu_P(\infty) = \deg(P)$, so, in the same way, we obtain an estimate of the number of critical points of P in $\mathbb{C}$. These two estimates are given in

Corollary 2.18.1 (Bound on amount of critical points of a rational)**.** *A rational map of positive degree* d *has at most* $2d - 2$ *critical points in* $\mathbb{C}_\infty$*. A polynomial of positive degree* d *has at most* $d - 1$ *critical points in* $\mathbb{C}$*.*

Proof. Given a non-constant rational map R of degree d, for a point to be critical we must have $\nu_R(z) > 1$, so $\nu_R(z) - 1 \geqslant 1$. Thus, the number of critical points of R in $\mathbb{C}_\infty$ is at most $2d - 2$.
In the case of polynomials, we have $\nu_P(\infty) = \deg(P) = d$, so the contribu-

tion of ∞ to the sum in the Riemann-Hurwitz relation is $d-1$. Thus, the number of critical points of P in $\mathbb{C}$ is at most $(2d-2)-(d-1) = d-1$. □

We shall see later that these estimates play a vital role in iteration theory.

Definition 2.27 (Multiplicity of a critical point)**.** Usually, we allocate a multiplicity to a critical point z: this multiplicity is $\nu_R(z) - 1$ and with this convention, a rational map of degree d has exactly $2d-2$ critical points.

A topological proof of Theorem 2.18, using the Euler characteristic is available and has the advantage that, suitably modified, it is valid for any analytic map between compact Riemann surfaces: this is given in Chapter 5. However, at this point some readers may prefer the previous more elementary proof.

We end this section with a simple result which locates the critical values of the iterate R^n in terms of the critical points of R.

Theorem 2.19 (Critical values of R^n)**.** *Given a rational map* R *and its set of critical points* C*. Then, the set of critical values of* R^n *is*

$$R(C) \cup \cdots \cup R^n(C).$$

Proof. See [7, p. 45]. □

2.9 A Topology on the Rational Functions

To study the global dynamics and stability of families of rational maps, we must equip the set of rational functions with a robust topological structure.

Definition 2.28 (Spaces $\mathcal{C}$ and $\mathcal{R}$)**.** Let $\mathcal{C}$ denote the class of all continuous maps from the extended complex plane into itself:

$$\mathcal{C} = \{f : \mathbb{C}_\infty \longrightarrow \mathbb{C}_\infty \mid f \text{ is continuous}\}.$$

We denote by $\mathcal{R}$ the specific subclass consisting of all non-constant and constant rational functions.

Definition 2.29 (Uniform Metric ρ)**.** The *metric of uniform convergence* $\rho : C \times C \longrightarrow \mathbb{R}^+ \cup \{0\}$ on the complex sphere is defined by:

$$\rho(f, g) = \sup_{z \in \mathbb{C}_\infty} \sigma_0(f(z), g(z)),$$

where σ_0 is the spherical metric.

Since the chordal metric σ and the spherical metric σ_0 only differ by a bounded factor, we could equally well use the chordal metric here (the metric would change, but the topology would not).

Now $\mathcal{R}$ is a closed subset of C because if the rational functions R_n converge uniformly to R on the complex sphere, then R is analytic on the sphere and so it too is rational. More generally (and again we could use either the chordal metric or the spherical metric), we have

Theorem 2.20 (Uniform Limits Preserve Analyticity)**.** *Suppose that each function in a sequence* $\{f_n\}$ *is analytic in a domain* $D \subseteq \mathbb{C}_\infty$*, and that* f_n *converges uniformly on* D *to* f *with respect to the spherical metric* σ_0*. Then* f *is also analytic in* D*.*

Proof. See [7, p. 46]. □

Corollary 2.20.1 ($\mathcal{R}$ is Closed)**.** *The space of rational functions* $\mathcal{R}$ *is a closed subset of the continuous function space* C *under the topology induced by the uniform metric* ρ*.*

With the topology established, we now investigate the topological behavior of the degree function. Crucially, the degree of a rational map cannot change abruptly under small uniform perturbations.

Theorem 2.21 (Continuity of the Degree Function)**.** *The degree map* $\deg : \mathcal{R} \longrightarrow \{0, 1, 2, \dots\}$ *is continuous. In particular, if a sequence of rational functions* R_n *converges uniformly on the Riemann sphere to a function* R*, then* R *is rational and* $\deg(R_n) = \deg(R)$ *for all sufficiently large* n*.*

Proof. See [7, p. 46]. □

Definition 2.30 (Space of Degree n Maps)**.** We denote by $\mathcal{R}_n$ the collection of all rational maps of degree exactly equal to n:

$$\mathcal{R}_n = \{R \in \mathcal{R} \mid \deg(R) = n\}.$$

As a consequence of this theorem (Theorem 2.21), the class $\mathcal{R}_n$ of rational maps of degree n is an open subset of $\mathcal{R}$ for it is the inverse image of the open subset $\{n\}$ of $\mathbb{Z}$ under the continuous map deg. It is easy to see that each $\mathcal{R}_n$ is connected (for if R and S lie in $\mathcal{R}_n$, we can simply "slide" the zeros and poles of R to those of S while maintaining the same degree) so we obtain

Corollary 2.21.1 (Components of $\mathcal{R}$)**.** *The subspaces $\mathcal{R}_n$ are the components of $\mathcal{R}$.*

There are a few remarks that I've not included in the notes, because Beardon points out that they can be safely ignored, in [7, p. 47].

Chapter 3

The Fatou and Julia Sets

The informal discussion in Chapter 1 suggested that the dynamics of a rational map R induces a subdivision of the complex sphere into two sets, namely the Fatou set F and the Julia set J of R. Here, we introduce the notion of equicontinuity of a family of maps, and then formally define the Fatou and Julia sets in terms of the equicontinuity of the family $\{R^n\}$. In addition, we consider conditions which imply that a family of analytic maps is equicontinuous (or a normal family) in a domain.

3.1 The Fatou and Julia Sets

Definition 3.1 (Continuous function)**.** Given two metric spaces (X, d_X) and (Y, d_Y). A map $f : X \to Y$ is *continuous at* x_0 in X if and only if, for every positive ε there exists a positive δ such that for all x in X,

$$d_X(x_0, x) < \delta \quad \text{implies} \quad d_Y(f(x_0), f(x)) < \varepsilon.$$

The function f is *continuous on a subset* X_0 *of* X if it is continuous at each point x_0 of X_0.

Clearly, in this definition, δ depends on f, x_0 and ε. For equicontinuity we will look for a δ that works for all f in a family of functions.

Definition 3.2 (Equicontinuous family)**.** Given two metric spaces (X, d_X) and (Y, d_Y). A family $\mathcal{F}$ of maps of (X, d_X) into (Y, d_Y) is *equicontinuous at* x_0 in X if and only if, for every positive ε there exists a positive δ such

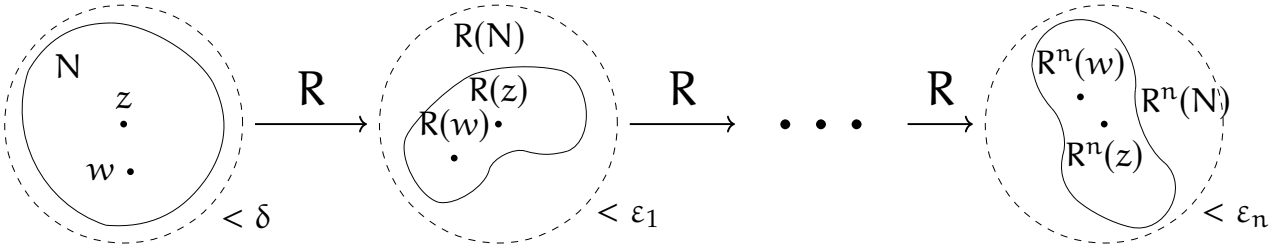


Figure 3.1: Equicontinuity for the family $\mathcal{F} = \{R^n\}_{n\in\mathbb{N}}$.

that for all x in X, and for all f in $\mathcal{F}$,

$$d_X(x_0, x) < \delta \quad \text{implies} \quad d_Y(f(x_0), f(x)) < \varepsilon.$$

The family $\mathcal{F}$ is *equicontinuous on a subset* X_0 *of* X if it is equicontinuous at each point x_0 of X_0.

It is important to realize that the equicontinuity of $\mathcal{F}$ at x_0 is the formal expression for the idea of preservation of proximity which was introduced informally in Chapter 1: indeed, it implies that every function in $\mathcal{F}$ maps the open ball with center x_0 and radius δ into a ball of radius at most ε.

Suppose $\mathcal{F} = \{R^n \mid n \in \mathbb{N}\}$ is an equicontinuous family, then given $z \in \mathbb{C}_\infty$, points w close to z will remain close after n iterations ($R^n(w)$ will be close to $R^n(z)$). Indeed, if we take ε_n to be the maximum distance we want between the n-th iterates, we can obtain a maximum distance between the $(n-1)$-th iterates (ε_{n-1}). Doing this recursively we can obtain a maximum distance $\varepsilon_0 = \delta$ between w and z so that their n-th iterates are as close as we want, see Figure 3.1.

Remark 3.1 (Equicontinuity is preserved under union). If the family $\mathcal{F}$ is equicontinuous on each of the subsets D_α of X, then it is automatically equicontinuous on the union $\cup D_\alpha$.

Taking the collection $\{D_\alpha\}$ to be the class of all open subsets of X on which $\mathcal{F}$ is equicontinuous, this leads to the following general principle.

Theorem 3.1 ($\exists$ a maximal open subset where $\mathcal{F}$ is equicontinuous)**.** *Let $\mathcal{F}$ be any family of maps, each mapping (X, d_X) into (Y, d_Y). Then there is a maximal open subset of X on which $\mathcal{F}$ is equicontinuous.*

Corollary 3.1.1 (∃ a maximal open subset where $\{f^n\}$ is equicontinuous)**.** *If f maps a metric space (X, d_X) into itself, then there is a maximal open subset of X on which the family of iterates $\{f^n \mid n \in \mathbb{N}\}$ is equicontinuous.*

This (at last) provides us with a formal definition for the Fatou and Julia sets of a rational map R, and the following terminology is chosen to honor the creators of the theory.

Definition 3.3 (Fatou and Julia sets)**.** Let R be a non-constant rational function. The *Fatou set* of R is the maximal open subset of $\mathbb{C}_\infty$ on which $\{R^n \mid n \in \mathbb{N}\}$ is equicontinuous, and the *Julia set* of R is its complement in $\mathbb{C}_\infty$.
We denote the Fatou set of a rational map R by F, and the Julia set by J, although sometimes, when we need to mention R explicitly, we use $F(R)$ and $J(R)$ instead.

Remark 3.2 (F is open and J is compact). Note that, by definition, $F(R)$ is *open*, and $J(R)$ is *compact*.

Although the term "Julia set" is standard, the use of "Fatou set" was suggested as late as 1984 (in [9]). It seems appropriate, but the reader should be familiar with the common alternatives, namely *the stable set*, and *the set of normality*. This latter term is a reference to normal families of analytic functions, but we have preferred to base our definition on equicontinuity because of its more immediate geometric appeal.

We end this section with two simple, but important, properties of F and J. A rational map, and in particular, a Möbius map and its inverse, satisfy a Lipschitz condition with respect to the spherical (and chordal) metric on $\mathbb{C}_\infty$ (Theorem 2.5). Thus, if we conjugate R with respect to a Möbius map, equicontinuity is transferred in the obvious way, and we have

Theorem 3.2 (Equicontinuity is preserved under conjugation)**.** *Let R be a non-constant rational map, let g be a Möbius map, and let $S = gRg^{-1}$. Then $F(S) = g(F(R))$ and $J(S) = g(J(R))$.*

A similar argument leads to

Theorem 3.3 (F and J are preserved under iteration)**.** *For any non-constant rational map* R*, and any positive integer* p*,* $F(R^p) = F(R)$ *and* $J(R^p) = J(R)$*.*

Proof. See [7, p. 51]. □

3.2 Completely invariant sets

Definition 3.4 (Invariant set under g)**.** Given a map g of a set X into itself, a subset E of X is:

(a) *forward invariant* if $g(E) = E$;

(b) *backward invariant* if $g^{-1}(E) = E$;

(c) *completely invariant* if $g(E) = E = g^{-1}(E)$.

Remark 3.3 (g surjective $\implies$ backward invariant = completely invariant). If g is surjective (that is, if $g(X) = X$) then the concepts of backward invariance and complete invariance coincide. Indeed, by surjectivity (but not necessarily in the general case, because there may be some points in E where $g^{-1}(x) = \emptyset$),

$$g(g^{-1}(E)) = E$$

which, with backward invariance, yields $g(E) = E$. The requirement of surjectivity here is crucial and without it, there is a difference between backward and complete invariance.

Example 3.1. The map $z \mapsto \exp(z)$ shows how these concepts can be different. Under this map $\mathbb{C}$ is backward invariant, but not completely invariant.

As our main concern is rational maps, we recall that these map $\mathbb{C}_\infty$ onto itself: thus for rational maps, backward invariance coincides with complete invariance.

We illustrate these ideas with the following result which will be of impor-

tance later.

Theorem 3.4 (Completely invariant under R and finite $\implies$ at most two elements)**.** *Let* R *be a rational map of degree at least two and suppose that a finite set* E *is completely invariant under* R*. Then* E *has at most two elements.*

Proof. See [7, p. 52]. □

It is convenient to collect together several, general, but elementary, properties of completely invariant sets and throughout, *we shall assume that* $g : X \to X$ *is surjective* so that we may use complete invariance and backward invariance interchangeably.

Proposition 3.5 (Complete invariance is preserved under conjugation)**.** *Given a surjective map* g *of a set* X *into itself, and a completely invariant subset* E *of* X *under* g*. Then, if* h *is a bijection of* X *onto itself, then* h(E) *is completely invariant under* hgh^{-1}*.*

Proposition 3.6 (Complete invariance is preserved under intersection)**.** *Given a surjective map* g *of a set* X *into itself, and a family of completely invariant subsets* $\{E_\alpha\}$ *of* X *under* g*. Then,*

$$E = \bigcap E_\alpha$$

is completely invariant.

Proof. The proof can be obtained by applying the standard construction which is used to generate, for example, subgroups, subspaces and topologies. The operator g^{-1} commutes with the intersection operator, that is, for any collection $\{E_\alpha\}$ of sets,

$$g^{-1}\left(\bigcap E_\alpha\right) = \bigcap g^{-1}(E_\alpha)$$

□

This means, that we can take any subset E_0 and form the intersection, say E, of all those completely invariant sets which contain E_0: obviously, E is then

the *smallest completely invariant set that contains* E_0.

Definition 3.5 (Generation of a completely invariant set)**.** We say that E_0 *generates* E if E is the intersection of all the completely invariant sets that contain E_0. That is, E is then the smallest completely invariant set that contains E_0.

We now introduce an equivalence relation which greatly facilitates the discussion of completely invariant sets.

Definition 3.6 (Relation $\sim$ (under g))**.** Given a map g of a set X into itself, for any x and y in X, we define the relation $\sim$ on X by $x \sim y$ if and only if there exist non-negative integers n and m with

$$g^n(x) = g^m(y). \tag{3.1}$$

Remark 3.4 ($\sim$ is an equivalence relation). Obviously the relation $\sim$ is symmetric and reflexive, and it is also transitive, since (3.1) and $g^p(y) = g^q(z)$ imply that

$$g^{n+p}(x) = g^{m+q}(z);$$

thus $\sim$ is an equivalence relation on X.

Definition 3.7 (Orbit of x (under g))**.** We denote the equivalence class containing x by $[x]$, and we call this the *orbit* of x (some authors call this the "great orbit").

It is easy to identify the orbit $[x]$, and we have

Theorem 3.7 ($[x]$ is the completely invariant set generated by $\{x\}$)**.** *Let* $g : X \to X$ *be any map of* X *onto itself. Then* $[x]$ *is the completely invariant set generated by the singleton* $\{x\}$.

Proof. See [7, p. 53]. □

Corollary 3.7.1 (Completely invariant $\iff$ union of orbits)**.** *A set* E *is completely invariant if and only if it is a union of equivalence classes* $[x]$.

This suggests we should also look at the interior, closure and boundary operators.

Theorem 3.8 (Complete invariance is preserved under the complement, interior and boundary operators)**.** *Let* g *be any continuous, open map of a topological space* X *onto itself and suppose that* E *is completely invariant. Then so are the complement* $X - E$*, the interior* E^o*, the boundary* ∂E *and the closure* $\overline{E}$*, of* E.

Proof. See [7, p. 53]. □

We now prove that the division of the sphere into Fatou and Julia sets is completely invariant: this is fundamental.

Theorem 3.9 (F and J are completely invariant)**.** *Let* R *be any rational map. Then the Fatou set* F *and the Julia set* J *are completely invariant under* R.

Proof. See [7, p. 54]. □

We can say more than Theorem 3.9 when R is a polynomial, and we have

Theorem 3.10 (F_∞ is completely invariant for polynomials)**.** *Let* P *be a polynomial of degree at least two. Then* ∞ *is in* $F(P)$*, and the component* F_∞ *of* F *containing* ∞ *is completely invariant under* P.

Proof. See [7, p. 54]. □

We end this section with a simple observation which will be used several times later.

Proposition 3.11 (Finite number of components $\implies$ they are periodic (under a continuous map $f : X \to X$))**.** *Let f be a continuous map of a topological space X onto itself, and suppose that X has only a finite number of components X_j. Then for some integer m, each X_j is completely invariant under f^m.*

Proof. See [7, p. 55]. □

This, of course, was the essential idea in the proof of Theorem 3.4.

3.3 Normal families and equicontinuity

We have defined the Fatou and Julia sets in terms of the equicontinuity of the family $\{R^n\}$. However, equicontinuity is closely related to normal families of analytic functions, and once we have understood this connection (which is expressed in the Arzelà-Ascoli Theorem), we can exploit the more powerful results about normal families of analytic functions to derive further information about the Fatou and Julia sets. This is our program for this section.

We begin with the definitions relevant to normal families.

Definition 3.8 (Pointwise convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges pointwise on* X to some map f if for each x in X: for every positive ε there exists a natural number N such that for all $n \geqslant N$,

$$d_Y(f_n(x), f(x)) < \varepsilon.$$

Definition 3.9 (Uniform convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges uniformly on* X to some map f if for every positive ε there exists a natural number N such that for all x in X, and for all $n \geqslant N$,

$$d_Y(f_n(x), f(x)) < \varepsilon.$$

Remark 3.5 (Uniform convergence $\implies$ pointwise convergence). Note that for a family to be uniformly convergent is a stronger statement than for it to be pointwise convergent. Indeed, if we write the definitions in terms of limits we have:

$$\lim_{n\to\infty} f_n = f \text{ pointwise} \iff \forall x \in X \lim_{n\to\infty} d_Y(f_n(x), f(x)) = 0;$$

$$\lim_{n\to\infty} f_n = f \text{ uniformly} \iff \lim_{n\to\infty} \sup\{d_Y(f_n(x), f(x)) \mid \forall x \in X\} = 0.$$

This last expression may also clarify why in section 2.9, we called ρ the *metric of uniform convergence*. Since, using that metric, convergence in C is the same as uniform convergence.

Definition 3.10 (Local uniform convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges locally uniformly on* X to some map f if each point x of X, has a neighborhood on which f_n converges uniformly to f.

Remark 3.6 (Local uniform convergence $\implies$ uniform convergence in compacts). Given a sequence (f_n) of maps from a uniform metric space (X, d_X) to a metric space (Y, d_Y) that converges locally uniformly on X. Then, it converges uniformly on each compact subset of X.

Definition 3.11 (Normal family)**.** A family $\mathcal{F}$ of maps from a metric space (X, d_X) to a metric space (Y, d_Y) is said to be *normal*, or a *normal family*, in X if every infinite sequence of functions from $\mathcal{F}$ contains a subsequence which converges locally uniformly on X.

We give the usual reminder that the limit of the convergent subsequence need not lie in $\mathcal{F}$. Although we have formulated these definitions for general metric spaces, we shall only be concerned with subdomains of the complex sphere with the chordal (or spherical) metric, and we come now to a statement of the Arzelà-Ascoli Theorem.

Theorem 3.12 (In $\mathbb{C}_\infty$ with continuous functions: equicontinuous family $\iff$ normal family)**.** *Let* D *be a subdomain of the complex sphere, and let* $\mathcal{F}$ *be a family of continuous maps of* D *into the sphere. Then* $\mathcal{F}$ *is equicontinuous in* D *if and only if it is a normal family in* D*.*

Proof. See [3, p. 222]. □

As an illustration of the direct use of normality rather than equicontinuity, we consider Vitali's Theorem: this exploits analytic continuation and so enables us to derive information about the limit f of a sequence (f_n) throughout a domain D simply from a knowledge of f on only a small part of D.

Theorem 3.13 (Vitali's Theorem)**.** *Suppose that the family* $\{f_1, f_2, \ldots\}$ *of analytic maps is normal in a domain* D*, and that* f_n *converges pointwise to some function* f *on some non-empty open subset* W *of* D*. Then* f *extends to a function* F *which is analytic on* D*, and* $f_n \to F$ *locally uniformly on* D*.*

Proof. See [7, p. 56]. □

We turn now to obtain conditions which guarantee the normality of a family of analytic functions.

Proposition 3.14 (In $\mathbb{C}_\infty$ with continuous functions: holomorphic and uniformly bounded $\iff$ equicontinuous family $\iff$ normal family)**.** *Given a family of functions* $\mathcal{F}$ *which are holomorphic and uniformly bounded in* D*. Then,* $\mathcal{F}$ *is equicontinuous, and hence normal in* D*.*

Proof. It is an elementary consequence of Cauchy's Integral Formula that if f is holomorphic and satisfies $|f| \leqslant M$ in a domain D in $\mathbb{C}$, then $|f'|$ is bounded above on each compact subset K of D, the upper bound depending on M, K, and D *but not on* f. □

It is possible to obtain a variety of mild generalizations of this result without much effort, but all such results are hopelessly inadequate for our purposes. Instead, we pass these by and go straight to one of the most powerful results concerning normality: this result is absolutely essential for the study

of iteration of rational maps, yet its substantial proof can safely be omitted without prejudicing an understanding of iteration theory.

Theorem 3.15 (Montel's fundamental normality criterion)**.** *Let* D *be a domain on the complex sphere* $\mathbb{C}_\infty$*, and let* Ω *be the domain* $\mathbb{C}_\infty - \{0, 1, \infty\}$*. Then the family* $\mathcal{F}$ *of all analytic maps* $f : D \to \Omega$ *is normal in* D*.*

Proof. See [7, p. 59]. □

The hypothesis of Theorem 3.15 is that each function in $\mathcal{F}$ fails to take the values 0, 1 and ∞ at any point of D. Let's see how Theorem 3.15 can be used to produce two variations on the same theme. In the first one of these, we assume only that each function f fails to take three values which now *may depend on* f. It is too much to expect that three values can be completely arbitrary, and we must add the requirement that they are uniformly separated on the sphere.

Theorem 3.16 (Montel's theorem with separated values)**.** *Let* $\mathcal{F}$ *be a family of maps, each analytic in a domain* D *on the complex sphere. Suppose also that there is a positive constant* m *and, for each* f *in* $\mathcal{F}$*, three distinct points* a_f*,* b_f *and* c_f *in* $\mathbb{C}_\infty$ *such that:*

(i) f *in* $\mathcal{F}$ *does not take the values* a_f*,* b_f *and* c_f *in* D*; and*

(ii) $\min\{\sigma(a_f, b_f), \sigma(b_f, c_f), \sigma(c_f, a_f)\} \geqslant m$*.*

Then $\mathcal{F}$ *is normal in* D*.*

Proof. See [7, p. 58]. □

Of course, Theorem 3.16 contains Theorem 3.15 as a special case, for we can take $a_f = 0$, $b_f = 1$ and $c_f = \infty$ for every f. There is another variation of Theorem 3.15 that we shall need.

Theorem 3.17 (Montel's theorem for omitted functions)**.** *Let* D *be a domain, and suppose that the functions* φ_1*,* φ_2 *and* φ_3 *are analytic in* D*, and are such that the closures of the domains* $\varphi_j(D)$ *are mutually disjoint. Now let* $\mathcal{F}$

be a family of functions, each analytic in D*, such that for every* z *in* D*, and every* f *in* $\mathcal{F}$*,* $f(z) \neq \varphi_j(z)$*,* $j = 1, 2, 3$*. Then* $\mathcal{F}$ *is normal in* D*.*

Proof. See [7, p. 58]. □

Chapter 4

Properties of the Julia Set

This chapter is devoted to developing the basic and most easily available properties of the Julia set of a rational map. Other properties of the Julia set occur throughout the text.

4.1 Exceptional Points

We begin now to exploit the fundamental Theorem 3.15. Given a rational map R, we have the induced equivalence relation $\sim$ discussed in section 3.2, and the equivalence class $[z]$ is the smallest completely invariant set which contains z. It is easy to see that $[z]$ can be finite only in rather special circumstances, so we enshrine this idea in a formal definition.

Definition 4.1 (Exceptional point)**.** A point z is said to be *exceptional* for R when $[z]$ is finite, and the set of such points is denoted by $E(R)$.

First, we justify the terminology by showing that such points are indeed exceptional.

Theorem 4.1 (Rational maps have at most two exceptional points)**.** *A rational map R of degree at least two has at most two exceptional points.*

(i) If $E(R) = \{\zeta\}$, then R is conjugate to a polynomial with ζ corresponding to ∞.

(ii) If $E(R) = \{\zeta_1, \zeta_2\}$, then R is conjugate to some map $z \mapsto z^d$, where d is an integer ($d \in \mathbb{Z}$), and ζ_1 and ζ_2 correspond to 0 and ∞.

Proof. See [7, p. 66]. □

From this and Theorems 3.3 and 3.10, we have immediately

Corollary 4.1.1 (Exceptional points of rational maps lie in F)**.** *If* $\deg(R) \geqslant 2$*, then the exceptional points of* R *lie in* $F(R)$*.*

We turn now to another characterization of exceptional points.

Definition 4.2 (Backward orbit and predecessors)**.** For any z, the *backward orbit* of z is the set

$$\begin{aligned} O^-(z) &= \{w \mid \text{for some } n \geqslant 0, R^n(w) = z\} \\ &= \bigcup_{n \geqslant 0} R^{-n}\{z\}, \end{aligned}$$

and we call the points in $O^-(z)$ the *predecessors* of z.

Remark 4.1 (Exceptional $\implies$ finite backward orbit)*.* As $O^-(z) \subset [z]$, any exceptional point z has a finite backward orbit.

We show now that the converse is true.

Theorem 4.2 (Characterization of exceptional points)**.** *The backward orbit* $O^-(z)$ *is finite if and only if* z *is exceptional.*

Proof. See [7, p. 66]. □

As an application of this result we can obtain

Theorem 4.3 (R^k is conjugate to a polynomial $\implies$ R^2 is conjugate to a polynomial)**.** *If for some positive integer* k*,* R^k *is conjugate to a polynomial, then* R^2 *is also conjugate to a polynomial.*

Proof. Suppose that some iterate R^k of R is conjugate to a polynomial. Then there is some w with $\left(R^k\right)^{-1}\{w\} = \{w\}$ (see Theorem 2.9) and using Theorem 4.2 this implies that w is exceptional for R. Now such a map need not be conjugate to a polynomial (consider, for example, $z \mapsto z^{-2}$) but from Theorem 4.1, R^2 must be. Thus, we have proved the theorem. □

4.2 Properties of the Julia Set

We shall continue to exploit Theorem 3.15 as we explore the structure of the Julia set. We have not yet considered whether the Julia set can be non-empty, and our first task will be to settle this issue. The case when the rational map R has degree one is trivial and of little interest, but in all other cases, J is not empty.

Theorem 4.4 (J is infinite)**.** *If* $\deg(R) \geqslant 2$, *then* $J(R)$ *is infinite.*

Proof. See [7, p. 67]. □

It is clear from Theorem 3.15 that the integer three has a significant role to play, and it appears yet again in a decisive way in the next result.

Theorem 4.5 (J is minimal)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$, *and suppose that* E *is a closed, completely invariant subset of the complex sphere. Then either:*

(i) E *has at most two elements and* $E \subset E(R) \subset F(R)$*; or*

(ii) E *is infinite and* $E \supset J(R)$.

Proof. See [7, p. 68]. □

It is usual, and convenient, to express the conclusion of Theorem 4.5 in the form: J *is the smallest, closed, completely invariant set with at least three points*, and we shall refer to this property as the *minimality* of J. The proof of Theorem 4.5 seems surprisingly short in view of the interesting corollaries which

follow from it but, of course, this is merely a reflection of the potency of Theorem 3.15. We now begin to develop these corollaries.

Remark 4.2 ($\mathbb{C}_\infty = J^o \cup \partial J \cup F$). First, the extended plane $\mathbb{C}_\infty$ is the disjoint union of the interior J^o of J, the boundary ∂J of J, and the Fatou set F.

Theorem 4.6 ($J = \mathbb{C}_\infty$ or $J^o = \emptyset$)**.** *Either* $J = \mathbb{C}_\infty$ *or* J *has empty interior.*

Proof. We know that F is completely invariant: likewise, J is and hence so are J^o and ∂J (Theorem 3.8). If F is not empty then $F \cup \partial J$ is an infinite, closed, completely invariant set and so, by the minimality of J, it contains J. In these circumstances, $J \subset \partial J$ and this yields the result. □

We shall consider the case when $J = \mathbb{C}_\infty$ in the next section.

Before examining the structure of the Julia set J, let us recall a fundamental definition from topology. Intuitively, a perfect set is one that is solid, containing no gaps and no lonely, isolated points.

Definition 4.3 (Perfect set)**.** A set E is called *perfect* if it is closed and contains no isolated points. Equivalently, E is perfect if it equals its own derived set, meaning $E = E'$ (where E' is the set of all accumulation points of E).

Next, we show that the Julia set fits this description perfectly by analyzing its behavior under the rational map R.

Proposition 4.7 (J is a perfect set)**.** *The derived set* J' *of* J *is an infinite, closed, and completely invariant subset of* J*. By the minimality of* J*, it follows that* $J \subset J'$*, which implies* $J = J'$*. Consequently,* J is a perfect set (and thus has no isolated points).

Proof. First, since J is infinite (by Theorem 4.4), it must contain accumulation points, so J' is non-empty. By definition, any derived set is automatically closed.
Next, we check invariance under the rational map R (which is continu-

ous and of finite degree):

- Since R is continuous, it maps accumulation points to accumulation points, so $R(J') \subset J'$, which means $J' \subset R^{-1}(J')$ (forward invariance).
- Since R is an open map, the preimage of an accumulation point is also an accumulation point, giving us $R^{-1}(J') \subset J'$ (backward invariance).

Combining these, we see that J' is completely invariant. Finally, by the minimality of the Julia set (Theorem 4.5), any such set within J must be dense and infinite, verifying that $J = J'$. By our definition, J is perfect. □

Theorem 4.8 (J is uncountable)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$*. Then the Julia set* J *is uncountable.*

Proof. By our previous proposition, the Julia set J is a nonempty, closed, and complete metric space that contains no isolated points.
To see why J must be uncountable, suppose for contradiction that J is countable. This means we can list its points as a sequence:

$$J = \{x_1, x_2, x_3, \dots\} = \bigcup_{i=1}^{\infty} \{x_i\}$$

Let $E_i = \{x_i\}$ be the singleton set containing only the point x_i. We analyze its topological properties within J:

(i) Each singleton $\{x_i\}$ is closed because finite points are closed in metric spaces.

(ii) Because J has no isolated points, no single point can form an open neighborhood by itself. Therefore, the interior of each $\{x_i\}$ is completely empty.

A closed set with an empty interior is, by definition, *nowhere dense*.
Thus, we have written J as a countable union of nowhere dense sets $\bigcup_{i=1}^{\infty} E_i$. In topology, such a union is called a set of the *first category*.

However, Baire's Category Theorem states that no nonempty complete metric space can be written as a countable union of nowhere dense sets. This directly contradicts our setup.
Because this contradiction arises solely from assuming J is countable, we conclude that J must be uncountable. □

In the next chapter (Chapter 5) we shall prove a stronger, but similar, result for the components of J (Theorem 5.18).

We continue to explore the consequences of Theorem 3.15, and the next result says that the successive iterates of R increasingly expand a neighborhood of any point J (we have seen this phenomenon before in some of the examples studied in Chapter 1). Given an open set W which meets J, the result describes the extent to which the images $R^n(W)$ cover the sphere. We recall $E(R)$ has at most two elements.

Theorem 4.9 (W open and $W \cap J \neq \emptyset \implies \bigcup R^n(W)$ explodes to $\mathbb{C}_\infty - E(R)$ and $R^n(W)$ converges to a set that contains J)**.** *Let* R *be a rational map of degree at least two, and let* W *be any non-empty open set which meets* J*. Then:*

(i) $\cup_{n=0}^{\infty} R^n(W) \supset \mathbb{C}_\infty - E(R)$*; and*

(ii) for all sufficiently large integers n, $R^n(W) \supset J$.

Proof. See [7, p. 69]. □

The next three results show how the Julia set J is realized as the limit, or the closure, of some countable set.

Definition 4.4 (Periodic point)**.** A point ζ is a *periodic point* of R if it is fixed by some iterate R^m.

Remark 4.3 (The number of periodic points is countable). There are only a countable number of periodic points.

Theorem 4.10 ($J \subset \overline{\{\text{periodic points}\}}$)**.** *Suppose that* $\deg(R) \geqslant 2$*. Then* J *is contained in the closure of the set of periodic points of* R*.*

Proof. See [7, pp. 70–71]. □

Remark 4.4 (z non-exceptional point $\implies$ $J \subset \overline{[z]}$). We now consider the relationship between J and an orbit $[z]$ of a non-exceptional point z. As the closure $\overline{[z]}$ of $[z]$ is both closed and completely invariant, the minimality of J implies that *if* z *is not exceptional, then* $J \subset \overline{[z]}$.

Remark 4.5 (z non-exceptional point and $z \in J$ $\implies$ $J = \overline{[z]}$). If z is in J, then it is not exceptional, so $[z]$, and hence also $\overline{[z]}$, is contained in J: thus *if* z *is in* J*, then* $J = \overline{[z]}$.

In fact, we can replace $[z]$ by the smaller set $\mathcal{O}^-(z)$ and still prove

Theorem 4.11 (z non-exceptional point $\implies$ $J \subset \overline{\mathcal{O}^-(z)}$ and if $z \in J$ they are equal)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$*.*

(i) If z *is not exceptional, then* J *is contained in the closure of* $\mathcal{O}^-(z)$*.*

(ii) If $z \in J$*, then* J *is the closure of* $\mathcal{O}^-(z)$*.*

Proof. See [7, p. 71]. □

Remark 4.6 (Using a computer to illustrate J). This shows that for any non-exceptional z the backward orbit $\mathcal{O}^-(z)$ is a countable set which accumulates at every point of J, and this is often used as the basis of a computer program to illustrate J. Roughly speaking, one selects a non-exceptional z and then computes successive inverse images of z to plot the set J where these accumulate. There are difficulties, however; for example, if $\deg(R) = d$, then the number of inverse images at the n-th

stage is d^n and this grows too rapidly for convenient computations, so some choices need to be made (see [21] and [6] for more details).

If z is in F (and is non-exceptional), then the closure of $O^-(z)$ contains J but, of course, is strictly larger than J. Instead of looking at the entire inverse orbit $O^-(z)$ we can also look at the n-th inverse image $R^{-n}(z)$ (which contains at most d^n points) and ask when (in some sense) do these points converge to J? The same question can be asked of a set E instead of the singleton $\{z\}$, and this is a more useful result to have available.

Theorem 4.12 ($R^{-n}(E)$ converges to J for E compact and non-F-limit)**.** *Let R be a rational map of degree at least two, and let E be a compact subset of the complex sphere with the property that for all z in $F(R)$, the sequence $(R^n(z))_{n\in\mathbb{N}}$ does not accumulate at any point of E. Then given any open set U which contains $J(R)$, $R^{-n}(E) \subset U$ for all sufficiently large n.*

Proof. See [7, p. 72]. □

Remark 4.7 (Notation $R^{-n}(E) \to J$). If the conclusion of the theorem holds we say that $R^{-n}(E)$ *converges to* J *and write* $R^{-n}(E) \to J$. Examples of such sets E are easy to find: for example, if F_0 is a component of the Fatou set that contains an attracting fixed point ζ, then any compact subset of $F_0 - \{\zeta\}$ has the required property. In Chapters 7 and 8, we shall analyze completely the possible forward images of $\{R^n(z) \mid n \geqslant 1\}$ of a point z in F and after that, Theorem 4.12 will be even more significant.

We end this section with the following generalization of Theorem 3.3.

Theorem 4.13 (J is preserved between commuting functions)**.** *Suppose that R and S are rational maps, each of degree at least two. If R and S commute, then $J(R) = J(S)$.*

Proof. See [7, pp. 72–73]. □

4.3 Rational Maps with Empty Fatou Set

Example 4.1. Our first objective is to show that for the rational map

$$z \mapsto \frac{(z^2+1)^2}{4z(z^2-1)}, \tag{4.1}$$

the Julia set is the entire complex sphere: this example (the first of its kind was given by Lattès in 1918, [17]). After this, we shall make some general observations about other rational functions with this property.
We shall use the technique illustrated in section 1.4, although here, we use a group of translations with two generators, and we require considerably more complex analysis. Let λ and μ be complex numbers that are not real multiples of each other, and let Λ be the corresponding lattice, that is,

$$\Lambda = \{m\lambda + n\mu \mid n, m \in \mathbb{Z}\}. \tag{4.2}$$

A *period parallelogram* for Λ is any closed parallelogram of the form

$$\Omega = \{z + s\lambda + t\mu \mid 0 \leqslant s \leqslant 1, 0 \leqslant t \leqslant 1\}.$$

A non-constant function f is an *elliptic function* for Λ if it is meromorphic on $\mathbb{C}$, and if each ω in Λ is a period of f; that is, if for all z in $\mathbb{C}$ and all ω in Λ, $f(z+\omega) = f(z)$. Of course, any such function maps $\mathbb{C}$ into $\mathbb{C}_\infty$ and, because $f(\mathbb{C}) = f(\Omega)$, we see that $f(\mathbb{C})$ is a compact subset of $\mathbb{C}_\infty$. However, by the Open Mapping Theorem, $f(\mathbb{C})$ is also an open subset of $\mathbb{C}_\infty$: thus

$$f(\Omega) = f(\mathbb{C}) = \mathbb{C}_\infty.$$

Our argument is based on the properties of the Weierstrass elliptic function

$$\wp(z) = \frac{1}{z^2} + \sum{}^{*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right]$$

where $\sum^*$ denotes summation over non-zero elements ω in Λ (see for example, [3, pp. 272–276], [12] or [16]). We shall only give the main steps in the argument but, for the convenience of some readers, we give an expanded version in the Appendix to this chapter. Now $\wp$ and its derivatives satisfy certain algebraic identities, and among these there is the *addition formula*

$$\wp(2z) = R(\wp(z)), \tag{4.3}$$

where R is the rational function given by

$$R(z) = \frac{z^4 + g_2 z^2/2 + 2g_3 z + (g_2/4)^2}{4z^3 - g_2 z - g_3}, \tag{4.4}$$

and where g_2 and g_3 are known quantities defined in terms of the lattice Λ. Note that (4.3) is analogous to the double angle formulae in elementary trigonometry.

Now let D be any disc in $\mathbb{C}$, let $U = \wp^{-1}(D)$, and define $\varphi(z) = 2z$. As U is open, and as $\varphi^n(U)$ is the set U expanded by a factor 2^n, it is clear that for sufficiently large n, $\varphi^n(U)$ contains a period parallelogram Ω of $\wp$. Using this with (4.3), we deduce that for these n,

$$R^n(D) = R^n(\wp(U)) = \wp(2^n U) = \mathbb{C}_\infty :$$

roughly speaking, this implies that the functions R^n explode any small disc D onto the whole sphere. Clearly, as D is arbitrary, (4.4) implies that the family $\{R^n\}$ is not equicontinuous on *any* open subset of the complex sphere, and so we deduce that $J(R) = \mathbb{C}_\infty$.

This argument gives a family of rational maps whose Julia set is the sphere; indeed, we always have

$$g_2^3 - 27g_3^2 \neq 0, \tag{4.5}$$

and any pair (g_2, g_3) satisfying this is realized by some lattice (see [16, p. 39]). Using this, we can construct Λ so that $g_2 = 4$ and $g_3 = 0$; then R given by (4.4) is the function (4.1). However, there is an easier construction than this. First, we let τ be a positive number, and we construct the square lattice Λ, and the corresponding function $\wp$, by taking $\lambda = \tau$ and $\mu = i\tau$. This choice leads easily to $g_3 = 0$ (see the Appendix in [7, pp. 77–79]), and hence to a simplification of (4.4), namely

$$R(z) = \frac{16z^4 + 8g_2 z^2 + (g_2)^2}{16z(4z^2 - g_2)}.$$

Now from (4.5), as $g_3 = 0$, we have $g_2 \neq 0$, so by taking $h(z) = 2z/\sqrt{g_2}$, we find that hRh^{-1} (which also has $\mathbb{C}_\infty$ as its Julia set) is the function given in (4.1). For further information and related examples, see references in [7, p. 75].

There are other interesting dynamical aspects of this example and briefly, we shall discuss some of these. For each positive integer k, we denote the set of points $\omega/2^k$, $\omega \in \Lambda$, by $2^{-k}\Lambda$, and observe that if z is in $2^{-k}\Lambda$, then

$$R^k(\wp(z)) = \wp(2^k z) = \infty.$$

This shows that the inverse orbit $\mathcal{O}^-(\infty)$ under R is $\wp(\cup_{k\in\mathbb{N}} 2^{-k}\Lambda)$. As $\cup_{k\in\mathbb{N}} 2^{-k}\Lambda$ is dense in $\mathbb{C}$, its $\wp$-image is dense in $\mathbb{C}_\infty$, and so by Theorem 4.11, we find again that $J(R) = \mathbb{C}_\infty$.
Next, the formula (4.3) shows that if $m = 2^k - 1$, then

$$R^k(\wp(\omega/m)) = \wp(2^k\omega/m) = \wp(\omega + \omega/m) = \wp(\omega/m),$$

and so the points $\wp(\omega/m)$ are periodic points of R. By considering all ω and all k, we now see that *the periodic points of* R *are dense in* $\mathbb{C}_\infty$. Although we are not yet in a position to claim that this shows that $J(R) = \mathbb{C}_\infty$, it is intuitively clear that it must (and it will follow from some of our later results).

Definition 4.5 (Pre-periodic point)**.** A point ζ is *pre-periodic* under R if it is not periodic, but a forward image of it is.

Theorem 4.14 ($\forall z \in C$, z pre-periodic $\implies J = \mathbb{C}_\infty$)**.** *If every critical point of the rational map* R *is pre-periodic, then* $J = \mathbb{C}_\infty$*.*

Proof. See Chapter 9 of [7]. □

Example (Continuation)**.** The rational map R in (4.4) is of degree four, so it has six critical points, say $c_1, \ldots, c_6$. We shall show now that each c_j is finite, and that each is mapped to ∞ (a fixed point of R) by R^2: then each c_j is pre-periodic, and we have $J = \mathbb{C}_\infty$. Again, we only sketch the main ideas and refer the reader to the Appendix for more details.
We shall now find the critical points of R in (4.4). Let $\zeta_1, \ldots, \zeta_6$ be the six points

$$\lambda/4, \quad \mu/4, \quad (\lambda+\mu)/4, \quad (3\lambda+\mu)/4, \quad (2\lambda+\mu)/4, \quad (3\lambda+2\mu)/4,$$

and let $\eta_1, \ldots, \eta_6$ be the six points

$$3\lambda/4, \quad 3\mu/4, \quad (3\lambda+3\mu)/4, \quad (\lambda+3\mu)/4, \quad (2\lambda+3\mu)/4, \quad (\lambda+2\mu)/4 :$$

note that these twelve points are distinct. As $\wp$ is an even function, and as $\eta_j + \zeta_j \in \Lambda$, we find that

$$\wp(\eta_j) = \wp(-\zeta_j) = \wp(\zeta_j),$$

and from this, and the fact that modulo Λ, each point of $\mathbb{C}_\infty$ has precisely two pre-images under $\wp$, we see that the six values $\wp(\zeta_j)$ are distinct. Now $\wp'(z) = \infty$ if and only if $\wp(z) = \infty$, that is, if and only if z is in Λ, and it is also known that $\wp'(z) = 0$ if and only if z is in $2^{-1}\Lambda$ but not in Λ. It follows that

$$\wp'(\zeta_j) \neq 0, \infty, \quad \wp'(2\zeta_j) = 0,$$

and as

$$R'(\wp(z))\wp'(z) = 2\wp'(2z),$$

we see that the six values $\wp(\zeta_j)$ are the six critical points of R. Finally, it is immediate from (4.3) that R^2 maps each of these to ∞.

Example 4.2. Theorem 4.14 enables us to produce other examples of maps whose Julia set is the sphere: for example, the map

$$R(z) = \frac{(z-2)^2}{z^2}$$

has critical points at 0 and 2, and as $2 \to 0 \to \infty \to 1 \to 1$ under an application of R, Theorem 4.14 implies that for this function too, $J(R)$ is the complex sphere. It is easy to see that this R is conjugate to the map

$$z \mapsto 1 - 2/z^2$$

and in fact, many functions in the family $1 + \omega/z^2$ have this same property: see reference in [7, p. 76] for further details.

We end this section with a characterization of those rational maps whose Julia set is the entire sphere, reference [53] in [7].

Definition 4.6 (Forward orbit)**.** For any z, the *forward orbit* of z is the set

$$\mathcal{O}^+(z) = \{R^n(z) \mid n \geqslant 1\}.$$

Theorem 4.15 ($J = \mathbb{C}_\infty \iff \exists z$ with dense forward orbit in $\mathbb{C}_\infty$)**.** *Let R be a rational map. Then $J(R) = \mathbb{C}_\infty$ if and only if there is some z whose forward orbit is dense in the complex sphere.*

Proof. See [7, pp. 76–77]. □

Chapter 5

The Structure of the Fatou Set

We begin with the simple properties of the Fatou set of a rational map R. As R is a branched covering map of the sphere onto itself with the critical points as branch points, the positions of the critical points influences the structure of the Fatou set F through the Euler characteristic. We discuss these ideas and apply them to study the structure of F.

Remark 5.1 (Rational maps as branched covering maps). A rational map $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ of degree $d \geqslant 2$ can be geometrically interpreted as a *branched covering map* of the sphere onto itself. This structural behavior underpins why the positions of critical points directly influence the topology of the Fatou set.

To see this from a local perspective, recall from Chapter 2 that if $w \in \mathbb{C}_\infty$ is a non-critical value, its preimage $R^{-1}\{w\}$ consists of exactly d distinct points $z_1, \ldots, z_d$. Around these points, there exist neighborhoods N of w and N_j of z_j such that each restriction $R : N_j \to N$ is a bijection (and hence a homeomorphism). Topologically, if we remove the finite set of critical values from the target sphere and their preimages from the domain sphere, R acts as a local homeomorphism. This is precisely the definition of an unbranched d-fold *covering map*.

This uniform covering behavior breaks down exclusively at the critical points of R. At any critical point z_0 where the valency satisfies $\nu_R(z_0) = k > 1$, the map fails to be a local injection. Instead, under an appropriate local change of coordinates, R behaves exactly like the standard analytic power map $z \mapsto z^k$. This means a neighborhood of z_0 is wrapped around

a neighborhood of the critical value $R(z_0)$ exactly k times.
Because the total number of critical points is finite (bounded globally by $\delta(\mathbb{C}_\infty) = 2d - 2$), R is a covering map punctured by a finite number of these singular *branch points*. This rigid geometric framework is what justifies using the Euler characteristic χ to track how domains pull back and split across the Riemann sphere.

5.1 The Topology of the Sphere

Some results on the structure of the Fatou and Julia sets are simple consequences of facts concerning the topology of the complex sphere and so require almost nothing of the theory of iteration. It seems best to present these results in this light and so our first task is to collect together the relevant topological results. Throughout, the underlying space is the complex sphere $\mathbb{C}_\infty$, and the complement of a set E is denoted by $\mathbb{C}_\infty - E$.

Definition 5.1 (Simply connected domain)**.** A domain D is *simply connected* if every closed curve in D can be deformed (in D) to a point of D.

In more intuitive terms, being simply connected is the same as not having any "holes" (otherwise a closed curve that surrounded a hole could not be deformed to a point).

Definition 5.2 (Domain connectivity operator)**.** The *connectivity* $c(D)$ of a domain D is the number of components of ∂D.

In more intuitive terms, this operator measures how many "holes" domain has. A simply connected domain, has 0 "holes"; thus, it has connectivity 1.

Definition 5.3 (k-connected domain)**.** A domain D is k*-connected* if its connectivity satisfies $c(D) = k$ for some integer k.

In more intuitive terms, a k-connected domain has exactly $k - 1$ "holes" (for instance, a doubly connected domain like an annulus has $c(D) = 2$ and contains a single hole).

Definition 5.4 (Infinitely connected domain)**.** A domain D is *infinitely connected* if its connectivity satisfies $c(D) = \infty$.

In more intuitive terms, being infinitely connected is the same as having infinitely many "holes".

The first three results are standard, and we omit their proofs (for more details see, for example, [3] or [20]).

Proposition 5.1 (Connectedness is preserved under closure)**.** *The closure of a connected set is connected.*

We will use Jordan curves as topological "knives" (to separate compact sets).

Definition 5.5 (Separation by a Jordan Curve)**.** Let $K \subseteq \mathbb{C}_\infty$ be a compact set. We say a Jordan curve γ *separates* K if K is disjoint from γ and meets both components of the complement of γ.

Proposition 5.2 (Characterization of disconnected for compact sets)**.** *A compact set K on the sphere is disconnected if and only if there exists a Jordan curve γ which separates K.*

Proposition 5.3 (Simply connected domain $\iff$ complement connected)**.** *A domain D is simply connected if and only if its complement is connected.*

We will also find the following two variations of Proposition 5.3 useful.

Proposition 5.4 (Simply connected domain $\iff$ boundary connected $\iff$ $c(D) = 1$)**.** *A domain D is simply connected if and only if its boundary ∂D is connected.*

Proposition 5.5 (Closed set is connected $\iff$ complement's components are simply connected)**.** *Let D be an open subset of the complex sphere. Then $\mathbb{C}_\infty - D$ is connected if and only if each component of D is simply con-*

nected.

Proof. See [7, p. 81]. □

To illustrate our earlier remarks, we record an immediate (though elementary) consequence of Proposition 5.5.

Proposition 5.6 (J connected $\iff$ every component of F is simply connected)**.** *Let* R *be a rational map. Then* J(R) *is connected if and only if each component of* F(R) *is simply connected.*

Proof. See [7, p. 81]. □

Finally (because we use this idea often), we include this result.

Proposition 5.7 (The union of a family of simply connected sets that is linearly ordered by inclusion is simply connected)**.** *Let* $\{D_\alpha\}$ *be a collection of simply connected domains that is linearly ordered by inclusion. Then* $\bigcup_\alpha D_\alpha$ *is a simply connected domain.*

Proof. See [7, p. 82]. □

For finite families this is obvious, it becomes of interest when considering infinite or uncountable families.

5.2 Completely Invariant Components of the Fatou Set

We begin our discussion of the structure of the Fatou set F by studying the completely invariant components of F. The basic result in this section is

Theorem 5.8 (Connectedness properties when there is a completely invariant component of F)**.** *Suppose that* $\deg(R) \geqslant 2$ *and that* F_0 *is a completely invariant component of* F*. Then:*

(i) $\partial F_0 = J$;

(ii) F_0 *is either simply connected or infinitely connected;*

(iii) all other components of F *are simply connected; and*

(iv) F_0 *is simply connected if and only if* J *is connected.*

Proof. See [7, p. 83]. □

Applying this directly to the case when the Fatou set itself is connected, we obtain this result.

Corollary 5.8.1 (F connected $\implies$ F simply (infinitely) connected and J connected (has an infinite number of components))**.** *If* $\deg(R) \geqslant 2$ *and if* F *is connected, then either:*

(i) F *is simply connected and* J *is connected; or*

(ii) F *is infinitely connected and* J *has an infinite number of components.*

In the case of a polynomial P, $\{\infty\}$ is a completely invariant set and hence too is the unbounded component of the Fatou set $F(P)$ (see Theorem 3.10). With this, another direct application of Theorem 5.8 yields

Theorem 5.9 (For polynomials: connectedness properties of components of F)**.** *Let* F *be the Fatou set of a non-linear polynomial. Then:*

(i) the unbounded component of F *is either simply connected or infinitely connected; and*

(ii) each bounded component of F *is simply connected.*

Examples of the two possibilities described in Theorem 5.9 (i) occur in Chapters 1 and 9 ([7, pp. 192–245]), and Theorem 5.9 verifies a visually obvious feature of the illustrations of Julia sets given earlier.

5.3 The Euler Characteristic

We devote this section to a brief and informal introduction to the Euler characteristic (a detailed account can be found in any one of numerous standard texts on the subject). Let S be a compact, or a bordered, surface: for our applications, it is enough to consider S to be either the sphere, or a plane domain together with its boundary, provided that this boundary consists only of a finite number of simple closed curves.

Definition 5.6 (Triangulation, vertices, edges and faces)**.** A *triangulation* T of S is a partition of S into a finite number of mutually disjoint subsets called *vertices*, *edges* and *faces* with the following properties:

(i) each vertex is a point of S;

(ii) for each edge e, there is a homeomorphism φ of a closed interval $[a, b]$ in $\mathbb{R}$ into S which maps the open interval (a, b) onto e, and the end-points a and b to vertices of T;

(iii) for each face f, there is a homeomorphism φ of a closed triangle Q in $\mathbb{C}$ into S which maps the edges and vertices of Q (in the usual sense) to edges and vertices of T, and such that f is the φ-image of the interior of Q.

We stress that T partitions S into mutually disjoint subsets of S.

Definition 5.7 (Simplex)**.** Each such subset is either a vertex, an edge or a face, and we call each of these a *simplex* of T (of dimension 0, 1 and 2 respectively).

Definition 5.8 (Euler characteristic)**.** For any simplex s of dimension m we define the *Euler characteristic* $\chi(s)$ as $(-1)^m$: more generally, if S_0 is any subset of S comprising a union of simplices, say $s_1, \ldots, s_r$ where s_j has dimension n_j, we define

$$\chi(S_0) = \sum_j \chi(s_j) = \sum_j (-1)^{n_j}.$$

Remark 5.2 ($\chi(S) = F - E + V$). In particular, if the triangulation T contains F faces, E edges and V vertices respectively, then the Euler characteristic $\chi(S)$ of S is, by definition,

$$\chi(S) = F - E + V.$$

Theorem 5.10 (The Euler characteristic is a topological invariant)**.** *The crucial, and well-known, fact (which we accept here without proof) is that* $\chi(S)$ *is a topological invariant which is independent of the particular triangulation* T *used.*

Thus, we can compute $\chi(S)$ by using any convenient triangulation we choose. When we refer to a triangulation and the Euler characteristic of a domain, we are implicitly assuming that the closure of the domain can be triangulated so that the domain itself is a union of simplices.

Calculations of χ can often be simplified by using the following simple idea. Suppose that S_1 and S_2 are subsets of S, each comprising a union of some of the simplices in T: then, simply by counting the contribution of each simplex to each side of the equation, we obtain

$$\chi(S_1 \cup S_2) + \chi(S_1 \cap S_2) = \chi(S_1) + \chi(S_2). \tag{5.1}$$

Let us illustrate this with some simple examples.

Example 5.1 ($\chi(S) = \chi(S^o) + \chi(\partial S)$)**.** The boundary ∂S of S must be a union of edges and vertices (for clearly, no face can meet ∂S) so if S^o denotes the interior of S, we have

$$\chi(S) = \chi(S^o) + \chi(\partial S).$$

Example 5.2 ($\chi(\gamma) = 0$)**.** It is easy to see that a simple closed curve γ is a union of the same number of edges and vertices, so $\chi(\gamma) = 0$.

Remark 5.3 (If the boundary of S is a finite number of mutually disjoint simple closed curves: $\chi(S) = \chi(S^o)$). From this, we see that if the boundary of a surface S consists of a finite number of mutually disjoint simple closed curves, then S and its interior S^o have the same Euler characteris-

tic.

Remark 5.4 ($\chi(\mathbb{C}_\infty) = 2$ and $\chi(D) = 1$ for any open or closed disc D). Next, by constructing explicit triangulations it is easy to see that $\chi(\mathbb{C}_\infty) = 2$, and that $\chi(D) = 1$ for any open or closed disc D: these computations are simple but important.

Now suppose that D is the complement (in $\mathbb{C}_\infty$) of k mutually disjoint topological closed discs $Q_1, \ldots, Q_k$, each being bounded by a Jordan curve, so D is of connectivity k (has $k-1$ "holes"). We can triangulate the sphere such that each of D, $Q_1, \ldots, Q_k$ is a union of simplices, and then (5.1) yields

$$2 = \chi(\mathbb{C}_\infty) = \chi(D) + \sum_{j=1}^{k} \chi(Q_j)$$

so $\chi(D) = 2 - k = 2 - c(D)$.

Remark 5.5 (Euler characteristic and connectedness). Note that for any such domain D of the sphere:

(i) $\chi(D) = 2$ if and only if D is the sphere $\mathbb{C}_\infty$;

(ii) $\chi(D) = 1$ if and only if D is simply connected, but not $\mathbb{C}_\infty$;

(iii) $\chi(D) = 0$ if and only if D is doubly connected;

while in all other cases, $\chi(D)$ is negative.

Later, when we apply these ideas to a component of the Fatou set, we shall see that there is an important distinction between the cases $\chi(D) \geqslant 0$ and $\chi(D) < 0$.

5.4 The Riemann-Hurwitz Formula for Covering Maps

We are going to obtain the Riemann-Hurwitz formula for the situation described briefly by

$$R : U \to V,$$

where R is a rational map, and U and V are domains in $\mathbb{C}_\infty$ (and so generalize the Riemann-Hurwitz formula described in section 2.8 which is the case $U = V = \mathbb{C}_\infty$).

Remark 5.6 (For any domain U, $V = R(U)$: $\partial V \subset R(\partial U)$). First, for any domain U, we have

$$\partial R(U) \subset R(\partial U)$$

so, if R maps U onto V, then

$$\partial V \subset R(\partial U); \tag{5.2}$$

of course, there may be strict inclusion here (for example, imagine U is a small disc, and R wraps its boundary around itself like a loop that crosses through the interior of V. The boundary points of U can map inside V, not just to its boundary).

Proof. Let U be a domain in $\mathbb{C}_\infty$ and let $V = R(U)$. Since R is a non-constant rational map on the Riemann sphere $\mathbb{C}_\infty$, it is both an open map and a closed map.
Because R is a closed map, the image of the closure of U is equal to the closure of its image:

$$R(\overline{U}) = \overline{R(U)} = \overline{V}.$$

Since R is an open map and U is open, $V = R(U)$ is also open, which implies that the interior of V is V itself. Recall that the boundary of any set is its closure minus its interior. Thus, we can express the boundary of V as:

$$\partial V = \overline{V} \setminus V = R(\overline{U}) \setminus R(U).$$

By writing $\overline{U} = U \cup \partial U$, we can expand the image of the closure as $R(\overline{U}) = R(U) \cup R(\partial U)$. Substituting this into the boundary equation yields:

$$\partial V = \big(R(U) \cup R(\partial U)\big) \setminus R(U).$$

Any point belonging to this set must lie in $R(\partial U)$ because it explicitly cannot lie in $R(U)$. Therefore, we conclude that:

$$\partial V \subset R(\partial U).$$

□

Remark 5.7 (For U a component of $R^{-1}(V)$: $R(U) = V$ and $R(\partial U) = \partial V$). Suppose now that U is a component of $R^{-1}(V)$. Then each point ζ of ∂U must map to ∂V (for certainly, $R(\zeta)$ must be in the closure of V, and it cannot lie in V unless ζ lies in U) so

$$R(\partial U) \subset \partial V. \tag{5.3}$$

In addition to this, R maps U onto V. Indeed, $R(U) \subset V$ and if there is not equality here, we could join a point of $R(U)$ to a point of $V - R(U)$ by a curve in V and such a curve would have to meet $\partial R(U)$ in V, contradicting the fact that $\partial R(U) \subset R(\partial U) \subset \partial V$. It follows that if U is a component of $R^{-1}(V)$, then both (5.2) and (5.3) hold and so in this case,

$$\partial V = R(\partial U).$$

These facts suggest that instead of considering a map R of U into V, we should (as is so often the case) emphasize R^{-1} and study the map R of a component U of $R^{-1}(V)$ onto V. This change of emphasis is important.

In this section, our objective is to use R to relate the Euler characteristics $\chi(U)$ and $\chi(V)$ of domains U and V as above. Any such relationship will involve a contribution from the critical points of R and in order to quantify this, we introduce the deficiency of R at z.

Definition 5.9 (Deficiency of R at z)**.** The *deficiency* of R at z is

$$\delta_R(z) = \nu_R(z) - 1, \tag{5.4}$$

where $\nu_R(z)$ is the valency as defined in section 2.5.

Definition 5.10 (Total deficiency of R over A)**.** For any set A we define the *total deficiency of* R *over* A as

$$\delta_R(A) = \sum_{z \in A} \delta_R(z).$$

Remark 5.8 (Total deficiency is additive). The total deficiency is additive,

that is, for disjoint sets A and B we have

$$\delta_R(A \cup B) = \delta_R(A) + \delta_R(B).$$

Frequently, but only when R is understood, we omit the suffix R and use δ instead of δ_R.

The reader will recall that (5.4) is the expression arising in the Riemann-Hurwitz relation (Theorem 2.18) and that $\delta_R(z) = 0$ for all but a finite set of z. In our new notation, the Riemann-Hurwitz relation is

$$\delta(\mathbb{C}_\infty) = 2\deg(R) - 2,$$

and, of course, for a polynomial P, $\delta(\mathbb{C}) = \deg(P) - 1$.

We are now ready to relate the quantities $\chi(U)$, $\chi(V)$, $\deg(R)$ and $\delta_R(U)$ as suggested above. We shall assume:

(1) V is a domain bounded by a finite number of mutually disjoint Jordan curves;

(2) U is a component of $R^{-1}(V)$; and

(3) there are no critical values of R on ∂V;

and with these assumptions we prove

Theorem 5.11 (Riemann-Hurwitz relation for subdomains)**.** *With the assumptions above, there is an integer* m *such that* R *is an* m*-fold map of* U *onto* V *and*

$$\chi(U) + \delta_R(U) = m\chi(V). \tag{5.5}$$

Proof. See [7, pp. 87–89]. □

In order to obtain a better appreciation of this result, we briefly consider some consequences of it. First, taking U and V to be $\mathbb{C}_\infty$, we have $\chi(U) = \chi(V) = 2$ and $m = \deg(R)$ and this recaptures the Riemann-Hurwitz relation given in section 2.8. Second, if we take V to be a simply connected domain other than $\mathbb{C}_\infty$, then $\chi(V) = 1$ and (5.5) yields

$$\delta_R(U) = (m - 1) + (1 - \chi(U)).$$

Written this way, both terms on the right are non-negative, and we can conclude that either:

(i) R has critical points in U (that is, $\delta_R(U) > 0$); or

(ii) U is simply connected and R is a homeomorphism of U onto V.

Indeed, the only way a map has no critical points in U ($\delta_R(U) = 0$) is if $\chi(U) = 1$ (U is simply connected) and $m = 1$ (the map is a homeomorphism). If, in addition to this, $R^{-1}(V)$ is connected, then we must have $U = R^{-1}(V)$ and $m = \deg(R)$, so in this case,

$$\delta_R(R^{-1}(V)) \geqslant \deg(R) - 1$$

with equality only when $R^{-1}(V)$ is simply connected (for example, the map $z \mapsto z^d$ of the unit disc onto itself). For future reference, we state this as

Corollary 5.11.1 (If $R^{-1}(V)$ is connected $\delta_R(R^{-1}(V)) \geqslant d - 1$)**.** *Let R be a rational map of degree d, suppose that V is simply connected, and that $R^{-1}(V)$ is connected. Then*

$$\delta_R(R^{-1}(V)) \geqslant d - 1$$

with equality only when $R^{-1}(V)$ is simply connected.

We end this section with another simple consequence of Theorem 5.11, and we recall that for a domain D bounded by $c(D)$ Jordan curves,

$$\chi(D) = 2 - c(D). \tag{5.6}$$

We now have

Proposition 5.12 ($c(U) \geqslant c(V)$)**.** *With the same assumptions as in Theorem 5.11, $c(U) \geqslant c(V)$.*

Proof. See [7, p. 89]. □

5.5 Maps Between Components of the Fatou Set

The Euler characteristic of a domain D on the complex sphere has been defined whenever its boundary ∂D consists of a finite number of Jordan curves. In general, however, the boundary of a domain D is much more complicated than this; we may not be able to triangulate the closure of D and in these circumstances, $\chi(D)$ is as yet undefined. As this is likely to be so in the case of

major interest to us (when D is a component of the Fatou set), this presents us with a problem which we must now tackle. We propose to show that given any domain D, we can define $\chi(D)$ as the limiting value of the Euler characteristic of subdomains which exhaust D, and once this has been done, we can then use the Euler characteristic as a tool to study the way in which R maps one component of the Fatou set onto another. We shall restrict our discussion to subdomains of $\mathbb{C}_\infty$; nevertheless, the following development is closely related to the construction of the ideal boundary components of a Riemann surface (see [2, pp. 25–26, 81–87]).

Definition 5.11 (Regular subdomain)**.** Let D be any domain on the sphere. A subdomain Ω of D is said to be a *regular subdomain* of D if:

(1) Ω is bounded by a finite union of mutually disjoint Jordan curves, say $\gamma_1, \ldots, \gamma_n$, all of which lie in D; and

(2) the complement of Ω consists of n topological discs, say $W_1, \ldots, W_n$ (bounded by $\gamma_1, \ldots, \gamma_n$ respectively) and each W_j meets the complement of D.

Intuitively, (2) says that a regular subdomain Ω is a piece of a domain that is not allowed to have any "new holes" that the parent domain D didn't already have.

Example 5.3. For example, $\{|z| < 1\}$ is a regular subdomain of $\mathbb{C}$, whereas $\{1 < |z| < 2\}$ is not.

Observe that $\chi(\Omega)$ is defined for each regular subdomain of D.

Of course, if a subdomain Ω of D satisfies (1) but not (2), we can adjoin to Ω those sets $W_j \cup \gamma_j$ which do not meet the complement of D to form a regular subdomain Ω_0 of D. Obviously, Ω has a higher connectivity than Ω_0, and $\chi(\Omega_0) \geqslant \chi(\Omega)$.

We want to consider D as the limit of regular subdomains, and as no canonical sequence of subdomains of D presents itself, it is best to reject the idea of a sequential limit and to consider instead convergence with respect to the *directed set* (or *net*) of regular subdomains. There is no need for great generality here and the details are quite simple and explicit. First we prove

Lemma 5.13 (Exhaustion by regular subdomains)**.** *Let* D *be a proper subdomain of the complex sphere. Then:*

(i) any compact subset of D *lies in some regular subdomain of* D*; and*

(ii) if Ω_1 *and* Ω_2 *are regular subdomains of* D*, then there is a regular subdomain* Ω *of* D *which contains both* Ω_1 *and* Ω_2*.*

Proof. See [7, p. 91]. □

Our next task is to show that $\chi(\Omega)$ is a monotonic function on the class $\mathcal{R}(\mathrm{D})$ of all regular subdomains D, and so tends to a limit (which may be $-\infty$) as Ω increases to D through $\mathcal{R}(\mathrm{D})$. The monotonicity property is expressed in

Lemma 5.14 (The Euler Characteristic is monotone decreasing)**.** *The Euler characteristic* χ *is a decreasing function on* $\mathcal{R}(\mathrm{D})$*: explicitly, if* Ω_1 *and* Ω_2 *are regular subdomains of* D *such that* $\Omega_1 \subset \Omega_2$ *then* $\chi(\Omega_2) \leqslant \chi(\Omega_1)$*.*

Proof. See [7, p. 91]. □

Lemma 5.13 says that the class $\mathcal{R}(\mathrm{D})$ of regular subdomains of D is a *net* and Lemma 5.14 is the monotonicity of the function $\chi : \mathcal{R}(\mathrm{D}) \to \{2, 1, 0, \ldots, -\infty\}$: thus these lead to

Definition 5.12 (Euler Characteristic generalization for subdomains in $\mathbb{C}_\infty$)**.** For any subdomain D of $\mathbb{C}_\infty$,

$$\chi(\mathrm{D}) = \inf\{\chi(\Omega) : \Omega \text{ a regular subdomain of } \mathrm{D}\}.$$

Quite explicitly, either:

(1) $\chi(\mathrm{D}) = -\infty$, and there are regular subdomains Ω_n with $\chi(\Omega_n) \to -\infty$ or, equivalently, with $c(\Omega_n) \to +\infty$; or

(2) there is some regular subdomain Ω_0 of D such that

$$\chi(\Omega_0) = \chi(\mathrm{D}) > -\infty,$$

and then (from Lemma 5.14) $\chi(\Omega) = \chi(D)$ whenever Ω is a regular subdomain which contains Ω_0. Note that when $\chi(D)$ is finite, it is necessarily attained by some Ω_0 as χ is integer valued.

If D is a simply connected domain, then ∂D is connected and each regular subdomain Ω can only have one complementary component: thus $\chi(D) = 1$ for a simply connected domain D, regardless of the nature of ∂D. More generally, if D has connectivity k, then $\chi(\Omega) = 2 - k$ for all sufficiently large regular subdomains Ω, and so $\chi(D) = 2 - k$, again irrespective of the complexity of ∂D.

As culmination of these ideas, we have

Theorem 5.15 (Riemann-Hurwitz relation for related Fatou components)**.** *Let* F_0 *and* F_1 *be components of the Fatou set* F *of a rational map* R *and suppose that* R *maps* F_0 *into* F_1*. Then, for some integer* m*,* R *is an* m*-fold map of* F_0 *onto* F_1 *and*

$$\chi(F_0) + \delta_R(F_0) = m\chi(F_1) \tag{5.7}$$

Proof. See [7, pp. 92–93]. □

Remark 5.9 (Riemann-Hurwitz relation in our context). We call (5.7) the Riemann-Hurwitz relation.

5.6 The Number of Components of the Fatou Set

Theorem 5.16 (Number of completely invariant components of F)**.** *The Fatou set* F *of* R *contains at most two completely invariant components, and if there are two, then each is simply connected.*

Proof. We suppose that R has degree d, where $d \geqslant 2$, and we shall now obtain information on the number of components of F. Suppose first that each of $F_1, \ldots, F_k$ is a completely invariant component of F, where $k \geqslant 2$. Applying Theorem 5.8 to each of the components F_j in turn, we find (as $k \geqslant 2$) that every $F_j$is simply connected. Next, we apply Theorem 5.15 to

each F_j and, noting that F_j is simply connected, $\chi(F_j) = 1$, we obtain

$$\delta_R(F_j) = (d-1)\chi(F_j) = d-1.$$

The Riemann-Hurwitz relation now yields

$$\begin{aligned} k(d-1) &= \sum_{j=1}^{k} \delta_R(F_j) \\ &\leqslant \delta_R(\mathbb{C}_\infty) = 2d-2, \end{aligned}$$

and so $k \leqslant 2$. □

Example 5.4. Both of the possibilities in Theorem 5.16 can arise: if $R(z) = z^2 - 2$, then F has a single completely invariant component, and if $R(z) = z^2$, then F has two completely invariant components.

Theorem 5.16 does not by itself exclude the possibility that F may have two completely invariant components *and* other additional components: however, it will be a consequence of Sullivan's No Wandering Domain Theorem that *if* F *has two completely invariant components, then these are the only components of* F.

Remark 5.10 (For any m, $\exists$ simply connected domains $D_1, \ldots, D_m$ with $\partial D_1 = \cdots = \partial D_m$). With reference to Theorem 5.16, it is of interest to note that for any positive integer m, there exist simply connected domains $D_1, \ldots, D_m$ with

$$\partial D_1 = \partial D_2 = \cdots = \partial D_m;$$

see [15, p. 143] (for this shows that we cannot prove Theorem 5.16 from the single topological fact that all the completely invariant components of F have a common boundary, namely J).

An argument similar to that used above enables us to obtain information on the total number of components of F. First, Chapter 1 contains a discussion of the examples

$$z \mapsto z^2 - 2, \quad z \mapsto z^2, \quad z \mapsto z^2 - 1,$$

and in these, F has one, two and infinitely many components respectively. In

addition, we have seen (in section 4.3) that it is possible for F to be empty. It is possible, then, for F to have exactly 0, 1, 2, or infinitely many components, and our last result in this section says that these are the only possibilities.

Theorem 5.17 (F has either 0, 1, 2 or ∞ components)**.** *The Fatou set F of a rational map R has either 0, 1, 2 or infinitely many components.*

Proof. See [7, p. 94]. □

Remark 5.11 (Key idea of the proof: certain components of $F(R)$ contain critical points). The proof of Theorem 5.16 is based on the idea that certain components of $F(R)$ contain critical points and this, in conjunction with the Riemann-Hurwitz relation, bounds the number of components. This idea is important and will be used again later to good effect.

5.7 Components of the Julia Set

As usual, R is a rational map of degree d, where $d \geqslant 2$.

Remark 5.12 (J disconnected $\implies$ it has infinitely many components). We have seen that no point of J is isolated (Proposition 4.7), and it is easy to show that if J is disconnected, then it has infinitely many components. Indeed, if J has only a finite number of components, say $J_1, \dots J_n$, then some J_k, say J_1, is infinite (as J is). Moreover, by Proposition 3.11, there is some integer m such that each J_k is completely invariant under R^m so, by the minimality of the Julia set,

$$J(R) = J(R^m) = J_1,$$

and $J(R)$ is connected.

We now prove a stronger result than this, namely

Theorem 5.18 (J disconnected $\implies$ it has uncountably many components and its points are accumulation points of an infinite number of components)**.** *If J is disconnected, then it has uncountably many components*

and each point of J *is an accumulation point of infinitely many distinct components of* J.

Proof. See [7, pp. 95–96]. □

In Chapter 11 of [7], we can see an example of a rational map R with the properties (i) J is disconnected, and (ii) each component F_j of F has finite connectivity. It follows that only a countable number of components of J can meet the boundary of some F_j; thus in this example, *there are components of* J *which do not meet the closure of any component of* F.

In order to prove Theorem 5.18, we need

Lemma 5.19 (For K compact and connected, $R^{-1}(K)$ has at most $\deg(R)$ components and each is mapped by R onto K)**.** *Let* K *be a compact, connected subset of the complex plane. Then* $R^{-1}(K)$ *has at most* $\deg(R)$ *components and each is mapped by* R *onto* K.

Proof. See [7, pp. 96–98]. □

Chapter 6

Periodic Points

This chapter is devoted to an extensive discussion of the fixed and periodic points of a rational map and their role in iteration theory.

6.1 The Classification of Periodic Points

We have already used elementary facts about fixed points, and we turn now to a systematic and detailed discussion of the periodic points of a rational map R. A fixed point ζ of R is classified according to the *multiplier* $m(R, \zeta)$ of R at ζ (see 2.7) and as this is conjugation invariant, we may assume that ζ is in $\mathbb{C}$ and so $m(R, \zeta) = R'(\zeta)$. It is important to realize that as the classification of fixed points is a purely local matter, it applies to any analytic function and, in particular, to the local inverse (when it exists) of a rational map.

Definition 6.1 (Types of fixed points based on the multiplier)**.** Suppose that ζ in $\mathbb{C}$ is a fixed point of an analytic function f. Then ζ is:

(a) *super-attracting* if $f'(\zeta) = 0$;

(b) *attracting* if $0 < |f'(\zeta)| < 1$;

(c) *repelling* if $|f'(\zeta)| > 1$;

(d) *rationally indifferent* if $f'(\zeta)$ is a root of unity;

(e) *irrationally indifferent* if $|f'(\zeta)| = 1$, but $|f'(\zeta)|$ is not a root of unity.

Some explanatory remarks may be helpful. First, the distinction between (a) and (b) is that a super-attracting fixed point is a critical point of f, whereas

an attracting fixed point is not; thus f has a local inverse near ζ if ζ is attracting but not if it is super-attracting. The reader may recall that in section 1.1 no such distinction was made and, of course, there are still occasions when we wish to embrace both cases; we do this by saying that ζ is *(super)attracting*. Similarly, it is convenient to combine cases (d) and (e), and such a fixed point is said to be *indifferent*. For an indifferent fixed point ζ, the best linear approximation of f near ζ is a rotation about ζ and the distinction between (d) and (e) is simply whether this rotation has finite or infinite order.

The notion of periodic points, and the extension of the classification to these, is straightforward. As we already defined in Definition 4.4, a point ζ is a *periodic point* of a rational function R if it is a fixed point of some iterate R^m. For such a point ζ there is a positive integer n for which

$$\zeta, R(\zeta), R^2(\zeta), \ldots, R^{n-1}(\zeta) \tag{6.1}$$

are distinct but $R^n(\zeta) = \zeta$.

Definition 6.2 (Cycle of a periodic point)**.** The finite set of points in (6.1) is the *cycle of* ζ.

Definition 6.3 (Period of a periodic point)**.** The integer n is the *period* of ζ.

Of course, the fixed points of R are points of period one: more generally, ζ has period n if and only if it is a fixed point of R^n but not of any lower-order iterate.

Remark 6.1 ($(R^n)'(\zeta_m) = \prod_{k=0}^{n-1} R'(\zeta_k)$). A periodic point ζ of period n is classified as a fixed point of R^n, but we can say a little more than this. By conjugation, we may assume that the cycle does not contain ∞, and we write

$$\zeta_m = R^m(\zeta), \quad m = 0, 1, 2, \ldots,$$

so $\zeta_{m+n} = \zeta_m$. By n applications of the Chain Rule, we now have

$$(R^n)'(\zeta_m) = \prod_{k=0}^{n-1} R'(R^k(\zeta_m)) = \prod_{k=0}^{n-1} R'(\zeta_k),$$

the product being a re-arrangement of the first. This argument shows that the derivative $(R^n)'$ has the same value at each point ζ_j of the cycle.

As a consequence of this, we can extend our classification to cycles and speak naturally of the *multiplier of the cycle*, *attracting cycles* and so on.

Definition 6.4 (Multiplier of R on a cycle)**.** We define the multiplier of a cycle as the multiplier of R on any point of the cycle (all of them have the same multiplier).

Definition 6.5 (Types of cycles based on the multiplier)**.** We say a cycle is *super-attracting*, *attracting*, *repelling*, *rationally indifferent* or *irrationally indifferent* depending on the type of its points (all of them are of the same type).

As we already defined in Definition 4.4, a point ζ is *pre-periodic* under R if it is not periodic but if some image, say $R^m(\zeta)$, is: in this case there exist positive integers m and n with

$$\zeta, R(\zeta), R^2(\zeta), \ldots, R^m(\zeta), \ldots, R^{m+n-1}(\zeta)$$

distinct but with $R^{m+n}(\zeta) = R^m(\zeta)$. Most periodic points have associated pre-periodic points.

Example 6.1. For example, the origin is a pre-periodic point of the polynomial $z^2 - 2$.

In the remainder of this chapter, we shall discuss how the attracting, repelling and indifferent cycles relate to the Fatou and Julia sets.

Remark 6.2 ((Super)Attracting points are in F and repelling points are in J). Briefly, the (super)attracting cycles are in the Fatou set, while the repelling cycles are in the Julia set.

The corresponding results for indifferent cycles are more delicate and require much more work.

6.2 The Existence of Periodic Points

In an attempt to find the periodic points of period n of a rational map R, we naturally consider the solutions of

$$R^n(z) = z. \tag{6.2}$$

Any solution of this equation will be periodic; however, it will not necessarily be of period n for it may already be fixed by some earlier iterate R^m, where $m < n$.

Example 6.2. This occurs, for example, when

$$R(z) = z^2 - z : \tag{6.3}$$

in this case every solution of $R^2(z) = z$ is also a solution of $R(z) = z$ and so for this R, *there are no periodic points of period* 2. The Julia set of $z^2 - z$ is shown in Figure 6.1.

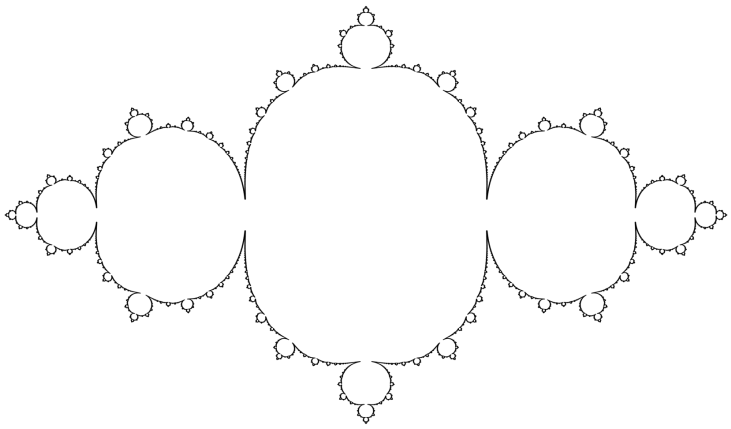


Figure 6.1: Julia set of $z \mapsto z^2 - z$.

This example raises a question of existence, namely: How often are periodic points absent? Could there exist, for example, a rational function with only finitely many periodic points? The answer is no. In fact, in the vast majority of cases, periodic points of a given period do exist and for polynomials, the situation is particularly striking. We have

Theorem 6.1 (Existence of periodic points for polynomials)**.** *Let* P *be a polynomial of degree at least two and suppose that* P *has no periodic points of period* n*. Then* $n = 2$ *and* P *is conjugate to* $z \mapsto z^2 - z$*.*

In short, the example given in (6.3) was the only example of this type (up to conjugacy) that we could possibly have given. There is also a corresponding (though slightly weaker) result for rational functions, namely

Theorem 6.2 (Existence of periodic points for rational maps)**.** *Let* R *be a rational function of degree* d*, where* $d \geqslant 2$*, and suppose that* R *has no periodic points of period* n*. Then* (d, n) *is one of the pairs*

$$(2, 2), (2, 3), (3, 2), (4, 2);$$

moreover, each such pair does arise from some R *in this way.*

This shows for example, that all rational functions have periodic points of period $4, 5, 6, \ldots$, and that every rational function of degree five or more has periodic points of all orders. These results are due to I.N. Baker [5], and we shall prove them in section 6.8. Roughly speaking, if a rational map fails to have any periodic points of period n, then every solution of $R^n(z) = z$ must also be a solution of some equation $R^m(z) = z$ for some m which divides, and is less than, n. This places a strong algebraic constraint on the coefficients of R which, according to the two results above, is rarely satisfied; nevertheless, the known proofs of these results are function-theoretic.

The reader should note that not only is Theorem 6.2 weaker than Theorem 6.1 in that several exceptional pairs (d, n) arise, but is also weaker in that an exceptional pair (d, n) does not necessarily determine a unique conjugacy class of exceptional maps R.

Remark 6.3 (Each rational map has periodic points with prime period p for all but at most $d + 1$ prime numbers p). Although we shall prove Theorem 6.1 and Theorem 6.2 later, we shall give a simple argument now which guarantees the existence of at least an infinite set of periodic points for each rational map (a fact that we shall need before we prove the two theorems). This also follows from Theorem 4.10, but the following argument is simple and direct. We suppose that R is a rational function of degree d, where $d \geqslant 2$, and that R has no points of period p,

where p is a prime. Then every solution ζ, say, of $R^p(z) = z$ is of some period k, where k is less than, and divides p: thus $k = 1$, and ζ is a fixed point of R itself. As R and R^p have divided $d+1$ and d^p+1 fixed points respectively, we deduce that there must be some fixed point ζ of R such that R^p has more fixed points at ζ than R does, and the only way that this can arise is that $R'(\zeta) \neq 1$ but $R'(\zeta)^p = 1$ (see Corollary 2.17.3). Finally, there are at most $d+1$ fixed points ζ fixed by R and for each of these, $R'(\zeta)^q = 1$ for at most one prime q. Thus, *each rational map has periodic points with prime period* p *for all but at most* $d+1$ *prime numbers* p.

If we consider this argument when R is a quadratic polynomial with no periodic points of period two, we find that up to conjugacy $R(0) = 0$ and $R'(0) = -1$. It follows that $R(z) = az^2 - z$, and just by changing the scale, this is easily seen to be conjugate to the polynomial $z^2 - z$ in (6.3).

6.3 (Super)Attracting Cycles

Suppose that ζ is a (super)attracting fixed point of a rational map R. Then $|R'(\zeta)| < 1$ and there is a number α, $|R'(\zeta)| < \alpha < 1$, and a disc D centered at ζ such that in D,

$$|R(z) - \zeta| = |R(z) - R(\zeta)| < \alpha|z - \zeta|.$$

This shows that each R^n maps D into itself, so D lies in the Fatou set $F(R)$ and $|R^n| \to \zeta$ uniformly on D. By Vitali's Theorem (Theorem 3.13), $R^n \to \zeta$ locally uniformly on the component $F(R)$ which contains ζ, and as $F(R^q) = F(R)$ for any iterate R^q, we easily obtain

Theorem 6.3 (Local basins of a (super)attracting cycle)**.** *Let* $\{\zeta_1, \ldots, \zeta_q\}$ *be a (super)attracting cycle of* R*. Then each* ζ_j *lies in a component, say* F_j*, of the Fatou set* $F(R)$*, and as* $n \to \infty$*,* $(R^q)^n \to \zeta_j$ *locally uniformly on* F_j*.*

Thus, each point of the cycle must lie in a distinct Fatou component, as every point in F_j converges to ζ_j. In view of this, we introduce some terminology.

Definition 6.6 (Local or immediate basin of a (super)attracting fixed point)**.** Given a (super)attracting fixed point ζ of R, the component of

$F(R)$ which contains ζ is called the *local*, or *immediate*, *basin* of ζ.

It follows that $R^n \to \zeta$ precisely when z lies in some inverse image $R^{-m}(B)$, $m \geqslant 0$, of the local basin B.

Definition 6.7 (Basin or stable set of a (super)attracting fixed point)**.** We call the set of such z the *basin* (or *stable set*) for ζ.

More generally, for cycles of any period,

Definition 6.8 (Local basin of a (super)attracting cycle)**.** The *local basin of a (super)attracting cycle* $\{\zeta_1, \ldots, \zeta_q\}$ is the union of the (necessarily distinct) components $F_1, \ldots, F_q$ of $F(R)$ (as in Theorem 6.3)

Definition 6.9 (Basin of a (super)attracting cycle)**.** And the *basin* of the cycle is the union of the local basins and all its inverse images.

Although a rational map R of degree d, $d \geqslant 2$, has infinitely many cycles, we shall see later that it can have *only finitely many (super)attracting* cycles (in fact, at most $2d - 2$). Thus, for example, a quadratic polynomial can have at most one (super)attracting cycle in $\mathbb{C}$ (the reader may recall that in section 1.6 we showed that a quadratic polynomial can have at most one attracting fixed point in $\mathbb{C}$).

Our next task is to analyze the behavior of an analytic function f in a neighborhood of an attracting fixed point ζ, and this analysis will enable us to understand completely the dynamics of f near ζ.

Definition 6.10 (Local analytic conjugate)**.** Let f be an analytic map in some neighborhood N of an attracting fixed point ζ, and we may assume that $f(N) \subset N$. Now let g be any function which is analytic and univalent on N with $g(\zeta) = 0$, and let $F = gfg^{-1}$. We call F a *local (analytic) conjugate* of f.

Remark 6.4 (Local analytic conjugates are analytic (on $g(N)$) and preserve fixed points and their multipliers). Note that, then, F is defined and analytic on the neighborhood $g(N)$ of the origin (see Figure 6.2) and, as

$F'(0) = f'(\zeta)$, we see that 0 is an attracting fixed point of F.

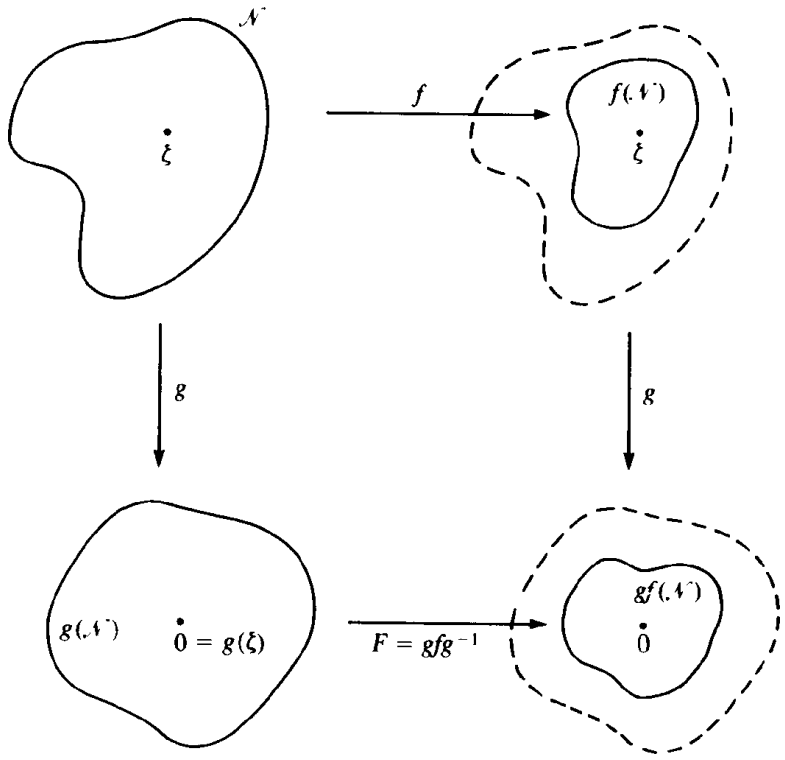


Figure 6.2

And we ask whether, by choosing g suitably, we can arrange for F to be considerably simpler (say, in terms of Taylor coefficients) than f is. The benefits that arise depend on how simple we can choose F to be but, in any case, the iterates of f (near ζ) are reflected in the behavior of the iterates of F for $F^n = gf^n g^{-1}$.

We shall assume that $\zeta = 0$: this simplifies the formulae, and the corresponding results for a general fixed point ζ can easily be obtained afterward by conjugation. The function f has a Taylor expansion at 0, say

$$f(z) = az + bz^q + \ldots, \quad 0 < |a| < 1,$$

and because $F'(0) = f'(\zeta)$, we must have

$$F(z) = az + b_2 z^2 + b_3 z^3 + \ldots.$$

We shall now show that by choosing a suitable g, we can achieve the greatest possible simplification, namely that each b_j is zero and F is the map $z \mapsto az$.

Theorem 6.4 (Koenigs' local linearization theorem)**.** *Suppose that* f *is analytic in a neighborhood of the origin, that* $f(0) = 0$*, and that* $f'(0) = a$ *where* $0 < |a| < 1$*. Then there exists a unique function* g *which is analytic in some disc* $\{|z| < r\}$ *with* $g(0) = 0$*,* $g'(0) = 1$*, and which satisfies* $gfg^{-1} = az$ *for all* z *sufficiently near the origin.*

Proof. See [7, p. 106]. □

In fact, this result implies more than we have stated so far.

Remark 6.5 ($B = W \cup W_1 \cup W_2 \cup \cdots$). We consider the case when f is a rational function, which we now denote by R, and we let B be the local basin of an attracting fixed point ζ of R. According to Theorem 6.4, there is a disc W centered at ζ such that $R(W) \subset W$, and a power series g analytic in W such that $gR(z) = ag(z)$ there. Now let W_n be the component of $R^{-n}(W)$ that contains ζ: we claim that

$$B = W \cup W_1 \cup W_2 \cup \cdots . \tag{6.4}$$

Proof. First, as W_n is a connected subset of $F(R)$, it must lie within B. Next, given any z in B, join z to ζ by a curve γ in B. Then by Theorem 6.3, there is some n such that $R^n(\gamma) \subset W$, and hence γ, being connected, lies in W_n. This shows that z is in W_n and thus proves (6.4). □

Remark 6.6 (Analytic continuation of g to local basin). Now we define g_n on W_n by the formula

$$g_n(z) = a^{-n} g(R^n z).$$

Clearly, g_n is analytic on W_n and for all z sufficiently close to the origin, $g_n(z) = g(z)$. We deduce that the sequence g_n provides an analytic continuation of g to the entire local basin B, and by analytic continuation, $gR(z) = ag(z)$ throughout B.

Finally, we observe that

$$g(W_n) = a^{-n} g(R^n(W_n)),$$

and as R^n maps each component of $R^{-n}(W)$ onto W, we have

$$g(W_n) = a^{-n} g(W).$$

Now $g(W)$ is an open neighborhood of the origin, and as $|a| < 1$, we easily deduce that $g(B) = \mathbb{C}$, and we have proved

Theorem 6.5 (Koenigs' global linearization theorem)**.** *Let ζ be an attracting fixed point of the rational map R with $R'(\zeta) = a$, and let B be the local basin of ζ. Then there exists a unique function g that is analytic in B with $g(\zeta) = 0$, $g'(\zeta) = 1$, and $g(B) = \mathbb{C}$, and such that in B, $gR(z) = ag(z)$.*

Proof. See [7, pp. 108–109] □

Remark 6.7 (g^{-1} does not exist throughout B). Unlike the local conjugacy relation $gRg^{-1}(z) = az$, the functional equation $gR(z) = ag(z)$ does not require the existence of g^{-1} (indeed, by Liouville's Theorem, g^{-1} cannot exist throughout B). The analysis of these equations and their solutions is attributed to Poincaré, Schröder and Koenigs.

6.4 Repelling Cycles

We begin with

Theorem 6.6 (Repelling cycles are in J)**.** *For any rational map R, every repelling cycle of R lies in $J(R)$.*

Proof. See [7, pp. 109–110]. □

We can also interpret Theorem 6.5 in the context of repelling fixed points. Suppose that ζ is a repelling fixed point of the rational map R, which implies that $R(\zeta) = \zeta$ and $|R'(\zeta)| > 1$. Because the derivative $R'(\zeta)$ is non-zero, the Inverse Function Theorem (Theorem 2.11) guarantees that a local branch of the inverse map, R^{-1}, exists in a small neighborhood of ζ. This inverse branch also fixes ζ (since $R^{-1}(\zeta) = \zeta$), and its derivative at ζ is given by the reciprocal

of the original derivative:

$$(R^{-1})'(\zeta) = \frac{1}{R'(\zeta)}.$$

Because $|R'(\zeta)| > 1$, it follows that $|(R^{-1})'(\zeta)| < 1$. Consequently, ζ acts as an *attracting* fixed point for this local inverse branch R^{-1}.

We can now directly apply Theorem 6.4 to this attracting branch R^{-1}. This provides a local analytic change of coordinates g such that R^{-1} is conjugate to a strict linear contraction near the origin:

$$gR^{-1}g^{-1}(z) = \frac{1}{R'(\zeta)}z.$$

By taking the algebraic inverse of both sides of this coordinate equation, we can undo the effect of R^{-1} and return to our original map. This demonstrates that R itself is locally conjugate to a linear expansion, namely

$$gRg^{-1}(z) = R'(\zeta)z,$$

valid in a sufficiently small neighborhood of the repelling fixed point ζ.

6.5 Rationally Indifferent Cycles

Although it is not easy to decide whether a given indifferent cycle lies in the Fatou or the Julia set, the question is easily settled for rationally indifferent cycles.

Theorem 6.7 (Rationally indifferent cycles are in J)**.** *If* $\deg(R) \geqslant 2$ *then every rationally indifferent cycle of* R *lies in* $J(R)$.

Proof. See [7, p. 111]. □

The remainder of this section is devoted to a detailed description of the dynamics of the iterates R^n near rationally indifferent cycles. This is a long and difficult task, and in order to obtain a clear understanding of the global structure of the arguments, we have provided a succession of results, each (generally speaking) being an improvement on the preceding results. One of the reasons why this description is much harder to obtain than, say, the

description of the dynamics near an attracting fixed point, is that a rational map R is *not* locally conjugate to its (linear) derivative near a rationally indifferent point ζ. Because the derivative is a rotation of finite order about ζ ($|R'(\zeta)| = 1 \implies R'(\zeta) = \exp(i\theta)$ because ζ is indifferent, and it is a root of the unity), and R can have finite order only if $\deg(R) = 1$ (because if R has finite order, there exists a natural number k such that $R^k = I$, thus, $\deg(R^k) = \deg(R)^k = \deg(I) = 1$). This observation means that we have to find another way to describe the dynamics of the R^n near such a cycle, and the answer lies in showing that R is conjugate to a translation not, of course, on the whole neighborhood of ζ, but on certain open sets which have ζ on their boundary. We shall include examples to illustrate the ideas as we proceed, and we begin with a simple lemma which isolates the most basic idea in this section.

Lemma 6.8 (Alternating attracting and repelling sectors for a rationally indifferent point)**.** *Suppose that* f *is analytic and satisfies*

$$f(z) = z - z^{p+1} + O(z^{p+2})$$

in some neighborhood N *of the origin. Let* $\omega_1, \ldots, \omega_p$ *be the* p*-th roots of unity and let* $\eta_1, \ldots, \eta_n$ *be the* p*-th roots of* -1*. Then for sufficiently small positive numbers* r_0 *and* θ_0*:*

(i) $|f(z)| < |z|$ *on each sector*

$$S_j = \left\{ z \colon 0 < \left| \frac{z}{\omega_j} \right| < r_0, \left| \arg\left(\frac{z}{\omega_j} \right) \right| < \theta_0 \right\};$$

and

(ii) $|f(z)| > |z|$ *on each sector*

$$\Sigma_j = \left\{ z \colon 0 < \left| \frac{z}{\eta_j} \right| < r_0, \left| \arg\left(\frac{z}{\eta_j} \right) \right| < \theta_0 \right\}.$$

Proof. See [7, p. 112]. □

The numbers ω_j and η_j in Lemma 6.8 are the $2p$-th roots of unity, and they

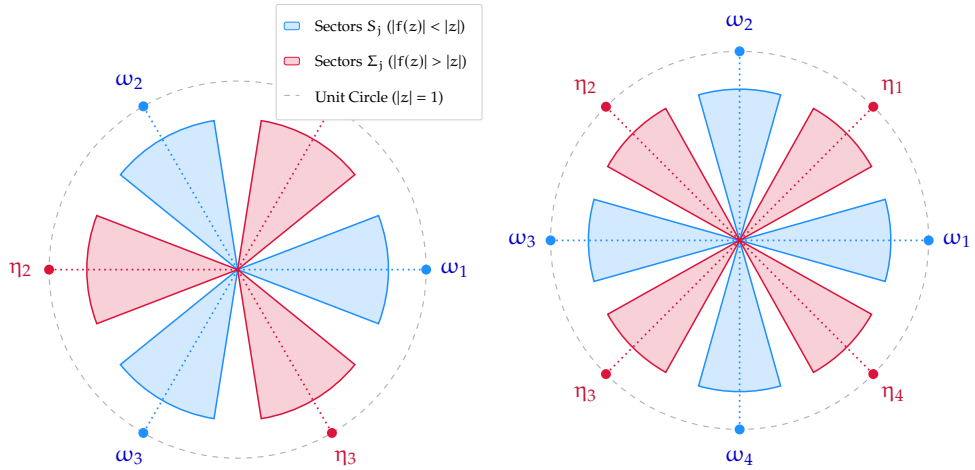
are alternately and equally spaced around the unit circle (see Figure 6.3). If

$$f(z) = z + az^{p+1} + O(z^{p+2}), \quad a \neq 0,$$

a similar result holds with the directions ω_j and η_j replaced by the directions of the solutions $az^p = -1$ and $az^p = 1$ respectively. Also with this f, f^{-1} exists near the origin and

$$f^{-1}(z) = z - az^{p+1} + O(z^{p+2})$$

so (with r_0 and θ_0 small enough) $|f^{-1}(z)| > |z|$ on each S_j and $|f^{-1}(z)| < |z|$ on each Σ_j.



(a) $p = 3$. (b) $p = 4$.

Figure 6.3: Alternating attracting and repelling sectors for a rationally indifferent point

Next, we consider a simple example and here our discussion contains the ideas that we shall use to develop the general theory.

Example 6.3. The polynomial

$$P(z) = z - z^2$$

has a rationally indifferent fixed point at the origin. If $0 < x < 1$, then $0 < P(x) < x$, and so $P^n(x) \to 0$ (for $P^n(x)$ can only converge to a fixed point of P): if $x < 0$, then $P(x) < x < 0$, and so $P^n(x) \to \infty$. These observations are consistent with Lemma 6.8, and we shall now investigate the behavior of $P^n(z)$ for all z near the origin.

Let $\sigma(z) = 1/z$ and replace P by its conjugate $Q = \sigma P \sigma^{-1}$: thus

$$Q(z) = z + 1 + \frac{1}{z-1}.$$

Now note that if $\mathrm{Re}\,[z] > 1$ then $\mathrm{Re}\left[(z-1)^{-1}\right] > 0$ so

$$\mathrm{Re}\,[Q(z)] \geqslant \mathrm{Re}\,[z] + 1.$$

In particular, if $t > 1$, then Q maps $\{x + iy \mid x > t\}$ into $\{x + iy \mid x > t + 1\}$, and rewriting this in terms of P, we see that $P^n \to 0$ uniformly in each sufficiently small disc $\{z\colon |z - r| < r\}$. In fact, P acts as illustrated in Figure 6.4 (which the reader should compare with Figure 1.1).

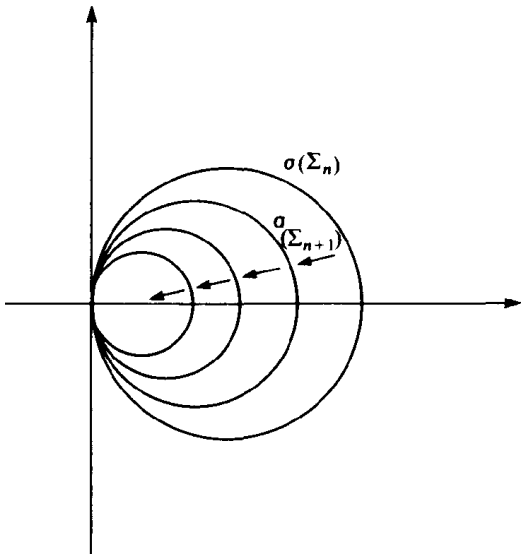


Figure 6.4

Example (Continuation)**.** Because of the particular simple form of Q, we can say more than this. Suppose that Π is any region that is disjoint from $\{|z| \leqslant 3\}$, and which is mapped into itself by Q. Then on Π,

$$\mathrm{Re}\,[Q(z)] \geqslant \mathrm{Re}\,[z] + 1 - \frac{1}{|z-1|} \geqslant \mathrm{Re}\,[z] + \frac{1}{2},$$

so

$$\mathrm{Re}\,[Q^n(z)] \geqslant \mathrm{Re}\,[z] + n/2$$

and $Q^n \to \infty$ on Π ("drifting" to the right). One possibility for this Π is the half-plane $\{\mathrm{Re}\,[z] > 3\}$, but it is easy to find a much larger region than this. Now $Q(z)$ lies in the shaded disc illustrated in Figure 6.5, and if $z \in \Pi$, the radius $r(z)$ of this disc is at most $1/2$. Furthermore, $r(z) \to 0$ as $z \to \infty$ and this suggests that we might be able to take Π to be bounded by some parabola. We shall now show that we can.

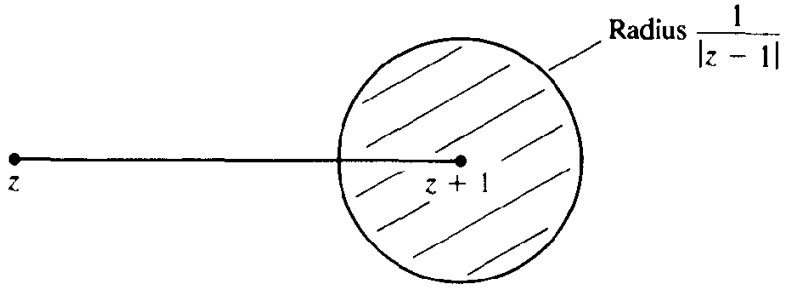


Figure 6.5

Example (Continuation)**.** Let

$$\Pi = \{x + iy \mid y^2 > 12(3 - x)\}.$$

The boundary of Π is a parabola which cuts the x-axis when $x = 3$, and it is easy to see that Π is disjoint from $\{|z| \leqslant 3\}$. Writing

$$z = x + iy, \quad Q(z) = X + iY, \quad \frac{1}{z-1} = a + ib$$

the expression of $Q(z)$ yields

$$X = x + 1 + a, \quad Y = y + b$$

and so for z in Π,

$$\begin{aligned} Y^2 - 12(3 - X) &= (y + b)^2 - 12(3 - [x + 1 + a]) \\ &= [y^2 - 12(3 - x)] + b^2 + 2by + 12(1 + a) \\ &\geqslant 12 - 2|by| - 12|a|. \end{aligned}$$

However, as $|z| \geqslant 3$, we have

$$|a| \leqslant |a + ib| \leqslant \frac{1}{|z| - 1} < \frac{1}{2}$$

and

$$|yb| \leqslant \frac{|z|}{|z|-1} < \frac{1}{3}.$$

It follows that if z is in Π, then $Y^2 > 12(3-X)$ and hence so is $Q(z)$: we deduced that Q maps Π into itself and so $P^n \to 0$ on $\sigma(\Pi)$. The Julia set of P is illustrated in Figure 6.6 where the black region is $\sigma(\Pi)$.

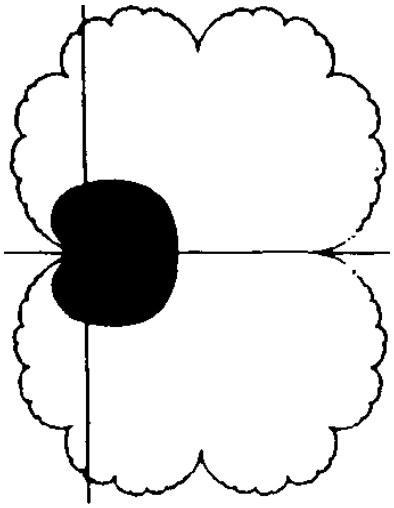


Figure 6.6: Julia set of $z \mapsto z - z^2$.

In order to analyze the situation for a general rationally indifferent fixed point, we need the notion of petals (which, roughly speaking, play the role of the region $\sigma(\Pi)$ in the example above).

Definition 6.11 (Petal)**.** For each positive t, each positive integer p and each k in $\{0, 1, \ldots, p-1\}$, we define the sets

$$\Pi_k(t) = \left\{ r\exp(i\theta) \middle| r^p < t(1+\cos(p\theta)); \left| \frac{2k\pi}{p} - \theta \right| < \frac{\pi}{p} \right\}.$$

These sets are called *petals* (at the origin).

We can see an illustration of these petals when $p = 6$ in Figure 6.7. Note that the petals are pairwise disjoint, and that each subtends an angle $2\pi/p$ at the origin (so the total angle subtended at the origin by all the petals is 2π).

Definition 6.12 (Axis of a petal)**.** We call the line of symmetry of $\Pi_k(t)$ (the ray $\theta = 2k\pi/p$) the *axis* of Π_k.

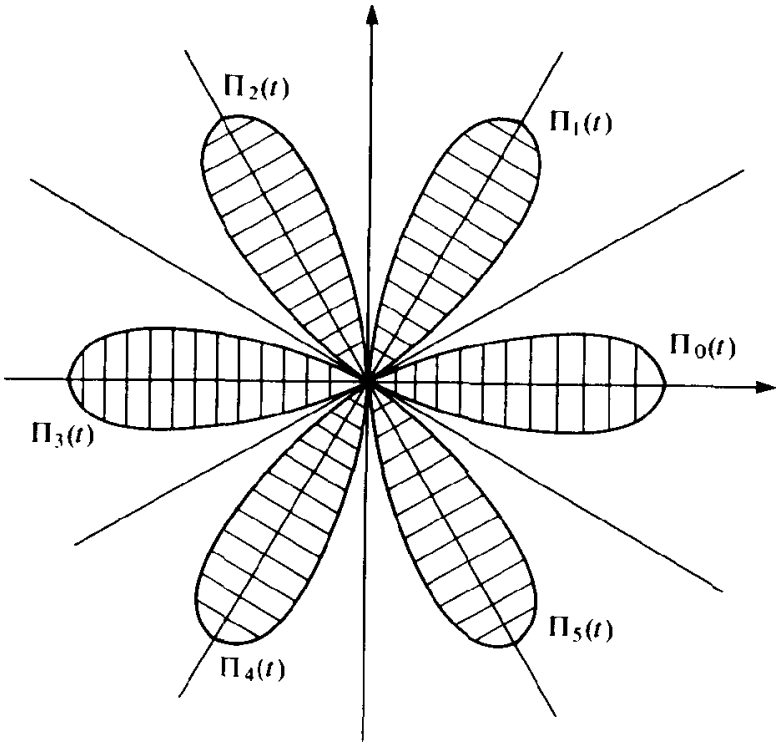


Figure 6.7: Julia set of $z \mapsto z - z^2$.

We come now to the first major result of this section and in this, we describe the dynamics of the iterates R^n in the petals at a fixed point ζ where $R'(\zeta) = 1$. By conjugation, we may assume that $\zeta = 0$ and, in addition, we shall assume that R has a Taylor expansion there of the form

$$R(z) = z(1 - z^p + bz^{2p} + cz^{2p+1} + \cdots).$$

Of course, we can always ensure that z^{p+1} has coefficient -1 simply by conjugating R with a map of the form $z \mapsto \alpha z$: thus *the significant part of this assumption is that there are no terms between* z^{p+1} *and* z^{2p+1} *in the Taylor series for* R. We emphasize that the role of this assumption is that it enables us to give an *explicit description of the petals*; without it, we can only establish the existence of petal shaped regions which are conformal images of $\Pi_k(t)$ described above.

Despite the fact that our interest centers on rational functions, it is desirable to prove the result for general analytic functions, for we can then apply it to branches R^{-1}. We now prove

Theorem 6.9 (The Petal Theorem)**.** *Suppose that the analytic map* f *has a Taylor expansion*

$$f(z) = z - z^{p+1} + O(z^{2p+1}) \tag{6.5}$$

at the origin. Then for all sufficiently small t*:*

(i) f *maps each petal* $\Pi_k(t)$ *into itself;*

(ii) $f^n(z) \to 0$ *uniformly on each petal as* $n \to \infty$*;*

(iii) $\arg(f^n(z)) \to 2k\pi/p$ *locally uniformly on* Π_k *as* $n \to \infty$*;*

(iv) $|f(z)| < |z|$ *on a neighborhood of the axis of each petal;*

(v) $f : \Pi_k(t) \to \Pi_k(t)$ *is conjugate to a translation.*

Proof. See [7, pp. 116–122].
In the proof the following lemmas are used.
Lemma[Forward Invariance and Translation of Π]:

(i) Π is forward invariant under g;

(ii) $\mathrm{Re}\,[g^n(w)] \to +\infty$ uniformly on Π;

(iii) $g : \Pi \to \Pi$ is conjugate to a translation.

Proof: [7, pp. 118–122]
Lemma[Local Uniform Convergence of u_n]: u_n converges locally uniformly on Π to some function u that is holomorphic and univalent on Π.
Proof: [7, p. 122]. □

By (i) the iterates f^n are defined on each $\Pi_k(t)$, and (iii) implies that for any z in $\Pi_k(t)$, the sequence $f^n(z)$ converges to 0 along a path which is asymptotic to the axis of the petal $\Pi_k(t)$: see Figure 6.8.

We recall that Theorem 6.9 has only been proved under the assumption (6.5) on the coefficients of R. The next step is to prove the corresponding result for functions f which satisfy $f(0) = 0$ and $f'(0) = 1$, but not the condition (6.5), and for this we need the following result.

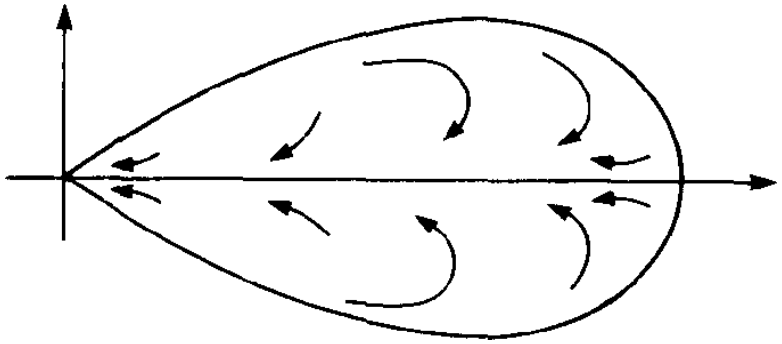


Figure 6.8: The dynamics on a petal.

Theorem 6.10 (Condition relaxation for the Petal Theorem)**.** *Suppose that* f *is analytic near* 0 *with*

$$f(z) = z + az^{p+1} + O(z^{p+2}), \quad a \neq 0,$$

as $z \to 0$*. Then* f *is conjugate near* 0 *to a function*

$$F(z) = z - z^{p+1} + O(z^{2p+1}).$$

Proof. See [7, pp. 123–124]. □

Remark 6.8 (First generalization of the Petal Theorem, still $R'(0) = 1$). With this available, the existence of petals (and the dynamics of f^n on these petals) for analytic maps of the form

$$f(z) = z + az^{p+1} + \cdots,$$

follows immediately. Indeed, by Theorem 6.10, such and f is conjugate to some F near the origin, and Theorem 6.9 is applicable to F; thus we can define the petals for f as the conformal images (under the conjugation map) of the petals of F. We shall not give an explicit statement of this result, and it suffices to remark that the conclusions of Theorem 6.9 are valid (using Lemma 6.8 to justify (iv)) in these more general circumstances except for the fact that we no longer have an explicit expression for the petals. We do, however, know that each petal for f subtends an angle $2\pi/p$ at the origin, and that the petals are pairwise disjoint (if taken small enough) for the function conjugating f to F is conformal there.

Theorems 6.9 and 6.10 describe the action of the iterates of any map of the form

$$f(z) = z + az^{p+1} + \cdots$$

on the p petals of f at the origin. We now consider the case when f is a rational map R and obtain information about the relationship of the petals to the Fatou set of R.

Theorem 6.11 (Relation between petals and the Fatou set)**.** *Let* R *be a rational map and suppose that*

$$R(z) = z + az^{p+1} + \cdots, \quad a \neq 0,$$

near the origin. Let Π_j *be the petals of* R *and, for each* j*, let* F_j *be the component of* $F(R)$ *that contains* Π_j*. Then:*

(i) $R^n(z) \to 0$ *and* $\arg(R^n(z)) \to 2\pi k/p$ *on* F_k*; and*

(ii) $F_0, \ldots, F_{p-1}$ *are distinct components of* $F(R)$*.*

Proof. See [7, p. 126]. □

Proof. See [7, pp. 126–128].
In the proof the following lemmas are used.
Lemma[Normality of Iterates in W]: $\{\varphi_n\}$ is normal in W.
Proof: See [7, p. 127]. □

Proof. See [7, pp. 128–130]. □

The geometric picture, then, is as follows. The rationally indifferent fixed point 0 lies on the boundary of p components F_k of $F(R)$, and each F_k contains a petal Π_k which makes an angle $2\pi/p$ at 0. Next, R maps each F_k into itself, and $R^n(z) \to 0$ locally uniformly on each F_k in such a way that for any z in F_k, $R^n(z) \to 0$ on a path which is asymptotic to the axis of the petal Π_k. It is an immediate consequence of this that different petals lie in different components of $F(R)$. In addition, this convergence shows that for any z, $R^n(z)$ ultimately lies within Π_k, so the consecutive inverse images of the petal Π_k in F_k expand to fill the entire component F_k.

Finally, we note that a rationally indifferent fixed point may also lie on the boundary of components of F other than the F_j: indeed, if $P(z) = z + \cdots$ is any non-linear polynomial, then the origin is not only on the boundary of each of the p components F_j, but it is also on the boundary of the completely invariant component F_∞ of F which contains ∞ (and which is distinct from each F_j because $P^n \to \infty$ on F_∞). As F_∞ is connected, it must therefore contain a thin "tongue" which stretches to the origin *between* the petals: see Figure 6.9 (where $p = 5$) and Figure 6.10 (where $p = 3$) in which the petals are colored black.

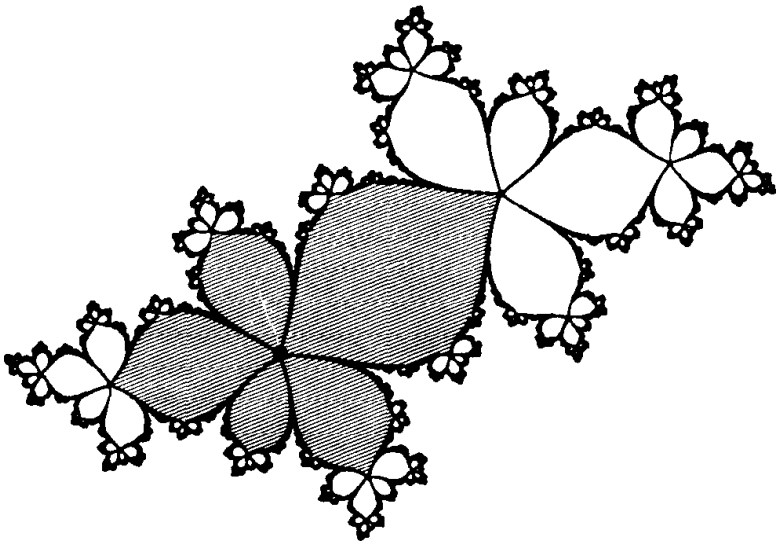


Figure 6.9

Remark 6.9 (Second generalization of the Petal Theorem). There is still one more result that we must consider before we can give the Petal Theorem in its full generality, namely the extension of 6.11 to the case when $R(0) = 0$ and $R'(0)$ is a root of unity other than 1. We assume, then, that

$$R(z) = az + bz^{p+1} + \cdots, \tag{6.6}$$

where $a \neq 1$ but $a = \exp(2\pi ir/q)$ where r and q are coprime (so $a^n = 1$ when $n = q$ but not for any smaller n). In this case, R^q is of the form

$$R^q(z) = z + cz^{t+1} + \cdots$$

for some t, so R^q has t petals at the origin. This means that there are components $F_1, \ldots, F_t$ of $F(R^q)$, and hence of $F(R)$, each containing a petal of R^q. As R maps each F_j to some component of $F(R)$ which makes an

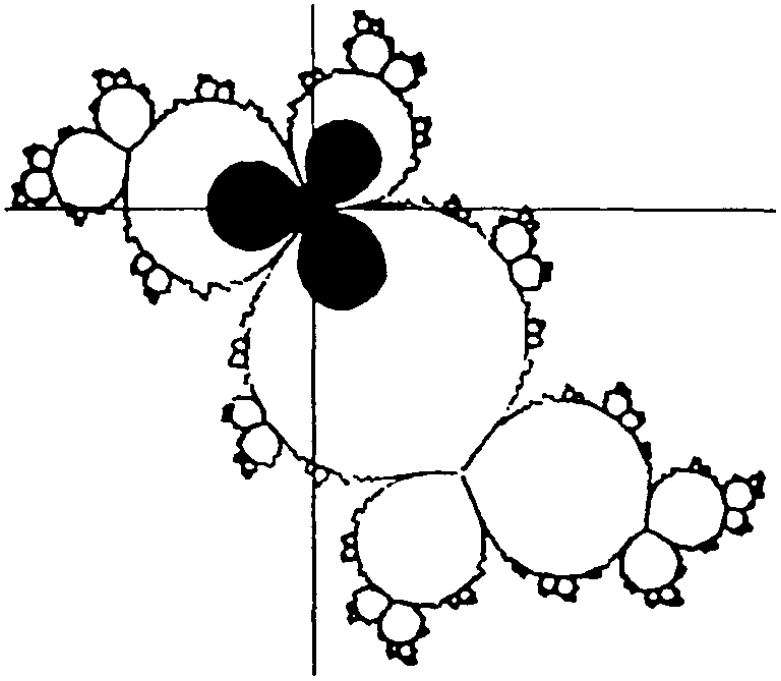


Figure 6.10: $z \mapsto z\exp(2\pi i/3) + z^2$.

angle $2\pi/t$ at the origin, we see that R must act as a permutation of the set $\{F_1, \ldots, F_t\}$. Further, as $a = \exp(2\pi ir/q)$ with $(r, q) = 1$, we find (from Theorem 6.9 applied to R^q) that R (viewed as a permutation of the F_j) is a composition of, say, k disjoint cycles each of length q, and if follows that $t = kq$. Thus, given R in the form (6.6), there is an integer k such that R has kq petals at the origin, these dividing naturally into k sets of q petals such that R acts as a cycle of length q on each such set. We shall not give a formal statement of this as it is a special case of our final version of the Petal Theorem which follows very shortly.

Observe that if R is given by (6.6), then

$$R^n(z) = a^n z + \cdots,$$

so if q does not divide n, then R^n has only one fixed point at the origin. If q does divide n, say $n = mq$, then

$$R^q(z) = z + cz^{t+1} + \cdots$$

so

$$\begin{aligned} R^n(z) &= (R^q)^m(z) \\ &= z + mcz^{t+1} + \cdots \end{aligned}$$

and as R^n has $t+1$ $(= kq+1)$ fixed points at the origin.

We turn now to the final and most comprehensive version of the Petal Theorem. Let $\{\zeta_1, \ldots, \zeta_m\}$ be a rationally indifferent cycle of R and write

$$\zeta_{n+i} = R^n(\zeta_i)$$

for all n, $n \geqslant 1$, so that $\zeta_i = \zeta_j$ if and only if $i \equiv j \mod m$. Essentially the only difference between this case and the previous one is that now, R maps each component of $F(R)$ that contains a petal at ζ_j to some other component containing a petal at ζ_{j+1}.

Of course, the previous version applies to the map R^m at each point ζ_j of the cycle, so we now put $S = R^m$ and suppose that

$$S(z) = R^m(z) \tag{6.7}$$

$$= \zeta_j + \exp\left(\frac{2\pi i r}{q}\right)(z-\zeta_j) + b(z-\zeta_j)^{p+1} + \cdots, \tag{6.8}$$

where $(r, q) = 1$. Note that the coefficient $\exp(2\pi i r/q)$ is independent of the choice of j here as it is the multiplier of the cycle. In keeping with the previous case, we also assume that

$$S^q(z) = R^{mq}(z) \tag{6.9}$$

$$= \zeta_j + (z-\zeta_j) + c(z-\zeta_j)^{t+1} + \cdots, \tag{6.10}$$

where (as before, but using S instead of R) $t = kq$ for some k. Note that S^q acting near ζ_j is conjugate (by the branch of R^{-1} which maps ζ_{j+1} to ζ_j) to RS^qR^{-1} $(= S^q)$ acting near ζ_{j+1}, and so S^q has the same number of fixed points at ζ_j as it has at ζ_{j+1}: this proves that t in (6.9), and hence k also, is independent of the choice of ζ_j.

Now S has qk $(= t)$ petals at each ζ_j, hence associated with the cycle $\{\zeta_1, \ldots, \zeta_m\}$ are mkq components $F_1, \ldots, F_{mkq}$ of $F(R)$. The action of R is to map F_1, say, with a petal at ζ_1, to F_2, say with a petal at ζ_2, then F_2 to F_3 with a petal at ζ_3, and so on until F_{m+1} $(= \zeta_1)$. However, $F_{m+1} \neq F_1$ (unless $q = 1$) for S maps F_1 to F_{m+1}. In this way, R acts as a permutation of $\{F_1, \ldots, F_{mkq}\}$ as the product of k cycles each of length mq. Thus, we have the next (and final) version of the Petal Theorem.

Theorem 6.12 (The Petal Theorem for cycles)**.** *Let $\{\zeta_1, \ldots, \zeta_m\}$ be a rationally indifferent cycle for R, and let the multiplier of R^m at each point of the cycle be $\exp(2\pi i r/q)$ where $(r, q) = 1$. Then there exists an integer k, and*

mkq *distinct components* $F_1, \dots, F_{mkq}$ *of* $F(R)$ *such that at each* ζ_j *there are exactly* kq *of these components containing a petal of angle* $2\pi/kq$ *at* ζ_j. *Further,* R *acts as a permutation* τ *on* $\{F_1, \dots, F_{mkq}\}$, *where* τ *is a composition of* k *disjoint cycles of length* mq, *and a petal based at* ζ_j *maps under* R *to a petal based at* ζ_{j+1}.

More information can be found in section 6.9, especially in Theorem 6.25 and (6.15).

We end with a count of the number of fixed points of R in the cycle described in the Petal Theorem: this will be needed in section 6.10, and it is convenient to prove it here while the relevant ideas are still fresh in the reader's mind. We recall that $q \mid n$ means that q divides n, while $q \nmid n$ means that it does not.

Theorem 6.13 (Fixed points of R^n at ζ_j)**.** *Suppose that the hypotheses of Theorem 6.12 and also* (6.7), (6.9) *hold.*

(i) If $m \nmid n$, *then* R^n *has no fixed points at* ζ_j.

(ii) If $m \mid n$ *but* $mq \nmid n$, *then* R^n *has one fixed point at* ζ_j.

(iii) If $mq \mid n$, *then* R^n *has* $t + 1$ *fixed points at* ζ_j.

Proof. See [7, p. 132]. □

6.6 Irrationally Indifferent Cycles in F

We recall the following definition.

Definition 6.13 (Linearizable map)**.** An analytic map f is *linearizable* near a fixed point ζ if f is conjugate to

$$f_0 : z \mapsto \zeta + (z - \zeta)f'(\zeta)$$

in some neighborhood of ζ: explicitly, if there is some neighborhood N of ζ, and some map g which fixed ζ, such that f is analytic on N, g is analytic and injective in $N \cup f(N)$, and $gfg^{-1} = f_0$ on $g(N)$.

We have seen that a rational map R is linearizable near each attracting, and each repelling, fixed point. Near an indifferent fixed point ζ, R is or is not linearizable depending *only* on whether ζ is in F or J, and this is our next result.

Theorem 6.14 (R linearizable near ζ indifferent $\iff$ $\zeta \in F$)**.** *Let ζ be an indifferent fixed point of a rational map R. Then R is linearizable near ζ if and only if ζ lies in $F(R)$.*

Proof. See [7, p. 133]. □

With reference to Theorem 6.14, we know that there can be indifferent fixed points of R in $J(R)$ (for example, rationally indifferent fixed points), and we shall soon see that there can be indifferent fixed points in $F(R)$. Ignoring (for the moment) this question of existence, we proceed to characterize the indifferent fixed points in the Fatou set by a method which gives an alternative proof of Theorem 6.14. We prove

Theorem 6.15 (Fatou components containing indifferent fixed points are Siegel discs)**.** *Let R be a rational map of degree d, $d \geqslant 2$, and suppose that ζ is an indifferent fixed point lying in some component F_0 of $F(R)$. Then F_0 is simply connected, and $R : F_0 \to F_0$ is analytically conjugate to a rotation of infinite order of the unit disc Δ.*

Proof. See [7, pp. 134–136]. □

Definition 6.14 (Siegel disc)**.** Any component F_0 of this type is called a *Siegel disc*.

Siegel discs are named after C.L. Siegel who, in 1941, was the first to establish their existence (see [23] and [24]). We shall verify their existence here, and for this we need the following two results.

Theorem 6.16 (Condition for an analytic funtion to be linearizable)**.** *Suppose that $|\alpha| = 1$, and that for some positive M and k, and all positive integers*

n,

$$\frac{1}{|\alpha^n - 1|} \leqslant Mn^k. \tag{6.11}$$

Then any analytic function f *with Taylor expansion*

$$f(z) = \alpha z + a_2 z^2 + \cdots$$

is linearizable near the origin.

Proof. See [7, pp. 136–141].
The following lemmas are used in the proof.
Lemma[Action Scheme of Conjugating Maps]: The maps φ, f and φ^{-1} act according to the scheme

$$D_4 \xrightarrow{\varphi} D_3 \xrightarrow{f} D_2 \xrightarrow{\varphi^{-1}} D_1$$

Proof: [7, p. 139].
Lemma[Derivative Bound for G on D_5]: On D_5, $|G'(z)| \leqslant 2C\delta^2/\theta^{k+2}$.
Proof: [7, p. 139].
Or see Chapter 3 of [24] or the original proof in [23]. □

Theorem 6.17 (Measure of the set of α's that satisfy the condition for an analytic function to be linearizable)**.** *Let* S *be the set of numbers* α *which satisfy* $|\alpha| = 1$ *and* (6.11) *for some* M *and* k*, and all* n*. Then* S *has measure* 2π*.*

Proof. See [7, pp. 141–142]. □

These imply the existence of Siegel discs, for Theorem 6.17 shows that there is some α in S, and Theorem 6.16 then implies that $P : z \mapsto \alpha z + z^2$ is linearizable near the origin. It follows from Theorem 6.14 that $0 \in F(P)$ and finally, by Theorem 6.15, that the origin is the center of a Siegel disc for P. Much work has been done to find necessary and sufficient conditions on α for f to be linearizable, and we refer the reader to [14] for more details.

6.7 Irrationally Indifferent Cycles in J

Our earlier results give us a clear understanding of the dynamics of the iterates R^n in the vicinity of all attracting, repelling, and rationally indifferent cycles, and also irrationally indifferent cycles in F, and we shall consider superattracting cycles in Theorem 6.23. The one result in these sections shows that the remaining possibility, namely irrationally indifferent cycles in J, can occur although the geometry and dynamics of the iterates in a neighborhood of these cycles seems not to be understood at all. We prove

Theorem 6.18 (Condition for existence of irrationally indifferent fixed points in J for polynomials)**.** *Suppose that* $P(z) = \alpha z + \cdots + z^d$, *where* $d \geqslant 2$ *and* $|\alpha| = 1$ *but* α *is not a root of unity. If*

$$|\alpha^n - 1| \leqslant \left(\frac{1}{n}\right)^{d^n - 1} \tag{6.12}$$

for infinitely many n, *then the origin is an irrationally indifferent fixed point of* P *that lies in* $J(P)$.

Proof. See [7, p. 143]. □

Remark 6.10 (Compare conditions on α). The reader should compare (6.12) with (6.11)

Remark 6.11 (α's verifying (6.12) exist). Of course, Theorem 6.18 does not establish the existence of irrationally indifferent fixed points in J, for we have yet to show that number α satisfying (6.12) do exist. In fact, it is easy to see that such numbers α are dense in the unit circle, and we begin by showing that if $\alpha = \exp(2\pi i\theta)$, then the condition (6.12) is closely related to the approximation of θ by rational numbers.

Now for any positive integers m and n, we have

$$\begin{aligned} |\alpha^n - 1| &= |\exp(n\pi i\theta) - \exp(-n\pi i\theta)| \\ &= 2|\sin(n\pi\theta)| \\ &= 2|\sin \pi(n\theta - m)| \\ &\leqslant 2\pi|n\theta - m|. \end{aligned}$$

It follows that (6.12) holds if for infinitely many n there is some m (depending on n) such that

$$\left|\theta - \frac{m}{n}\right| \leqslant \frac{1}{n^{N+2}}, \tag{6.13}$$

for then (with $n > 2\pi$)

$$|\alpha^n - 1| \leqslant 2\pi n \left|\theta - \frac{m}{n}\right| \leqslant \frac{1}{n^N}.$$

The problem, then, reduces to showing that the set of θ satisfying (6.13) is dense in $\mathbb{R}$. Now each real number x has a continued fraction expansion and the convergents of the continued fractions are particularly good rational approximations to x. By defining θ in terms of a specified continued fraction expansion, we can construct many θ satisfying (6.13), and it is apparent from the construction that the set of such θ is dense in $\mathbb{R}$. A self-contained argument for the construction is given in the exercises [7].

6.8 The Proof of the Existence of Periodic Points

We recall Theorem 6.1, namely that if a polynomial P of degree d ($d \geqslant 2$) has no periodic points of period N, then $N = 2$ and P is conjugate to $z \mapsto z^2 - z$. In this section, we prove this result and also the corresponding result for rational functions (Theorem 6.2).

The Proof of Theorem 6.1. See [7, pp. 145–147]. □

Finally, we comment on the

The Proof of Theorem 6.2. See [7, pp. 147–148]. □

6.9 The Julia Set and Periodic Points

We are now in a position to relate the Julia set to periodic points. First, we prove

Theorem 6.19 ($J = \{\text{periodic points}\}'$)**.** *Let* R *be a rational map of degree* d*, where* $d \geqslant 2$*. Then* J *is the derived set of the periodic points of* R*.*

Proof. See [7, pp. 148–149].
In the proof Lemma 6.21 is used to prove that the derived set of the periodic points of R is contained in J. □

Remark 6.12 (Recall of definition of accumulation point)*.* We recall that z is in the derived set of E if and only if there are *distinct* points z_n in E such that $z_n \to z$.

Later, we shall see that R has at most $2d-2$ non-repelling cycles, and with this Theorem 6.19 implies that J is the derived set of the repelling cycles. As each repelling cycles lies in J (and anticipating the result in section 9.6 of [7]), this yields

Theorem 6.20 ($J = \overline{\{\text{repelling periodic points}\}}$)**.** J *is the closure of the repelling periodic points of* R*.*

Lemma 6.21 (Components of F have at most one periodic point)**.** *Any component of* F *contains at most one periodic point of* R*.*

Proof. See [7, p. 149]. □

We end this section with the following extension of Theorem 4.9

Theorem 6.22 (W domain and $W \cap J \neq \emptyset \implies R^n(W)$ contains any compact set that doesn't have exceptional points)**.** *Let R be a rational map of degree d, where $d \geqslant 2$, let W be a domain that meets J, and let K be a compact set which contains no exceptional points of R. Then for all sufficiently large n, $R^n(W) \subset K$.*

Proof. See [7, pp. 149–150]. □

6.10 Local Conjugacy

Let

$$f(z) = a_1 z + a_2 z^2 + \cdots$$

be a power series, analytic near the origin. In this final section of the chapter, we comment briefly on the possible forms that the local conjugates gfg^{-1} of f can take. We shall assume that $g(0) = 0$, and we shall discuss *analytic conjugacy* (when g is analytic), *topological conjugacy* (when g is a homeomorphism), and *formal conjugacy* (when g is a formal power series).

We have already discussed analytic conjugacy when $|a_1| \neq 0, 1$ (Theorem 6.4). The situation when $a_1 = 0$ is straightforward, and the most optimistic guess is that in this case f is conjugate to the first non-trivial term of its Taylor expansion, say to $a_q z^q$ where $q \geqslant 2$. This is correct, but as $q \geqslant 2$, the two maps $z \mapsto a_q z^q$ and $z \mapsto z^q$ are conjugate, and so we can state this result in the following form

Theorem 6.23 (Simple local analytic conjugate for super-attracting fixed point)**.** *Suppose that*

$$f(z) = b_0 z^q + b_1 z^{q+1} + \cdots$$

is analytic near the origin, and that $b_0 \neq 0$, $q \geqslant 2$. Then there exists a unique function g, analytic near the origin, such that:

(i) $g(0) = 0$, $g'(0) = 1$; *and*

(ii) for all z near the origin, $gfg^{-1}(z) = z^q$.

Proof. See [7, pp. 151–152] □

Lemma 6.24 (Principal root bound). *Suppose that* $n \geqslant 2$, *let* $W = \{w : |w - 1| < 1/2\}$, *and let* $z^{1/n}$ *denote the principal branch of the* n*-th root of* z *in* W. *Then*

$$\left|z^{\frac{1}{n}} - 1\right| < 2\frac{|z-1|}{n} < \frac{1}{n}.$$

Proof. See [7, p. 151]. □

We shall now briefly consider the local analytic conjugates of f when $a_1 = 1$. In this case, f has the form

$$f(z) = z(1 + \alpha_p z^p + \alpha_{p+1} z^{p+1} + \cdots) \tag{6.14}$$

and by the Residue Theorem, the function $\tau(f)$ defined by

$$\tau(f) = \frac{1}{2\pi i} \int_{|z|=r} \frac{1}{f(z) - z} dz$$

is independent of r for $r < r_0$, say. Further, if we expand $(f(z) - z)^{-1}$ in a Laurent series about the origin, we see that $\tau(f)$ is of the form

$$\frac{P(\alpha_p, \ldots, \alpha_{2p})}{(\alpha_p)^{p+1}}$$

for some polynomial P: for example, if $p = 1$, then

$$\tau(f) = -\frac{\alpha_2}{\alpha_1^2}.$$

We now prove that the function τ is invariant under analytic conjugation.

Theorem 6.25 (τ is invariant under analytic conjugation). *If* f *is of the form* (6.14), *and is analytically conjugate to* g *near the origin, then* $\tau(f) = \tau(g)$.

Proof. See [7, pp. 153–154]. □

As an application of Theorem 6.25, recall that if f is of the form (6.14), then f is locally conjugate to a function

$$z \mapsto z(1 + z^p + Az^{2p} + O(z^{2p+1})) \tag{6.15}$$

(Theorem 6.10), and in this case, a simple calculation gives

$$\tau(f) = -A.$$

Next, we have

Theorem 6.26 (Elimination of intermediate terms)**.** *Suppose that f is given by* (6.14)*: then for any integer q with $q > 2p$, f is analytically conjugate to some function*

$$z \mapsto z(1 + z^p + Az^{2p} + O(z^q)).$$

The proof is essentially the same as that for Theorem 6.10, and we omit the details. By contrast, we also have

Theorem 6.27 (Analytic Conjugation to a Linear Map Modulo z^N)**.** *Suppose that f has a Taylor expansion*

$$f(z) = a_1 z + a_2 z^2 + \cdots,$$

where a_1 is not a root of unity. Then given any positive integer N, there is some function g, analytic near and vanishing at the origin, such that near the origin

$$g^{-1} f g(z) = a_1 z + O(z^N).$$

Proof. See [7, p. 154]. □

We turn now to discuss topological conjugacy and here, of course, the conjugacy classes are much larger.

Remark 6.13 (Fixed points are preserved under topological conjugacy). First, observe that the dynamics of the iterates f^n near an attracting fixed point is invariant under topological conjugacy, for if $f^n(z) \to 0$ and $F = hfh^{-1}$, then $F^n(hz) \to 0$ also, but in this case we cannot talk of the multiplier. In fact, given any two functions f and g (not necessar-

ily analytic) which are orientation-preserving homeomorphisms of some neighborhood of the origin into itself such that $|f(z)| < |z|$ and $|g(z)| < |z|$, then f and g are topologically conjugate [25].

In the case of rationally indifferent fixed points, we have

Theorem 6.28 (Conjugation for finite order iterates)**.** *Suppose that*

$$f(z) = a_1 z + a_2 z^2 + \cdots,$$

where $a_1 = \exp(2\pi i p/q)$, $(p, q) = 1$, *and also that the iterate* f^n *is the identity. Then there is some integer* k *such that near the origin,* f *is analytically conjugate to*

$$z \mapsto a_1 z(1 + z^{kq} + \cdots),$$

and topologically conjugate to $z \mapsto z(1 + z^{kq})$.

The reader is referred to [11] for the details.

Finally, we comment briefly on the question of formal conjugacy, and we mention just one result which sheds more light on the work in this section.

Theorem 6.29 (Voronin's formal conjugacy classification)**.** *Any formal power series about the origin is formally conjugate to one and only one formal series of the form* $z(1 + z^p + Az^{2p})$.

The proof is by repeated applications of Theorem 6.25, and we refer the reader to [28] for more details.

Chapter 7

Forward Invariant Components

In Chapter 8 we shall show that each component of the Fatou set F of R has a forward image that is mapped onto itself by some iterate R^m. Anticipating this, we devote this chapter to the classification of the forward invariant components of the Fatou set of a rational map. Although this classification logically follows Chapter 8, by giving it first we emphasize that it does not depend on the more advanced material in Chapter 8.

7.1 The Five Possibilities

Our objective is to give a complete analysis of the possible forward invariant components of the Fatou set of a rational map. We shall show that such a component can arise in exactly one of five different ways, and we begin with the terminology used to describe these possibilities: as usual, R is a rational map of degree at least two with Fatou set F.

Definition 7.1 (Types of forward invariant components of F)**.** A forward invariant component F_0 of F is:

(a) an *attracting component* if it contains an attracting fixed point ζ of R;

(b) a *super-attracting component* if it contains a super-attracting fixed point ζ of R;

(c) a *parabolic component* (or a *Leau domain*) if there is a rationally indifferent fixed point ζ of R on the boundary of F_0, and if $R^n \to \zeta$ on F_0;

(d) a *Siegel disc* if $R : F_0 \to F_0$ is analytically conjugate to a Euclidean rotation of the unit disc onto itself;

(e) a *Herman ring* if $R : F_0 \to F_0$ is analytically conjugate to a Euclidean rotation of some annulus onto itself.

Of course, in (d) and (e), the rotations necessarily have infinite order for otherwise, some R^n would be the identity and then $\deg(R) = 1$. We know that attracting, super-attracting and parabolic components exist, and the existence of Siegel discs was established in Chapter 6. The proof that Herman rings exist is more difficult, and we comment on this in section 7.4.

The central, and most important, result in this chapter is that (a)-(e) are the only possibilities for a forward invariant component of F, and this is recorded in

Theorem 7.1 (There are only five possibilities for a forward invariant component of F)**.** *A forward invariant component of the Fatou set* $F(R)$ *is one of the types (a)-(d) in Definition 7.1.*

Proof. Theorem 7.1 is simply an amalgamation of several other theorems that occur later in this chapter, but we shall explain the general strategy of the proof now. The five possibilities in Definition 7.1 can almost be distinguished from each other by considering those functions which can be expressed as the limit of some subsequence of R^n in F_0. In (a) and (b), the only possible limit is the fixed point ζ and this is in F_0; in (c), the only limit function is ζ, and this is in ∂F_0; and finally, in (d) and (e), non-constant limit functions exist. To complete the description note that (a) and (b) are distinguished according to whether ζ is, or is not, a critical point, and (d) and (e) are distinguished by the connectivity of F_0, or by the existence of a fixed point in F_0. □

7.2 Limit Functions

We now consider the class of functions which arise as locally uniform limits of subsequences of $(R^n)_{n\in\mathbb{N}}$ in F_0. First (and regardless of whether F_0 is forward invariant or not) we have

Definition 7.2 (Limit function)**.** A function φ is a *limit function* on a component F_0 of $F(R)$ if there is some subsequence of $(R^n)_{n\in\mathbb{N}}$ which converges locally uniformly to φ on F_0. The class of limit functions on F_0 is denoted by $\mathcal{F}(F_0)$.

Remark 7.1 ($\mathcal{F}(F_0)$ is always non-empty and limit functions are analytic). As $\{R^n \mid n \geqslant 1\}$ is normal in F_0, $\mathcal{F}(F_0)$ is always non-empty, and each map in $\mathcal{F}(F_0)$ is analytic in F_0.

Remark 7.2 (F_0 forward invariant $\implies$ $\varphi(F_0) \subset \overline{F_0}$). If F_0 is forward invariant, and if φ is a limit function in F_0, then $\varphi(F_0)$ lies in the closure of F_0: in particular, if φ is a constant, with value ζ say, then $\zeta \in F_0 \cup \partial F_0$.

Lemma 7.2 (Constant limit functions are fixed points)**.** *If F_0 is a forward invariant component of F, and if there exists a constant limit function on F_0 with value ζ, then ζ is a fixed point of R.*

Proof. If z is in F_0, then so is $R(z)$, and so on some sequence of integers n tending to ∞,

$$R(\zeta) = R\left(\lim_{n\to\infty} R^n(z)\right) = \lim_{n\to\infty} R^n(R(z)) = \varphi(R(z)) = \zeta.$$

□

Of course, if the component F_0 is (super)attracting with fixed point ζ, say, then $R^n \to \zeta$ locally uniformly on F_0 and so $\mathcal{F}(F_0)$ contains a single function, namely the constant function with value ζ. The same is true (by definition) if F_0 is a parabolic component as in Definition 7.1(c): note that it is necessary to assume that $R^n \to \zeta$ in this case for a rationally indifferent fixed point can lie on the boundary of a super-attracting component F (for example, when R

is a polynomial).

The next two results enable us to separate (a), (b) and (c) from (d) and (e) but before giving these, we recall

Definition 7.3 (Automorphism)**.** An *automorphism* of a domain D is an analytic bijection of D onto itself. The group of automorphisms of D is denoted by $\mathrm{Aut}(D)$.

Theorem 7.3 (F_0 forward invariant and $\mathcal{F}(F_0)$ has only constant functions $\implies$ $\mathcal{F}(F_0) = \{\zeta\}$ fixed point and $R^n \to \zeta$ locally uniformly on F_0)**.** *Suppose that F_0 is forward invariant and that every function in $\mathcal{F}(F_0)$ is constant. Then $\mathcal{F}(F_0)$ contains exactly one function, with value ζ, say, where $R(\zeta) = \zeta$ and $R^n \to \zeta$ locally uniformly on F_0.*

Proof. See [7, p. 163]. □

Theorem 7.4 (F_0 forward invariant and $\mathcal{F}(F_0)$ has a non-constant function $\implies$ $R \in \mathrm{Aut}(F_0)$ and $I \in \mathcal{F}(F_0)$)**.** *Suppose that F_0 is forward invariant, and that $\mathcal{F}(F_0)$ contains some non-constant function. Then R is in $\mathrm{Aut}(F_0)$, and the identity map I is in $\mathcal{F}(F_0)$.*

Proof. See [7, pp. 163–164]. □

Remark 7.3 (φ is a non-constant limit function, then $\varphi \in \mathrm{Aut}(F_0)$). The same ideas enable one to show (in the proof of Theorem 7.4) that if φ is a non-constant limit function, then $\varphi \in \mathrm{Aut}(F_0)$.

Proof. Indeed, $R \in \mathrm{Aut}(F_0)$ and so has an inverse, say $S : F_0 \to F_0$. Now $\{S^n\}$ is a normal family in F_0 (because the values taken by S^n do not lie in J), thus given any sequence n_j with $R^{n_j} \to \varphi$ locally uniformly in F_0, we can take a subsequence m_j of n_j such that S^{m_j} converges locally uniformly to some function ψ on F_0. Now as φ maps F_0 into itself, it

follows that

$$\psi\varphi(z) = \lim S^{m_j} R^{m_j}(z) = z, \tag{7.1}$$

and hence that φ is injective on F_0. Further, (7.1) shows that ψ is not constant, and exactly as in the proof of Theorem 7.4, we see that ψ maps F_0 into itself. With this, we can reverse the roles of φ and ψ in (7.1) and deduce that $\varphi\psi(z) = z$ on F_0. It follows that φ is a bijection of F_0 onto itself and so is in $\mathcal{A}ut(F_0)$. □

7.3 Parabolic Domains

Suppose now that F_0 is a forward invariant component of $F(R)$ and for the moment, regard ∂F_0 as the ideal boundary point of the one-point compactification of F_0: explicitly,

$$R^n \to \partial F_0 \tag{7.2}$$

means that for each compact subset K of F_0, $R^n(K)$ is disjoint from K for all but a finite set of n. If this is so, then any limit function φ on F_0 satisfies $\varphi(F_0) \subset \partial F_0$, and as ∂F_0 has an empty interior (simply because F_0 is a domain), we deduce (from the Open Mapping Theorem) that φ is necessarily constant. Thus, if (7.2) holds, then by Theorem 7.3, there is some fixed point ζ of R in ∂F_0 such that $R^n \to \zeta$ locally uniformly on F_0. Our next result contains these conclusions and adds one other important fact.

Theorem 7.5 ($R^n \to \partial F_0 \implies F_0$ is a parabolic component)**.** *Suppose that F_0 is a forward invariant component of $F(R)$, and that $R^n \to \partial F_0$ as $n \to \infty$. Then there is some rationally indifferent fixed point ζ of R in ∂F_0 such that:*

(a) $R^n \to \zeta$ *locally uniformly on* F_0 *as* $n \to \infty$*; and*

(b) $R'(\zeta) = 1$.

Proof. See [7, pp. 165–166]. □

Remark 7.4 (If every limit function in $\mathcal{F}(F_0)$ is constant, F_0 is a (super)attracting or parabolic component). Suppose that F_0 is forward invariant and that every limit function in $\mathcal{F}(F_0)$ is constant. Then for some

fixed point ζ, $R^n \to \zeta$ in F_0 and $\zeta \in F_0 \cup \partial F_0$. If $\zeta \in F_0$, then F_0 is a (super)attracting component of $F(R)$ (see the proof of Theorem 7.3), while if $\zeta \in \partial F_0$, then by Theorems 7.3 and 7.5, F_0 is a parabolic component of $F(R)$. Thus, with reference to the proof of Theorem 7.1, we have shown that if every limit function in $\mathcal{F}(F_0)$ is constant, then F_0 is one of the types (a), (b) and (c) in Definition 7.1.

7.4 Siegel Discs and Herman Rings

In this section we prove

Theorem 7.6 (F_0 forward invariant and $\mathcal{F}(F_0)$ has non-constant functions $\implies$ F_0 is a Siegel disc or a Herman ring)**.** *Suppose that F_0 is a forward invariant component of $F(R)$, where $\deg(R) \geqslant 2$, and that $\mathcal{F}(F_0)$ contains non-constant functions. Then F_0 is either a Siegel disc or a Herman ring.*

Proof. See [7, pp. 167–170].
The idea of the proof of Theorem 7.6 is as follows. As J has at least three points, the universal covering space of F_0 is the unit disc Δ, and F_0 is conformally equivalent to Δ/Γ, where Γ is the cover group (of Möbius transformations) acting on Δ. Now the conformal automorphisms R^n of F_0 accumulate at the identity map (see Theorem 7.4), and if we rewrite this in terms of the group Γ and its normalizer, we find that Γ is Abelian. However, the only Abelian cover groups are cyclic, and if we now list all possible cyclic groups and compute the corresponding quotient spaces, we find that Δ/Γ, and hence F_0, is simply or doubly connected. It follows that F_0 is conformally equivalent to either a disc or an annulus and the rest of the argument is straightforward. □

Remark 7.5 (Completion of the proof of Theorem 7.1). With this, we will have completed the proof of Theorem 7.1.

We end this section with some brief remarks on the existence of Herman rings. Let

$$R(z) = \lambda z^2 \left(\frac{1 + \overline{\alpha} z}{z + \alpha} \right), \tag{7.3}$$

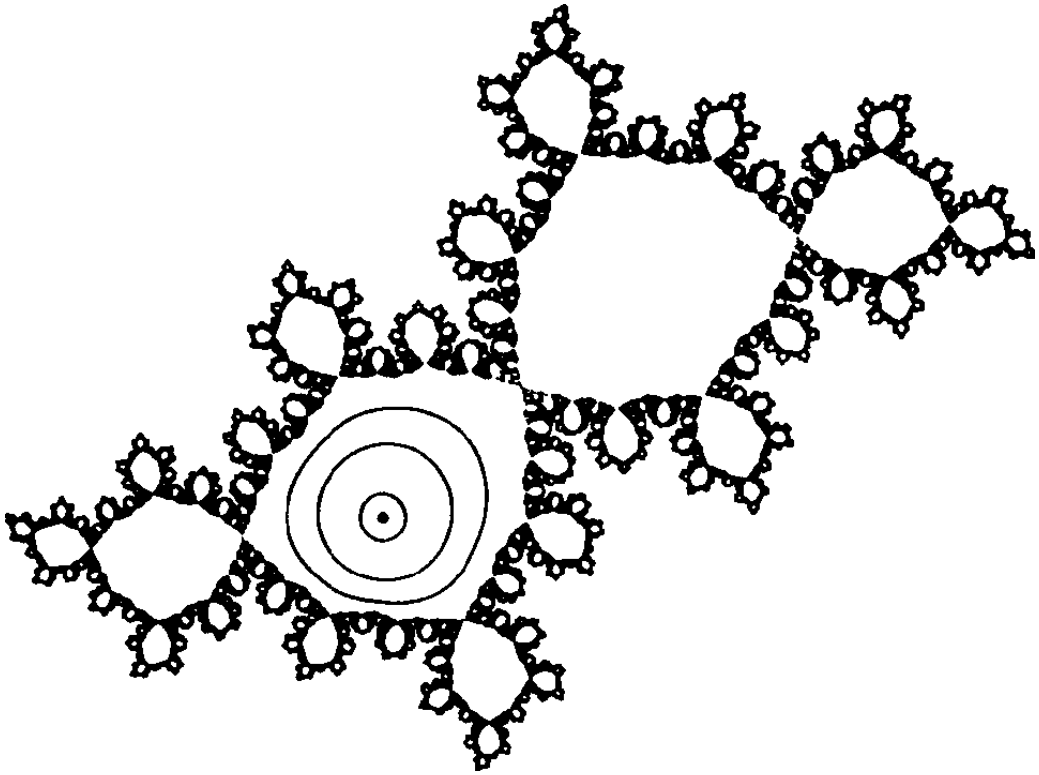


Figure 7.1: $z \mapsto \lambda z + z^2$, $\lambda = \exp(\pi i(\sqrt{5} - 1))$.

where $|\lambda| = 1$ and $0 < |\alpha| < 1$. If $|\alpha|$ sufficiently small, then R is a homeomorphism of the unit circle $\partial\Delta$ onto itself. Now it is known that for suitable choices of λ and α, $R : V \to V$ is analytically conjugate to a rotation (necessarily of infinite order) of $\partial\Delta$ and moreover, this conjugacy extends to an R-invariant neighborhood V of $\partial\Delta$ such that $R : V \to V$ is analytically conjugate to a rotation of an annulus (see [4] and [13]). Clearly, $\{R^n \mid n \geqslant 1\}$ is normal in V, and if F_0 is the component of $F(R)$ which contains V, then $\mathcal{F}(F_0)$ contains non-constant limit functions. We deduce that for these choices of λ and α, F_0 is either a Siegel disc or a Herman ring. As both 0 and ∞ are super-attracting fixed points of R, F_0 is not simply connected and so F_0 must be a Herman ring.

For an alternative construction of Herman rings, and quantitative information about them see [22]; this construction uses quasiconformal maps, and we consider these a part of material in [22] in Chapter 8 and section 9.8 [7].

For computer-generated illustrations of Siegel discs see Figure 7.1, [21], and for Herman rings, see [27]. For further information, see also references in [7, p. 171].

7.5 Connectivity of Invariant Components

This section contains results about the connectivity of a component F_0 of $F(R)$. To prove these, we need two results about critical points, the first of which is

Theorem 7.7 (F_0 contains a (super)attracting fixed point $\implies$ F_0 contains a critical point)**.** *Let* R *be a rational map of degree at least two. If a component* F_0 *of* $F(R)$ *contains a (super)attracting fixed point of* R*, then it also contains a critical point of* R*.*

Proof. See [7, pp. 172–173]. □

The second result about critical points is the corresponding result for Petal domains, namely

Theorem 7.8 (F_0 has a rationally indifferent fixed point on the boundary $\implies$ F_0 contains a critical point)**.** *Let* R *be a rational map of degree at least two, let* ζ *be a rationally indifferent fixed point lying on the boundary of some forward invariant component* F_0 *of* $F(R)$*, and suppose that* $R^n \to \zeta$ *locally uniformly in* F_0*. Then* F_0 *contains a critical point of* R*.*

Although we shall not prove this result until Chapter 9 [7] (which is devoted to the study of critical points), it is convenient (but not necessary) to use it in the proof of the next result. For a proof, readers may now consult Theorem 9.3.2 [7] and its proof, but for those who would prefer not to do this, we have also included an outline of a proof of the next result which does not use Theorem 7.8. We can now start to derive results about the connectivity of forward invariant components of $F(R)$. First, we have

Theorem 7.9 (F_0 forward invariant $\implies$ $c(F_0) \in \{1, 2, \infty\}$)**.** *Let* R *be a rational map of degree at least two. Then any forward invariant component of* $F(R)$ *is simply, doubly, or infinitely, connected.*

Proof. See [7, p. 173]. □

Theorem 7.9 and its proof contains the following results.

Corollary 7.9.1 (F connected $\implies$ $c(F) \in \{1, \infty\}$)**.** *If F is connected, then it is either simply or infinitely connected.*

Corollary 7.9.2 (F_0 has a fixed point $\implies$ $c(F_0) \in \{1, \infty\}$)**.** *If a component of $F(R)$ contains a fixed point of R, then it is either simply or infinitely connected.*

Corollary 7.9.3 (F_0 doubly connected $\iff$ F_0 Herman ring)**.** *A forward invariant component of R is doubly connected if and only if it is a Herman Ring.*

Remark 7.6 (F_0 (super)attracting or parabolic component $\implies$ $c(F_0) \in \{1, \infty\}$). If a forward invariant component F_0 is of one of the types (a), (b) or (c), then according to Theorem 7.9 and Corollary 7.9.3, it must be simply or infinitely connected: in fact, all six possibilities can occur.

We end this section with a sketch of a proof of Theorem 7.9 that does not use Theorem 7.8. This argument is closer in spirit to the proof of Theorem 7.7, it requires more background knowledge, and we omit the details.

For the sketch see [7, pp. 174–175].

Chapter 8

The No Wandering Domains Theorem

This chapter is devoted to the proof of Sullivan's important theorem, namely that for any component Ω of the Fatou set of R, the sequence $\Omega, R(\Omega), R^2(\Omega), \ldots$ is eventually periodic.

8.1 The No Wandering Domains Theorem

Consider the sequence

$$\Omega, R(\Omega), R^2(\Omega), \ldots, R^n(\Omega), \ldots \tag{8.1}$$

of successive images of a component Ω of the Fatou set of a rational map R. This chapter is devoted to the proof that this sequence is eventually periodic (the only alternative being that the regions in (8.1) are pairwise disjoint). We begin with the terminology which enables us to refer briefly to each possibility and, as usual, R is a rational map with $\deg(R) \geqslant 2$.

Definition 8.1 (Types of components of F)**.** A component Ω of the Fatou set $F(R)$ is:

(a) *periodic* if for some positive integer n, $R^n(\Omega) = \Omega$;

(b) *eventually periodic* if for some positive integer m, $R^m(\Omega)$ is periodic; and

(c) *wandering* if the sets $R^n(\Omega)$, $n \geqslant 0$, are pairwise disjoint.

A wandering component of $F(R)$ is called a *wandering domain* of R. Fatou asked whether such domains can exist and in 1985, [26], Sullivan proved that they cannot (for an alternative proof, see [8]).

Theorem 8.1 (The No Wandering Domains Theorem)**.** *Every component of the Fatou set of a rational map is eventually periodic.*

Sullivan's result is probably the most significant advance made in this subject in recent times, and we shall devote this entire chapter to its proof. It implies, for example, that the long-term dynamic of a rational map R on its Fatou set F is the action of R on the periodic components of F, and as a periodic component of F is forward invariant under some R^m, we have already analyzed this situation in Chapter 7. Further important applications arise by combining it with the results on critical points in Chapter 9 [7], for periodic components imply the existence of critical points (see Theorems 7.7 and 7.8), and from the Riemann-Hurwitz relation we know that there are at most $2d-2$ critical points. The rest of this chapter is concerned only with the proof of Theorem 8.1, and the reader who is primarily interested in applications can now pass safely on to Chapter 9.

Although we shall not consider entire functions in this text, we remark that wandering domains can exist for these: see references in [7, p. 177].

8.2 A Preliminary Result

The proof of Theorem 8.1 will be by contradiction so we suppose now that R does have a wandering domain. In this section we shall derive a consequence of this assumption which enables us to simplify Sullivan's original proof by restricting ourselves to simply connected components of $F(R)$: of course, this only contributes towards the required contradiction, and it will have no significance at all once Theorem 8.1 has been proved. We prove

Lemma 8.2 (Properties of wandering domains of R)**.** *Suppose that R has a wandering domain. Then for some component W of $F(R)$, the components*

$$W, R(W), R^2(W), \ldots, R^n(W), \ldots \tag{8.2}$$

of $F(R)$ are pairwise disjoint, simply connected, and contain no critical points of R. In particular, each is mapped by R homeomorphically onto the next.

The heart of the argument lies in showing that the existence of a wandering domain implies the existence of a simply connected wandering domain. We use $\mathrm{diam}[E]$ to denote the diameter of a set E with respect to the spherical metric, and as a first step, we prove

Lemma 8.3 (Compact subsets of wandering disappear)**.** *Suppose that W is a wandering domain. Then for any compact subset K of W, $\mathrm{diam}[R^n(K)] \to 0$ as $n \to \infty$.*

Proof. Suppose that this is false: then there is some compact subset K of W, some positive ε, and some increasing sequence n_j of integers such that for $j = 1, 2, \ldots,$

$$\mathrm{diam}[R^{n_j}(W)] \geqslant \varepsilon. \tag{8.3}$$

As $\{R^n\}$ is normal in W, there is a subsequence of R^{n_j} which converges locally uniformly on W to some analytic function g: for convenience, we relabel this subsequence and so assume that R^{n_j} itself has this property. Now if g is constant, with value α say, on W, then R^{n_j} converges uniformly to α on K and so, for large j, $R^{n_j}(K)$ lies in a $\varepsilon/3$-neighbourhood of α. This contradicts (8.3), and we conclude that R^{n_j} converges to a non-constant g locally uniformly on W.
We now take a point ζ in W where $g'(\zeta) \neq 0$ and draw a small circle C with center ζ which with its interior D lies in W, and which is such that $g(z) \neq g(\zeta)$ when z is on C. Then for $j \geqslant j_0$, say,

$$|R^{n_j}(z) - g(z)| < \inf_{w \in C} |g(w) - g(z)| < |g(z) - g(\zeta)|$$

on C and so by Rouché's Theorem, $R^{n_j}(D)$ contains the point $g(\zeta)$. This contradicts the fact that W is a wandering domain and so completes the proof. □

We can now give the

Proof of Lemma 8.2. First, R has a wandering domain, Ω say, so only a finite number of the sets $R^n(\Omega)$ contain critical points of R. Taking N to be sufficiently large, we let $W = R^N(\Omega)$ and $W_n = R^n(W)$, so W is a wandering domain and no W_j contains any critical points of R. As a

consequence of this, we have

$$\chi(W_n) = m_n\chi(W_{n+1}), \tag{8.4}$$

where R is an m_n-fold map of W_n onto W_{n+1} (Theorem 5.15). Now it is only necessary to prove that W (= W_0) is simply connected for then, from (5.7), $m_0\chi(W_1) = 1$; hence W_1 is also simply connected, $m_0 = 1$ and the argument then proceeds by induction in the same way to drive the simple connectivity of W_n for all subsequent n.
To complete the proof, then, we show that W is simply connected, and again we proceed by contradiction. Given that W is not simply connected, there is a simple closed curve, say γ, in W which is not homotopically trivial in W, and we define $\gamma_n = R^n(\gamma)$. Now

$$R^n : W \to R^n(W)$$

is a smooth unlimited covering map (for by construction, there are no critical points of any of the factors R in the composition R^n), and this ensures that γ_n is not homotopically trivial in $R^n(W)$, for if it were, the Monodromy Theorem would enable us to take the deformation of γ_n into a point of $R^n(W)$ and lift it to a deformation of γ into a point of W.
We now recall Theorem 2.8 and the δ defined in that result. By Lemma 8.3 (with $K = \gamma$) there is some m such that for $n \geqslant m$,

$$\mathrm{diam}[\gamma_n] \leqslant \delta.$$

Now according to Theorem 2.8, if $n \geqslant m$ then R maps the union of γ_n and its interior components into the union of γ_{n+1} and its interior components. Using Theorem 3.16, we deduce that $\{R^n \mid n \geqslant 1\}$ is normal in each interior component of γ_m and so each such component lies in F. It now follows that the union of γ_m and all its interior components is a connected, compact subset of F(R) whose complement is (by Proposition 5.3) simply connected. We deduce that γ_m is homotopic to a point in F(R) (see Exercise 8.2.1 [7]), and hence also in W_n, contrary to our earlier observation. It follows that W is simply connected, and the proof is complete. □

Remark 8.1 (Note about the proof). When (in section 8.6) we proceed to the proof of Theorem 8.1 by contradiction, we shall assume that the (hypothetical) wandering domain has those properties given in Lemma 8.2.

8.3 Conformal Structures

We shall need certain facts about quasiconformal maps and in order to avoid too great a disruption to the proof of Theorem 8.1, we shall adopt a rather axiomatic view and confine ourselves to a brief discussion of the few standard results that we need. The reader is referred to [1], [18] and [19] for the proofs and the background material.

We begin with a generalization of the Cauchy-Riemann equations. Given any function f with continuous partial derivatives in a domain D, we introduce the two differential operators

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right), \quad \frac{\partial f}{\partial z} = \frac{1}{2i}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right)$$

Observe that if f is analytic, then

$$\frac{\partial f}{\partial \bar{z}} = 0, \quad \frac{\partial f}{\partial z} = f'(z),$$

the first of these being the Cauchy-Riemann equations.

Definition 8.2 (Beltrami equation)**.** Our generalization of these equations is the *Beltrami equation*

$$\frac{\partial f}{\partial \bar{z}} = \mu \frac{\partial f}{\partial z}, \tag{8.5}$$

where μ is some suitable complex-valued function on D

The basic idea is that if $\mu = 0$ throughout D, then any sufficiently smooth solution f of (8.5) is analytic in D, while in the general situation, μ is taken as a measure of the deviation of a solution f from conformality. Of course, given a function f on D, we can use (8.5) to define μ.

Definition 8.3 (Complex dilation of f)**.** This μ (when it exists) is called the complex dilatation of f and is denoted by μ_f.

An essential feature of the theory is that it is developed for a very large class of μ; most notably, μ is not required to be continuous and as a result we have a tool of great flexibility. The functions $\mu : D \to \mathbb{C}$ are required to be Lebesgue measurable and to satisfy

$$\|\mu\|_\infty < 1 \tag{8.6}$$

in D, so $|\mu| \leqslant \|\mu\|_\infty$ almost everywhere (that is, except on a set of zero area).

Definition 8.4 (Beltrami coefficient)**.** We shall call any such function μ a *Beltrami coefficient* on D.

Definition 8.5 (Quasiconformal homeomorphism)**.** Given a domain D and a Beltrami coefficient μ on D, we say that a homeomorphism f on D is *quasiconformal* with *complex dilation* μ on D if f is an L^2 solution of (8.5) in D (see [18, p. 24]).

The reader need not worry about the technical nature of the definition as we shall not mention these ideas again; however, some explanatory remarks may be helpful.

Remark 8.2 (Quasiconformal maps are differentiable)*.* First, a quasiconformal map f is differentiable (in the sense that its derivative is a linear map from $\mathbb{R}^2$ to itself) almost everywhere (a.e.) in D, and we only require (8.5) to be satisfied a.e. in D.

Remark 8.3 (Quasiconformal maps move infinitesimal circles to infinitesimal ellipses with bounded eccentricity)*.* Second, the intuitive implication of (8.5) is that for almost all z, the function f maps an infinitesimal circle C centered at z to an infinitesimal ellipse E centered at $f(z)$, where both the eccentricity and the orientation of E are determined by $\mu(z)$. The significance of (8.6) is that it controls this distortion induced by f, and it implies that the eccentricity of E is bounded above throughout D.

We turn now to the basic existence and uniqueness theorem for solutions

of (8.5); see Theorems 4.2 and 4.4 in [18].

Theorem 8.4 (Existence of quasiconformal maps with a given complex dilation)**.** *Let* μ *be a Beltrami coefficient on a domain* D *on the complex sphere. Then:*

(i) there exists a quasiconformal map with complex dilation μ *on* D*; and*

(ii) if φ *and* ψ *are two such maps, then* $\varphi\psi^{-1}$ *is analytic.*

Taking μ to be identically zero and ψ to be the identity we obtain

Corollary 8.4.1 (Quasiconformal with dilation $0 \implies$ conformal)**.** *If* φ *is quasiconformal with dilation* 0 *on* D*, then it is conformal on* D*.*

Remark 8.4 (Measurable Riemann Mapping Theorem). Given a simply connected domain D (conformally equivalent to the unit disc Δ), and a Beltrami coefficient μ on D, then there is a quasiconformal map φ of D onto Δ with dilation μ almost everywhere in D: this is the *Measurable Riemann Mapping Theorem* for the case when μ is identically zero is the classical Riemann Mapping Theorem.

Theorem 8.4 leads naturally to the idea that each μ on D creates a Riemann surface, namely the space D together with a conformal structure on D determined by μ. For a given D and μ, this conformal structure is defined by the atlas $\mathcal{A}(\mu)$ (the totality of charts) consisting of the family of all maps $\varphi : D \to \mathbb{C}_\infty$ which are quasiconformal with dilation μ on D. Theorem 8.4(i) guarantees that each point of D is in the domain of some such φ, and Theorem 8.4(ii) guarantees that the transition map between two different co-ordinate charts is analytic; thus, $\mathcal{A}(\mu)$ defines a conformal structure on D and so converts D into a Riemann surface.

Definition 8.6 (μ-conformal structure)**.** We shall denote this Riemann surface by $D[\mu]$, and we call the structure the μ-*conformal structure* on D.

Of course, if $\mu \equiv 0$, then $D[\mu]$ is simply D with the usual structure as a subdomain of $\mathbb{C}_\infty$, and we usually use D rather than $D[0]$; however, where there may be some gain in clarity, we shall use $D[0]$.

Definition 8.7 (μ-analytic)**.** Given the Riemann surface $D[\mu]$, we say that the map

$$f : D \to \mathbb{C}_\infty$$

is μ-analytic if it is an analytic map between the Riemann surfaces $D[\mu]$ and $\mathbb{C}_\infty$ (with the usual structure).

Definition 8.8 (μ-conformal)**.** If, in addition, f is a homeomorphism, we say that it is μ-conformal on D.

Of course, we retain the usual usage of *analytic* and *conformal* when μ is identically zero.

Remark 8.5 (Analyticity is preserved under composition). Note that if f is μ-analytic on D, and g is analytic on $f(D)$, then (as the composition of analytic maps is analytic), fg is μ-analytic on D.

Quite generally, a homeomorphism φ form an open subset of a Riemann surface $\mathcal{R}$ to $\mathbb{C}_\infty$ is analytic if and only if it is a chart in the maximal atlas for $\mathcal{R}$; thus we record the different points of view in

Lemma 8.5 (Characterizations of μ-conformal)**.** *Let φ be a homeomorphism defined on D. Then the following are equivalent:*

(i) *φ is μ-conformal on D;*

(ii) *φ is a chart for $D[\mu]$;*

(iii) *$\varphi : D[\mu] \to \mathbb{C}_\infty[0]$ is analytic*

Our last task in this section is to formulate the transfer of a conformal structure from one domain to another in terms of complex dilatations. If g is a bijection of a Riemann surface X onto a set Y, then g can be used to transfer the conformal structure on X to Y in a natural, trivial, and unique, way so that $g : X \to Y$ is analytic. We need to express this transfer of structure in terms of Beltrami coefficients and we prove

Lemma 8.6 (μ-conformal structure transfer via Beltrami coefficients)**.** *Suppose that* μ *and* ν *are Beltrami coefficients on the domains* U *and* V *respectively and let* g *be an analytic map of* U *onto* V*. Then:*

(i) $g : U[\mu] \to V[\nu]$ *is analytic; if and only if*

(ii) $\nu(gz) = [g'(z)/\overline{g'(z)}]\mu(z)$ *a.e. in* U*.*

Proof. See [7, p. 182]. □

We shall use Lemma 8.6 to transfer structure in two different ways. First, if g is an analytic homeomorphism from U to V, and if μ is given on U, then we can define ν by (ii) and, as a result, (i) will hold: thus we have transferred the structure from U to V. If we no longer assume g is injective, this process may not be valid as (ii) may provide conflicting definitions of ν at some point of V. However, in this case, if ν is given, we can use (ii) to define μ and again force (i) to be true: thus we have transferred the structure from V to U. Thus, quite generally, we can pull back a conformal structure against the action of an analytic function so that the function remains analytic with respect to the two new structures. We shall use both of these ideas in the next section.

8.4 Quasiconformal Conjugates of Rational Maps

In this section we focus attention on the conjugate $\varphi R\varphi^{-1}$ of a rational map R by some μ-conformal map φ of the complex sphere onto itself. In general $\varphi R\varphi^{-1}$ will not be analytic, and our next result enables us to recognize when it is. We have

Lemma 8.7 (Preservation of rationality via quasiconformal conjugation)**.** *Suppose that* R *is a rational map, and that* φ *is a* μ*-conformal map of the complex sphere onto itself. Then* $\varphi R\varphi^{-1}$ *is rational if and only if*

$$\mu(Rz) = [R'(z)/\overline{R'(z)}]\mu(z) \tag{8.7}$$

a.e. on the sphere, and when this is so, $\deg(\varphi R\varphi^{-1}) = \deg(R)$*.*

Proof. By virtue of Lemma 8.6, condition (8.7) is equivalent to the statement that

$$R : \mathbb{C}_\infty[\mu] \to \mathbb{C}_\infty[\mu]$$

is analytic and this, in turn, is equivalent to the statement that $\varphi R \varphi^{-1}$ is an analytic map of the complex sphere onto itself with the usual structure (for φ is a chart of the Riemann surface $\mathbb{C}_\infty[\mu]$). This proves that (8.7) holds if and only if $\varphi R \varphi^{-1}$ is analytic throughout the sphere, and hence rational. Moreover, $\varphi R \varphi^{-1}$ must have the same degree as R because φ is a homeomorphism of the sphere onto itself and the degree of $\varphi R \varphi^{-1}$ is given by the cardinality of $(\varphi R \varphi^{-1})^{-1}\{w\}$ for most points w. □

We emphasize that (8.7) guarantees that we can conjugate a rational map by a *non-analytic* function φ and yet still remain analyticity.

We shall now incorporate this result into the situation in which a wandering domain exists, and in doing so, we take a significant step forward in our proof of Theorem 8.1. We prove

Lemma 8.8 (Extension of Beltrami coefficients on a wandering domain to $\mathbb{C}_\infty$)**.** *Let R be a rational map, and suppose that W satisfies the conditions of Lemma 8.2. Then any Beltrami coefficient μ on W extends to a Beltrami coefficient on the complex sphere with the property that if φ is any μ-conformal map of the sphere onto itself, then $\varphi R \varphi^{-1}$ is a rational map with the same degree as R.*

Proof. We begin with a Beltrami coefficient μ on W and our objective is to extend μ to the sphere so that the extension satisfies (8.5) a.e. on the sphere. As the sphere is the disjoint union of the completely invariant set $[W]$ and its (completely invariant) complement, say K, we can extend μ to $[W]$ and to K separately and then verify that (8.5) holds on each of these two sets. We define μ to be identically zero on K so (8.5) is certainly satisfied there.

The extension from W to $[W]$ is carried out as follows. First, using Lemma 8.2 and the ideas expressed at the end of the preceding section, we define μ on the (pairwise disjoint) W_n $(= R^n(W))$, $n \geqslant 0$, by ensuring that the map

$$R : W_n[\mu] \to W_{n+1}[\mu], \quad W_n = R^n(W),$$

is analytic. Next, we define μ on successive inverse images of the W_n by the requirement that R provides an analytic map between any component Ω and its image under R (with the μ-structures): this process determines μ uniquely on $[W]$ and is characterized by the fact that for every component Ω of $[W]$, the map

$$R : \Omega[\mu] \to R(\Omega)[\mu]$$

is analytic. As a consequence of this, the functional relation (8.5) holds a.e. on $[W]$ (this is simply Lemma 8.6) and this proves Lemma 8.8. □

8.5 Boundary Values of Conjugate Maps

Our proof of Theorem 8.1 requires a comparison between the boundary values of two functions which are conjugate to each other by a conformal map, so we devote this section to this topic. Let μ be a Beltrami coefficient on the unit disc Δ, let g be a conformal map of Δ onto a simply connected domain W in $\mathbb{C}$, and use g to transfer μ to a Beltrami coefficient ν on W (as in Lemma 8.6). Suppose now that Φ is a ν-conformal map of W onto itself, and that Φ extends to a homeomorphism of the closure of W onto itself such that $\Phi \equiv I$ (the identity map) on ∂W. Then the map

$$\varphi = g^{-1}\Phi g$$

is a μ-conformal map of Δ onto itself, and as such, it extends to a homeomorphism of the closed disc onto itself ([1, p. 47]). Our objective is to study the behavior of φ on $\partial\Delta$: one approach to this problem is through the theory of prime ends, but we shall take a different route.

When comparing maps which are conjugate by a conformal map, it is natural to try to exploit conformal invariants and one such invariant is the hyperbolic metric.

Definition 8.9 ((Hyperbolic) Displacement function of Φ)**.** For any homeomorphism Φ of a simply Ω onto itself, we define the (hyperbolic) *displacement function* of Φ by

$$z \mapsto \rho(\Phi z, z),$$

where ρ is the hyperbolic metric of Ω (which exists unless Ω is confor-

mally equivalent to $\mathbb{C}$ or $\mathbb{C}_\infty$).

Remark 8.6 (Having a bounded displacement is preserved under conformal conjugation). Returning to the discussion above, the conformal equivalence g is an isometry with respect to the hyperbolic metrics in Δ and W and so the property of having a bounded displacement is invariant under conjugation by g: thus if Φ has a bounded displacement o W, then φ has a bounded displacement on Δ (and conversely).

This, however, implies that $\varphi = I$ on $\partial\Delta$ for it implies that $\varphi(z)$ lies in the hyperbolic disc $\Delta(z, d)$ with center z and some (fixed) radius d, and as z converges to any ζ on $\partial\Delta$, the Euclidean radius of $\Delta(z, d)$ tends to zero (see Exercise 8.5.1 [7]).

We now seek a criterion for Φ to have a bounded displacement function in W, and for our purposes, it will be sufficient to assume that ν is a Beltrami coefficient on the entire sphere, and that Φ is a ν-conformal map of the sphere onto itself. With this additional information, we can readily obtain a criterion for Φ to have a bounded displacement function with respect to the chordal metric on the sphere, and we can then convert this to a criterion for Φ to have a bounded hyperbolic displacement on W. First, we have

Lemma 8.9 (Criterion for Φ to have a bounded displacement function with respect to the chordal metric on the sphere)**.** *For each positive ε there is a positive δ such that if the Beltrami coefficient ν on $\mathbb{C}_\infty$ satisfies $\|\nu\| < \delta$, then $\sigma(Fz, z) < \varepsilon$ in $\mathbb{C}_\infty$ for every function F that is a ν-conformal map of $\mathbb{C}_\infty$ onto itself which fixes 0, 1 and ∞.*

Proof. This is a non-sequential version of Theorem 4.6, [18]. If the conclusion is false there is a positive ε, a sequence of ν_n-conformal maps F_n, and a sequence z_n, such that:

(i) F_n fixes 0, 1 and ∞;

(ii) $\|\nu_n\| < 1/n$; and

(iii) $\sigma(F_n z_n, z_n) \geqslant \varepsilon$.

By Theorem 4.6, [18], (i) and (ii) imply that $F_n \to 1$ uniformly on $\mathbb{C}_\infty$,

and this contradicts (iii). □

Now let δ_0 be the value of δ corresponding to $\varepsilon = 1/8$ in Lemma 8.9. Then, we have

Lemma 8.10 (Hyperbolic displacement bound for ν-conformal maps)**.** *Let ρ be the hyperbolic metric of a simply connected subdomain W of $\mathbb{C}_\infty$. If the Beltrami coefficient ν on $\mathbb{C}_\infty$ satisfies $\|\nu\| < \delta_0$, then $\rho(\Phi z, z) < \log(2)$ in W for every function Φ which is a ν-conformal map of $\mathbb{C}_\infty$ onto itself and which satisfies $\Phi(W) = W$, and $\Phi = \mathrm{I}$ on ∂W.*

Proof. As the statement of Lemma 8.10 is invariant under conjugation by a Möbius map, we may assume that $\infty \in \partial W$. Now suppose that W, ρ, ν and Φ satisfy the hypotheses of Lemma 8.10 with $\|\nu\| < \delta_0$.
We take any ζ in W, let $r(\zeta)$ be the (Euclidean) distance between ζ and ∂W, and suppose that this is attained at α in ∂W. As ∂W is a connected set containing α and ∞, there is some point β in ∂W such that

$$|\beta - \alpha| = |\zeta - \alpha| = r(\zeta),$$

and we define the Euclidean similarity h by

$$h(z) = \frac{z - \alpha}{\beta - \alpha}.$$

The function F defined by $F = h\Phi h^{-1}$ is a ν-conformal map of $\mathbb{C}_\infty$ onto itself, and it fixed 0, 1 and ∞ (because Φ fixes α, β and ∞); thus by Lemma 8.9, $\sigma(Fz, z) < 1/8$ on $\mathbb{C}_\infty$. Putting $z = h(\zeta)$, we obtain

$$\sigma(h(\Phi\zeta), h\zeta) < \frac{1}{8} < \sigma(1, 2),$$

and as $|h(\zeta)| = 1$, this implies that $|h(\Phi\zeta)| \leqslant 2$, and hence that

$$\begin{aligned} 2|h(\Phi\zeta) - h\zeta| &= \sigma(h(\Phi\zeta), h\zeta)[(1 + |h(\Phi\zeta)|^2)(1 + |h(\zeta)|^2)]^{\frac{1}{2}} \\ &< \frac{1}{2}. \end{aligned}$$

Using the definition of h, this now yields

$$|\Phi(\zeta) - \zeta| < \frac{r(\zeta)}{4}.$$

Now the disc $\{z\colon |z-\zeta| < r(\zeta)\}$ lies in W, so if ρ_0 denotes the hyperbolic metric of this disc, then, by the Comparison Principle for hyperbolic metrics,

$$\rho(\Phi\zeta,\zeta) \leqslant \rho_0(\Phi\zeta,\zeta) \leqslant \log\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right) \leqslant \log(2)$$

as required. □

In conclusion, if $\varphi : \Delta \to \Delta$ and $\Phi : W \to W$ are as given above with $\varphi = g^{-1}\Phi\varphi$, and if Φ satisfies the hypotheses of Lemma 8.10, then $\varphi = I$ on $\partial\Delta$.

8.6 The Proof of The No Wandering Domains Theorem

In this final section of the chapter, we complete the proof of Theorem 8.1. We assume that R has a wandering domain, so there exists a component W of $F(R)$ with the properties expressed in Lemma 8.2. As W is simply connected (and J is infinite), there is a conformal equivalence g of Δ onto W. Now let μ be any Beltrami coefficient on Δ. Using g, we can transfer μ to ν on W (as in Lemma 8.6), and then extend ν to $\mathbb{C}_\infty$ as in Lemma 8.8. Solving the Beltrami equation with coefficient ν throughout the sphere, we obtain a ν-conformal map φ of the sphere onto itself such that $\varphi R\varphi^{-1}$ is rational (and of the same degree as R), and by combining φ with a Möbius map we may assume that φ fixed 0, 1 and ∞, and so is uniquely determined by μ. In this way we have created the composite map

$$\mu \mapsto \nu \mapsto \varphi \mapsto \varphi R\varphi^{-1} \tag{8.8}$$

of the space of Beltrami coefficients on Δ to the space of rational maps of degree $\deg(R)$.

Roughly speaking, the space of Beltrami coefficients on Δ is infinite dimensional while the space of rational maps of degree $\deg(R)$ is finite dimensional, so the map $\mu \mapsto \varphi R\varphi^{-1}$ must map a large subspace of Beltrami differentials onto a single rational function S. We shall justify this later, but to maintain the flow of ideas, we formulate the result we need as a lemma and continue with the proof of Theorem 8.1. For the moment, then, we assume the validity of

Lemma 8.11. *Suppose that* $\eta_0 > 0$. *For each* t *in* $[0,1]$ *we can construct a Beltrami coefficient* μ_t *in* Δ *such that:*

(i) $\|\mu\|_\infty < \eta_0$; *and*

(ii) (8.8) *maps each* μ_t *to the same rational function* S.

Further, this construction can be made so that for each z, *the map* $t \mapsto \varphi_t(z)$ *is continuous on* $[0,1]$, *where* φ_t *is the image of* μ_t *under the first two maps in* (8.8).

A consequence of this is that for all t in $[0,1]$,

$$\varphi_t R \varphi_t^{-1} = S = \varphi_0 R \varphi_0^{-1},$$

so if we put

$$\Phi_t = \varphi_0^{-1} \varphi_t$$

we find that:

(i) $\Phi_0(z) = z$ on $\mathbb{C}_\infty$;

(ii) for each t, the map $z \mapsto \Phi_t(z)$ commutes with R;

(iii) for each t, the map $z \mapsto \Phi_t(z)$ is a homeomorphism of $\mathbb{C}_\infty$ onto itself;

(iv) for each z, the map $t \mapsto \Phi_t(z)$ is continuous on $[0,1]$.

These properties lead to two striking facts.

Lemma 8.12 (Φ_t maps components to themselves)**.** *For each* t *in* $[0,1]$, $\Phi_t = I$ *on* $J(R)$. *Further,* Φ_t *maps each component of* $F(R)$ *onto itself.*

Proof. For $p = 1, 2, \ldots$, let F_p be the set of fixed points of R^p. By virtue of (ii), if z is in F_p then so is $\Phi_t(z)$; thus Φ_t maps each finite set F_p into itself. Now by (iv), for each z in F_p, the map $t \mapsto \Phi_t(z)$ is a continuous map of $[0,1]$ into the discrete set F_p; thus $\Phi_t(z)$ is independent of t and so, by (i), for all t in $[0,1]$ and all z in F_p, $\Phi_t(z) = z$. We deduce that for each t, Φ_t is the identity map on the union of the F_p, and hence by (iii), also on the closure of this union. As the closure of the periodic points of R contains $J(R)$ (Theorem 4.2.6), the restriction of Φ_t to $J(R)$ is the identity.

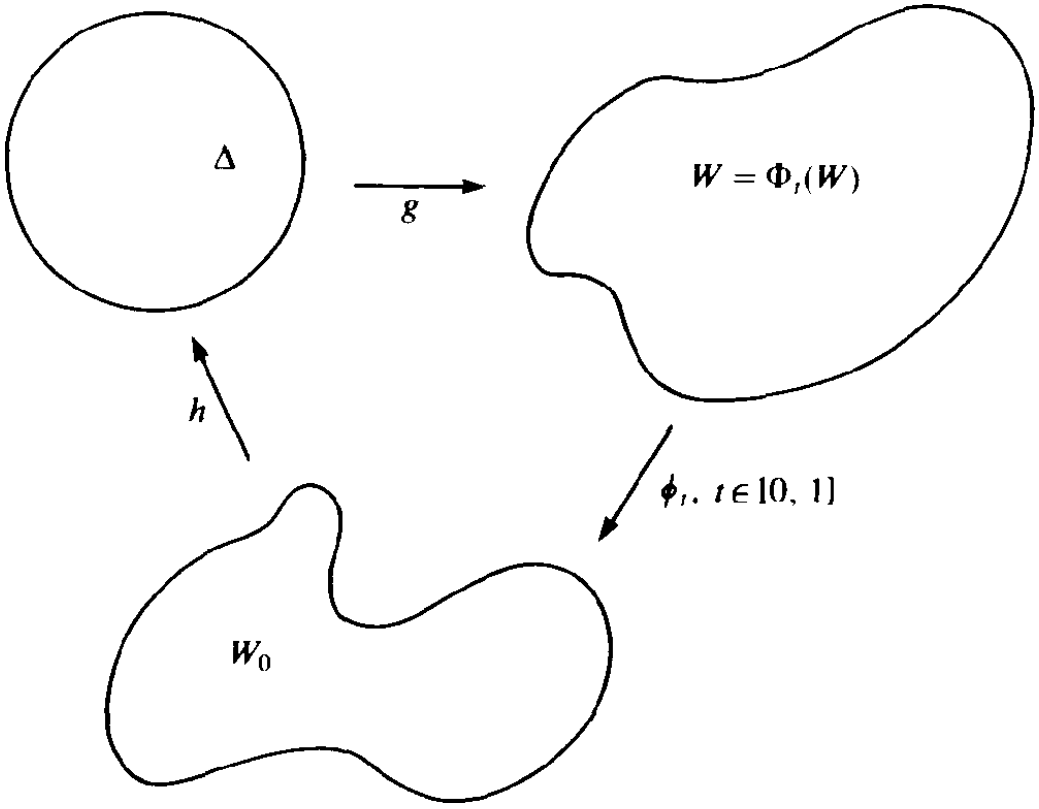


Figure 8.1

Because of this, and because Φ_t is a homeomorphism of the sphere onto itself, each Φ_t must permute the components of $F(R)$. Now take any z in any component F_0 of $F(R)$. By virtue of (i) and (iv), the image of $[0,1]$ under the map $t \mapsto \Phi_t(z)$ is a curve which lies in $F(R)$ and which starts at z. It follows that this curve lies entirely in F_0, and so for each t, Φ_t must map the arbitrary component F_0 into itself. This completes the proof of Lemma 8.12. □

As Φ_t maps each component F_0 of $F(R)$ onto itself, the homeomorphisms φ_t must map W onto the simply connected domain W_0 where $W_0 (= \varphi_0(W))$ is independent of t. It follows that there is a conformal equivalence h of W_0 onto Δ, and we have now reached the situation summarized in Figure 8.1.

Now in the construction of ν (on $\mathbb{C}_\infty$) from μ (on Δ) in (8.8) we have $\|\nu\|_\infty = \|\mu\|_\infty$ (see Lemmas 8.6 and 8.7), so, by virtue of Lemma 8.11(i), $\|\nu_t\|_\infty < \eta_0$ where, of course, μ_t maps to ν_t in (8.8). By taking η_0 small enough, we can ensure that the complex dilatation of $\Phi_t (= \varphi_0^{-1}\varphi_t)$ has L^∞-norm less than δ_0 as given in Lemma 8.10. By this lemma, and the remarks following it, we can now conclude that the map $g^{-1}\Phi_t g$ of Δ onto itself has bounded displace-

ment function and so extends to the identity on $\partial\Delta$.

Now observe that each of the maps in the composition

$$\Delta[\mu_t] \xrightarrow{g} W[\nu_t] \xrightarrow{\varphi_t} W_0[0] \xrightarrow{h} \Delta[0]$$

is analytic, whence $\psi_t(= h\varphi_t g)$ is a μ_t-conformal map of Δ onto itself and so extends to a homeomorphism of the closed disc onto itself. Moreover, in the open disc

$$\psi_0^{-1}\psi_t = g^{-1}\Phi_t g,$$

thus on $\partial\Delta$, $\psi_0^{-1}\psi_t = I$, and so $\psi_0 = \psi_t$.

Now let Ψ_t be any μ_t-conformal map of Δ onto itself. By Theorem 8.4, $\psi_t = M_t\Psi_t$ for some Möbius automorphism M_t of Δ, and so we deduce that

$$M_t\Psi_t = M_0\Psi_0 \tag{8.9}$$

on $\partial\Delta$. In proving Lemma 8.11, we shall construct maps Ψ_t so that $\Psi_t = \Psi_0$ on some open arc of $\partial\Delta$ and this will imply that $M_t = M_0$, and hence that $\Psi_t = \Psi_0$. However, it will be clear from our construction that this is not so, and this will be the contradiction that completes the proof of Theorem 8.1.

It remains, then, to prove Lemma 8.11 and to construct the maps Ψ_t with the properties used above. First, we take an integer N satisfying $N > 4d + 2$ (the space of rational maps of degree d, $d = \deg(R)$ having $4d + 2$ real degrees of freedom). We shall now construct, for each vector T lying in the cube $[0, \varepsilon_1]^N$ (where ε_1 is some small number yet to be chosen), a Beltrami coefficient μ_T on Δ and a μ_T-conformal map Ψ_T of Δ onto itself. In fact, we shall construct the maps Ψ_T first, and then define μ_T to be their complex dilatations.

Divide the interval $[0, 2\pi]$ into $2N$ equal and consecutive arcs $\sigma_1, \tau_1, \ldots, \tau_N, \sigma_N$, and for each of the arcs σ_j we construct a C^∞-function ω_j on $[0, 2\pi]$ such that:

(i) $\omega_j(x) > 0$ on the interior of σ_j while $\omega_j(x) = 0$ otherwise; and

(ii) $|\omega_j'(x)| < \frac{1}{2}$ for all x and j

(see Exercise 8.6.1). Now define the map

$$\Psi_T(z) = z \exp\left[i \sum_{j=1}^{N} t_j \omega_j(\theta) \right] \tag{8.10}$$

where $z = r\exp(i\theta)$ of Δ onto itself, and let the Beltrami coefficient μ_T on Δ be the complex dilatation of Ψ_T.

The complex dilatation μ_T can easily be computed in terms of the partial derivatives $\partial\Psi_T/\partial r$ and $\partial\Psi_T/\partial\theta$ (Exercise 8.6.2) and when this has been done we find that

$$\mu_T(r\exp(i\theta)) = \frac{\exp(2i\theta)\sum_{j=1}^{N} t_j\omega_j(\theta)}{2+\sum_{j=1}^{N} t_j\omega_j(\theta)}, \tag{8.11}$$

and hence for all T,

$$|\mu_T(r\exp(i\theta))| \leqslant \frac{\sum_{j=1}^{N}|t_j\omega_j(\theta)|}{2-\sum_{j=1}^{N}|t_j\omega_j(\theta)|} \leqslant \frac{N\varepsilon_1}{2-N\varepsilon_1}. \tag{8.12}$$

Note that:

(a) if $\arg z$ lies in any of the intervals τ_j, then $\Psi_T(z) = z$;

(b) distinct values of T lead to distinct functions Ψ_T; and

(c) given any η, $\|\mu_T\|_\infty < \eta$ if ε_1 is small enough.

The first two of these are the properties required of the function Ψ_t at the end of the proof of Theorem 8.1, and by choosing ε_1 small enough, (c) guarantees that we can satisfy Lemma 8.11(i) for all μ_T. We fix this value of ε_1, and it remains to complete the proof of Lemma 8.11.

Proof of Lemma 8.11. Given the Beltrami coefficient μ_T, let the composition (8.8) be

$$T \mapsto \mu_T \mapsto \nu_T \mapsto \varphi_T \mapsto R_T = \varphi_T R \varphi_T^{-1}. \tag{8.13}$$

The idea of the proof is to factor the map $T \mapsto R_T$ through the intermediate stage of a vector whose components are the zeros and poles of R_T, and then to show that the first factor, and hence the composition, is constant on some curve.
We may replace R by any conjugate with respect to a Möbius map, so we may assume that R is such that:

(1) R has distinct zeros $a_1, \ldots, a_d$ in $\mathbb{C}$;

(2) R has distinct poles $b_1, \ldots, b_d$ in $\mathbb{C}$; and

(3) $R(0) = 1$.

Now recall that φ_T in (8.13) fixes 0, 1 and ∞. It follows that R_T is the unique rational function with zeros $\varphi_T(a_j)$, with poles $\varphi_T(b_j)$, and with

$R_T(0) = 1$, and so we only have to show that the map

$$\Pi(T) = (\varphi_T(a_1), \ldots, \varphi_T(a_d), \varphi_T(b_1), \ldots, \varphi_T(b_d))$$

of $(0, \varepsilon_1)^N$ into $\mathbb{C}^{2d}$ is constant on some curve.
Now in the fundamental paper [2], Ahlfors and Bers examined the general question of how a solution $z \mapsto \varphi_t(z)$ of the Beltrami equation with coefficient $\mu_t(z)$ varies when the parameter t varies, and they showed that (in an appropriate sense) for each z, the map $t \mapsto \varphi_t(z)$ varies as smoothly as the map $t \mapsto \mu_t(z)$. We shall not enter into the details of this here, and we refer the interested reader to the original paper for the precise formulation of this result. For us, the important outcome is that because the map $T \mapsto \mu_T(z)$ is C^∞, the same is true of $T \mapsto \nu_T(z)$ (in the sense of [2]), and hence each of the maps

$$t \mapsto \varphi_T(a_j)$$
$$t \mapsto \varphi_T(b_j)$$

is C^∞. We conclude that Π is a C^∞-map of the open cube $\mathcal{C}_0 = (0, \varepsilon_1)^N$ into $\mathbb{C}^{2d}$.
The set of points where the rank of Π' is maximal, say k, is an open subset of $\mathcal{C}_0$ (for the rank of Π' is upper semi-continuous) and so the restriction of Π to $\mathcal{C}_0$ is a map of constant rank. It follows from the Implicit Function Theorem (see, for example, [10, p. 79]) that the inverse image of a point is a sub-manifold of $\mathcal{C}_0$ of dimension $N - k$ (which is positive) and so there is some curve in $\mathcal{C}_0$ on which Π is constant. The proof of the lemma, and of Theorem 8.1, is now finished. □

Notation

Basic Math

- fg: Composition of f and g ($f \circ g$).
- f^n: The composition of f with itself n times.
- $I = f^0$: Identity function.
- $f^{(n)}$: n-th derivative of f.
- $\sim$: Equivalence relation.
- $[x]$: Equivalence class of x (in some contexts orbit of x).
- $\mathbb{N}$: Naturals without 0.
- $\mathbb{N}_0$: Naturals with 0.
- $q \mid n$: q divides n.
- $q \nmid n$: q does not divide n.

Complex Analysis

- $\mathbb{C}_\infty$: Extended complex plane.
- Δ: Unit disk in $\mathbb{C}$.
- $R(z)$: Rational function in complex variable z.
- σ: Chordal metric.
- σ_0: Spheric metric.

- γ: A curve.
- $E(\gamma)$: Exterior component of a closed curve.
- $\nu_f(z)$: Valency of f at z.
- $\delta_R(z)$: Deficiency of R at z.
- $Aut(D)$: Group of automorphisms of D.

Topology

- N: Neighborhood of a point.
- $diam(E)$: Diameter of E.
- E^o: Interior of E.
- $\overline{E}$: Closure of E.
- ∂E: Boundary of E.
- E': Derived set of E.
- $\chi(S)$: Euler's characteristic of S.

Complex Dynamics

- ζ: Fixed, periodic or pre-periodic point.
- $\deg(R)$: Degree of R.
- $m(R, z_0)$: Multiplier of R at z_0.
- F or $F(R)$: Fatou set (of R).
- J or $J(R)$: Julia set (of R).
- C or $C(R)$: Set of critical points (of R).
- V or $V(R)$: Set of critical values (of R).

- $\mathbb{E}(R)$: Set of exceptional points of R.
- $\mathcal{O}(z)$ or $[z]$: Orbit of z.
- $\mathcal{O}^-(z)$: Backward orbit of z.
- $\mathcal{O}^+(z)$: Forward orbit of z.
- $\mathbb{F}(F_0)$: The class of limit functions on F_0.

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